Regression and Latent Variable Models for Bounded Outcomes

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This ebook includes worked examples and previously saved statistical results. Readers should consult the accompanying code, source publications, and model diagnostics when interpreting or reproducing the analyses.

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Chapter 1

Introduction

1.1 Why bounded outcomes need their own models

A proportion, a score out of twenty, a rating on a line segment, and a municipal development indicator share a basic feature: their possible values have known limits. Those limits carry information about how the observations were produced and what a fitted model should predict. They also help determine which scientific questions the data can answer.

This text introduces regression and latent variable models for such outcomes. Its four chapters move from continuous bounded regression to bounded counts, bounded item response theory (IRT), and bounded cognitive diagnosis models (CDMs). The examples connect these topics to health research, psychological measurement, and social indicators. Throughout, the central task is to match the response distribution and the model structure to both the measurement process and the research goal.

Consider four questions drawn from the chapters:

1. How does the distribution of a continuous proportion change with an observed covariate, beyond a change in its average?
2. How can a health count with a known maximum be modeled when the binomial distribution does not capture its variability or tails?
3. What can ten bounded personality ratings tell us about a person's position on an unobserved trait, including when some ratings are exactly zero?
4. What combinations of economic, health, and educational conditions are supported by a municipality's continuous indicators?

The common bounded scale connects these questions, but the intended answers require different models.

1.2 Four papers that motivate the text

The course summary in `ResumenJNE2026.tex` identifies four papers as the scientific basis of this text. Each motivates one chapter and connects a modeling problem to an applied example.

Continuous bounded regression. Piccirilli, De Bastiani, and Jorge Luis Bazán (2026), in *Flexible bounded mixed regression models for studies of longitudinal cancer mortality rate in Brazil*,

motivate flexible distributions for continuous bounded outcomes and mixed models for repeated measurements. Their cancer-mortality application connects distributional flexibility with longitudinal dependence. The first chapter develops the distributional construction and uses independent simulations to introduce estimation and interpretation before extensions to mixed models.

Bounded count regression. Bayes et al. (2024), in *A robust regression model for bounded count health data*, motivate the beta-2-binomial model for counts with a known upper limit. The additional shape parameter accommodates features beyond the beta-binomial mean and variance. The second chapter connects this proposal to simulated comparisons and the application to inappropriate days spent in hospital, illustrating separate mean, dispersion, and shape submodels.

Bounded item response theory. Molenaar et al. (2022), in *Zero and one inflated item response theory models for bounded continuous data*, motivate measurement models that retain continuous item scores and explicitly accommodate exact endpoint responses. The third chapter uses the Abasement ratings to connect interior response families and boundary probabilities to item parameters and continuous latent person positions.

Bounded cognitive diagnosis. Oliveira et al. (2026a), in *A cognitive diagnosis model for latent classification of bounded continuous variables*, motivate attribute classification from continuous bounded indicators. The fourth chapter presents B-DINA and its municipal application, where a Q-matrix links indicators to Economy, Health, and Education and the model estimates posterior probabilities for their latent combinations.

Together, these papers establish the text's emphasis on model formulation, parameter interpretation, estimation, and applied implementation. Their common bounded response setting supports different inferential goals: explaining observed outcomes, measuring a continuous trait, and classifying attribute profiles. The chapters also use these developments to motivate further questions about computation, dependence, and diagnostic uncertainty.

1.3 Start with the measurement scale

For a continuous measurement Y' with theoretical bounds $L < U$, the rescaling

$$Y = \frac{Y' - L}{U - L}$$

places observations on the unit interval. Bounds should come from the measurement definition, rather than the smallest and largest values observed in a particular sample. Rescaling preserves the ordering of observations, but it does not determine a probability distribution or turn a discrete score into a continuous measurement.

A continuous proportion strictly between zero and one can be modeled with an interior density, such as a beta distribution. Exact zeros or ones require additional consideration: an ordinary beta density assigns no probability mass to either endpoint. In the IRT chapter, endpoint observations are handled with a mixture that distinguishes boundary responses from interior responses.

A count $C \in \{0, 1, \ldots, m\}$ has a different support. Dividing by m produces a proportion on a finite grid, with the count denominator still carrying information. For example, 1/2 and 10/20 have the same value as proportions but different binomial sampling variances. The bounded count chapter keeps this discrete structure explicit. A known upper limit alone does not justify a binomial model: the interpretation of the score and its response mechanism must also be considered.

Chapter	Example developed in the chapter	Main inferential target
Continuous regression	Beta-generated and logistic skew-normal-generated responses with an observed covariate	Conditional location, variability, and distributional shape
Bounded counts	Binomial, beta-binomial, and beta-2-binomial simulations; an inappropriate-hospital-days application	Covariate effects on bounded counts and extra-binomial variation
Bounded IRT	Abasement ratings from 244 people on 10 adjectives	Continuous latent person positions and item characteristics
Bounded CDMs	19 indicators for 1,668 municipalities in Southeastern Brazil	Posterior probabilities of Economy, Health, and Education profiles

1.4 Continuous regression: explaining a bounded distribution

The first chapter, Bounded Continuous Regression, begins with continuous responses in $(0, 1)$. Brazilian longitudinal cancer mortality rates provide the substantive motivation in the course summary. The chapter's practical analysis uses simulated data, allowing the fitted models to be assessed against known generating mechanisms; it does not reproduce that mortality dataset.

In one simulation, an observed covariate x_i determines a beta conditional mean through

$$\text{logit}(\mu_i) = -0.40 + 1.20x_i.$$

In a second simulation, the same linear predictor controls the location of a skew-normal baseline on the logit-transformed scale. The resulting bounded response belongs to a Johnson-S_B type (JSBt) family. These two scenarios introduce an essential interpretation issue: a parameter called *location* does not always equal the response mean. In the beta model, μ_i is the conditional mean. In the JSBt construction, the baseline location lies on the real line, and its inverse-logit is a response-scale location index whose meaning depends on the chosen family.

Beta regression supplies a natural benchmark (Ferrari and Cribari-Neto 2004). JSBt models broaden the available distributional shapes by transforming real-valued baseline variables to the unit interval (Rodrigues et al. 2020). Within the GAMLSS framework, covariates can affect location, scale, and, where available, shape (Rigby and Stasinopoulos 2005). The motivating mixed regression work also addresses dependence in longitudinal or clustered observations (Piccirilli, De Bastiani, and Jorge Luis Bazán 2026); the chapter's fitted simulation examples focus on independent responses.

The practical comparison asks whether a candidate distribution captures the features of each simulated dataset. AIC and BIC assess penalized fit, while quantile residuals help reveal remaining discrepancies (Dunn and Smyth 1996). The lesson is to interpret coefficients through the actual parameterization and to examine the fitted distribution as well as its central tendency.

1.5 Bounded counts: retaining the known maximum

The second chapter, Bounded Count Regression, considers integer outcomes between zero and a known maximum m. Its first simulation uses $m = 20$ and

$$C_i \mid x_i \sim \text{Binomial}(20, \mu_i), \qquad \text{logit}(\mu_i) = -0.4 + 1.1x_i.$$

Here the baseline model implies

$$E(C_i \mid x_i) = 20\mu_i, \qquad \mathrm{Var}(C_i \mid x_i) = 20\mu_i(1-\mu_i).$$

This relationship makes the binomial a useful starting point and also a restriction to investigate. The chapter compares it with the beta-binomial (BB), which allows the success probability to vary across observational units. In the chapter's mean–dispersion parameterization, both BB and beta-2-binomial (B2B) regression have

$$E(C_i) = m_i\mu_i, \qquad \mathrm{Var}(C_i) = m_i\mu_i(1-\mu_i)\{1+(m_i-1)\delta_i\},$$

conditional on their covariates, where m_i is the known upper count for observation i. The parameter δ_i describes extra-binomial variation. B2B mixes the BB dispersion through a beta distribution and introduces a shape parameter ν_i. At fixed μ_i and δ_i, changing ν_i preserves the mean and variance while allowing differences in higher moments, tails, and boundary probabilities (Bayes et al. 2024). Thus, matching mean and variance alone does not establish that the full count distribution fits. Simulations from all three families illustrate when the added flexibility helps. The chapter also examines identification, monotonicity, skewness, and kurtosis, connecting distributional properties to regression interpretation.

The applied example uses `aep` from `gamlss.data` and concerns inappropriate days spent in hospital (Bayes et al. 2024). The repository reports a B2B fit to 1,383 observations. Its mean submodel includes `ward`, `loglos`, and `year`; its dispersion submodel includes `year` and `ward`; and its shape submodel includes `year`. This structure separates changes in the expected count from changes in extra-binomial variation and distributional shape. The reported mean coefficients associate wards 2 and 3 with lower expected counts than the baseline ward, and associate `loglos` and `year90` with higher expected counts, holding the other mean predictors fixed. These are model-based associations.

The practical workflow uses GAMLSS and `RobustDiscrete`: fit BB first, then use its fitted values, with the required dispersion conversion, to initialize B2B. The conversion matters because GAMLSS's BB scale is not the chapter's δ: the displayed B2B start uses $\delta = \sigma_{\mathrm{BB}}/(1+\sigma_{\mathrm{BB}})$. The B2B likelihood requires numerical integration. The chapter displays the repository application code without evaluating it during rendering and labels its numerical output as reported results. Its exercises extend the workflow to comparisons across R implementations, emphasizing that parameter names and scales may differ between packages.

Model assessment combines AIC and BIC with normalized randomized quantile residuals (Dunn and Smyth 1996) and convergence information. Mean, dispersion, and shape coefficients require separate interpretation on their link scales, followed by inverse-link transformations when reporting response-scale quantities. Unlike a continuous density on $(0,1)$, a bounded count distribution already assigns probabilities to zero and m_i. Their observed frequencies should be compared with the fitted probabilities.

1.6 Bounded IRT: measuring a continuous latent trait

The third chapter, Bounded IRT, changes the goal from explaining an outcome through observed covariates to measuring an unobserved person characteristic through several items. Its Abasement

example uses continuous ratings from 244 people on 10 adjectives. Participants rated how well each adjective described them on a 60-millimeter line segment, subsequently rescaled to $[0, 1]$ (Molenaar et al. 2022). This continuous response format differs from the original Adjective Check List's checklist format.

The example concerns self-critical attitudes and feelings of inadequacy. After the documented reverse scoring, larger responses indicate greater endorsement of those characteristics. Approximately 10.74% of the 2,440 responses are exact zeros, and none are exact ones. Consequently, boundary modeling is a practical part of this application.

For person i and item j, the models link an interior response distribution to a continuous latent trait θ_i through a predictor such as

$$\eta_{ij} = a_j\theta_i + b_j,$$

where $a_j > 0$ describes discrimination and b_j is an easiness parameter under this sign convention. The chapter compares Johnson-S_B, beta, and simplex interior models and introduces zero and one inflation (Molenaar et al. 2022). In a general mixture,

$$\Pr(Y_{ij} = 0) = \pi_{0,ij}, \qquad \Pr(Y_{ij} = 1) = \pi_{1,ij},$$

with weight $1 - \pi_{0,ij} - \pi_{1,ij}$ assigned to an interior density. The Abasement data motivate the zero-inflated case.

Using `BoundIRT`, the chapter develops Bayesian estimation, predictive checking, model comparison, and interpretation of item and person estimates. An overall score summarizes the responses, whereas IRT also describes how items differ and quantifies uncertainty about the latent trait. Reading the example prepares the reader to distinguish a response mean, an item parameter, and a person's latent position.

1.7 Bounded CDMs: learning attribute profiles

The fourth chapter, Bounded CDM, retains continuous bounded indicators but represents the latent structure through discrete attributes. Its municipal application uses 19 indicators and a supplied dataset of 1,668 municipalities in Southeastern Brazil (Oliveira et al. 2026a). The chapter documents a discrepancy with a published introductory count of 1,688 and uses the supplied data's row count.

The indicators describe economic, health, educational, and related living conditions. Coding is essential to interpretation: larger responses are intended to represent more favorable conditions, so an adverse indicator such as inequality is reversed using $1 - \text{GINI}$. A substantive Q-matrix links each indicator to Economy, Health, Education, or a combination of these domains (Oliveira et al. 2026a).

Each municipality has a latent profile

$$\alpha_i = (\alpha_{i,\text{Economy}}, \alpha_{i,\text{Health}}, \alpha_{i,\text{Education}}) \in \{0, 1\}^3.$$

There are eight possible profiles. For example, $(1, 0, 1)$ represents favorable economic and educational conditions and an unfavorable health condition under the model's interpretation. These are inferred latent conditions, rather than observed labels or externally validated policy thresholds.

The bounded DINA (B-DINA) model uses an indicator's Q-matrix row to choose between two beta response regimes. If an indicator requires both Economy and Health, its higher-mean regime applies only when both attributes are present. This conjunctive rule is expressed by

$$\xi_{ij} = \prod_{k:q_{jk}=1} \alpha_{ik}.$$

The two regimes can differ in both mean and precision. The observed indicators remain continuous; the attributes are discrete. Bayesian estimation in JAGS produces posterior probabilities for the profiles, item parameters, and population profile proportions. Predictive checks and MCMC diagnostics are needed before those classifications can be interpreted with confidence.

This example clarifies the difference from bounded IRT. IRT estimates positions on a continuous latent scale; B-DINA estimates probabilities for combinations of attributes. The choice follows the substantive measurement objective. A Q-matrix states proposed measurement relationships and does not establish causal effects among the domains.

1.8 A shared approach to analysis

Across the four chapters, a useful sequence of decisions is:

1. **Describe the observations.** Identify the observational unit, theoretical bounds, discrete or continuous support, endpoints, and any repeated or clustered measurements. Record rescaling and reverse scoring.
2. **State the target.** Decide whether the analysis concerns observed covariate effects, a continuous latent trait, or discrete attribute profiles.
3. **Choose the conditional model.** Match the response distribution and dependence structure to the data. Establish what each parameter means before interpreting its coefficient.
4. **Estimate and check.** Use the chapter's likelihood or Bayesian tools, examine numerical or sampling diagnostics, and compare predicted features with the observed data. Information criteria complement these checks.
5. **Communicate on the relevant scale.** Report fitted bounded responses, expected counts, latent trait estimates, or profile probabilities with appropriate uncertainty and substantive interpretation.

The chapters assume familiarity with regression and probability distributions and introduce the additional distributional and latent variable ideas through worked examples. The GAMLSS R packages support the continuous and count regression analyses (Stasinopoulos and Rigby 2007); the `BoundIRT` and `rjags` R packages, using JAGS, support the measurement applications. Each chapter supplies its own implementation details and bibliography.

Taken together, the examples prepare readers to select a model by its support, parameter interpretation, and inferential purpose. They also motivate further work on flexible distributions, dependence, latent structure, computation, and the assessment of diagnostic uncertainty.

Chapter 2

Bounded Continuous Regression

2.1 Introduction

Bounded continuous responses are observed when the outcome is measured on a finite interval, most commonly the unit interval $(0, 1)$. Examples include proportions, rates, indexes, and transformed mortality measures. Because the response cannot move outside the interval, standard Gaussian linear models may give fitted values outside the support and may fail to represent asymmetry, heavy tails, or strong concentration near the boundaries.

Regression models for bounded responses have been studied from several perspectives. Beta regression is the usual starting point for rates and proportions because the beta distribution is flexible and has an interpretable mean-precision parameterization (Ferrari and Cribari-Neto 2004). Extensions include mixed and mixture beta regression models for clustered continuous bounded responses (Verkuilen and Smithson 2012), Bayesian mixed beta regression (Figueroa-Zúñiga et al. 2013), augmented mixed beta regression for zero/one-inflated proportions (Galvis et al. 2014), and simplex mixed-effects models for longitudinal proportional data (Qiu et al. 2008). These models are important references, but they differ in their regression targets and distributional assumptions. For example, Figueroa-Zúñiga et al. (2013) allow regression structures for both the beta mean and precision. Piccirilli, De Bastiani, and Jorge Luis Bazán (2026) motivate a broader choice of baseline distributions and regression structures for location, scale, and shape, including alternatives for strongly skewed responses and bounded observations affected by outliers.

Rodrigues et al. (2020) discuss a general process for creating bounded response distributions on the unit interval. Their approach starts with a baseline distribution on the real line and combines it with a quantile transformation whose range is the baseline support. In the Johnson-SB type case, the transformation is the logit quantile function, so the bounded response is obtained by transforming a real-valued random variable through the inverse-logit map. Section 4.2 of Rodrigues et al. (2020) also outlines how these generated distributions can be embedded in regression models for bounded responses. That regression proposal is developed in greater detail by Piccirilli, De Bastiani, and Jorge Luis Bazán (2026), who extend the construction to flexible bounded mixed regression models for longitudinal data.

An earlier treatment of bounded mixed regression using Johnson-SB type distributions is given in the master's dissertation of Piccirilli (2021).

The paper by Piccirilli, De Bastiani, and Jorge Luis Bazán (2026) proposes a class of flexible

bounded mixed regression models based on Johnson-SB type distributions. The construction preserves the baseline distribution parameters as location, scale, and, when available, shape parameters. Regression models can then be specified for each distributional parameter using the GAMLSS framework, with random effects included in the location parameter to account for within-unit dependence in longitudinal or clustered data (Rigby and Stasinopoulos 2005).

This notebook summarizes the bounded continuous distributions proposed in Piccirilli, De Bastiani, and Jorge Luis Bazán (2026) and its supplementary material (Piccirilli, De Bastiani, and Jorge L. Bazán 2026), then presents regression models for bounded continuous responses. The practical section uses simulated data to compare the main JSBt regression models with the beta benchmark and with additional JSBt cases generated from logistic, normal, and Student-t baselines. The fitted models assume $0 < Y < 1$; explicit point masses at zero and one require an inflated extension, such as the augmented beta model of Galvis et al. (2014).

2.2 Bounded Continuous Distributions

Let Y be a continuous response variable with support $0 < Y < 1$. The Johnson-SB type (JSBt) construction used in the paper is a transformation model. Let W be a real-valued random variable with cumulative distribution function $G(w;\theta)$ and density $g(w;\theta)$, where θ is the vector of baseline distribution parameters. Let

$$Q_H(y) = \log\left(\frac{y}{1-y}\right), \qquad 0 < y < 1,$$

be the quantile function of the standard logistic distribution. Since $Q_H(y)$ maps $(0,1)$ onto the real line, a bounded distribution for Y is obtained by applying the density transformation

$$f(y;\theta) = g\left\{\log\left(\frac{y}{1-y}\right);\theta\right\}\frac{1}{y(1-y)}, \qquad 0 < y < 1.$$

The corresponding cumulative distribution function is

$$F(y;\theta) = G\left\{\log\left(\frac{y}{1-y}\right);\theta\right\}, \qquad 0 < y < 1.$$

Thus, if W follows one of the real-line baseline distributions, then

$$Y = \frac{\exp(W)}{1+\exp(W)}$$

follows the associated logistic bounded distribution. The location parameter of W controls the position of the density on $(0,1)$, the scale parameter controls dispersion, and additional shape parameters can control skewness, tail behavior, and kurtosis.

For any member of this family, the moments can be expressed through the baseline distribution. Let

$$s(w) = \frac{\exp(w)}{1+\exp(w)}$$

denote the inverse-logit transformation. Since $Y = s(W)$, the r-th moment is

$$\mathbb{E}(Y^r) = \int_{-\infty}^{\infty} \{s(w)\}^r g(w;\theta)\, dw = \int_0^1 y^r f(y;\theta)\, dy.$$

Closed-form expressions are not generally available for these moments, so numerical or Monte Carlo evaluations are used for selected parameter values (Piccirilli, De Bastiani, and Jorge Luis Bazán 2026; Piccirilli, De Bastiani, and Jorge L. Bazán 2026).

2.2.1 Logistic Gumbel Distribution

The logistic Gumbel distribution uses the Gumbel distribution as the baseline distribution. If W has density

$$g(w;\mu,\sigma) = \frac{1}{\sigma} \exp\left[\frac{w-\mu}{\sigma} - \exp\left\{\frac{w-\mu}{\sigma}\right\}\right], \qquad -\infty < w < \infty,$$

with $\mu \in \mathbb{R}$ and $\sigma > 0$, then $Y = \exp(W)/\{1+\exp(W)\}$ has density

$$f(y;\mu,\sigma) = \frac{1}{\sigma y(1-y)} \exp\left[\frac{\log\{y/(1-y)\}-\mu}{\sigma} - \exp\left\{\frac{\log\{y/(1-y)\}-\mu}{\sigma}\right\}\right].$$

The parameter μ is a location parameter on the logit scale. Larger values of μ shift the distribution toward larger values of Y. The parameter σ is a scale parameter and controls the spread of the distribution.

2.2.2 Logistic Reverse Gumbel Distribution

The logistic reverse Gumbel distribution uses the reverse Gumbel baseline distribution. If W has density

$$g(w;\mu,\sigma) = \frac{1}{\sigma} \exp\left[-\frac{w-\mu}{\sigma} - \exp\left\{-\frac{w-\mu}{\sigma}\right\}\right],$$

then the transformed response $Y = \exp(W)/\{1+\exp(W)\}$ has support $(0,1)$ and density

$$f(y;\mu,\sigma) = \frac{1}{\sigma y(1-y)} \exp\left[-\frac{\log\{y/(1-y)\}-\mu}{\sigma} - \exp\left\{-\frac{\log\{y/(1-y)\}-\mu}{\sigma}\right\}\right].$$

As in the logistic Gumbel model, μ controls location and σ controls dispersion. The reversed baseline changes the direction of skewness and provides a complementary bounded model.

2.2.3 Logistic Exponential Gaussian Distribution

The logistic exponential Gaussian distribution uses the exponential Gaussian baseline distribution. This baseline introduces a third parameter $\nu > 0$, in addition to $\mu \in \mathbb{R}$ and $\sigma > 0$. After the same logit transformation, the bounded density can be written as

$$f(y;\mu,\sigma,\nu) = \frac{1}{\nu y(1-y)} \exp\left[\frac{\mu - \log\{y/(1-y)\}}{\nu} + \frac{\sigma^2}{2\nu^2}\right] \Phi\left[\frac{\log\{y/(1-y)\} - \mu}{\sigma} - \frac{\sigma}{\nu}\right],$$

where $\Phi(\cdot)$ is the standard normal distribution function. In the supplementary material (Piccirilli, De Bastiani, and Jorge L. Bazán 2026), μ controls location, σ controls dispersion, and ν affects kurtosis and, to a smaller degree, skewness.

2.2.4 Logistic Power Exponential Distribution

The logistic power exponential distribution uses the power exponential baseline distribution. It is useful because the baseline distribution has a shape parameter $\nu > 0$ that can generate lighter or heavier tails. If

$$z = \frac{\log\{y/(1-y)\} - \mu}{c\sigma}, \qquad c^2 = \Gamma(1/\nu)\{\Gamma(3/\nu)\}^{-1},$$

then the bounded density is

$$f(y;\mu,\sigma,\nu) = \frac{\nu \exp(-|z|^\nu)}{2c\sigma\Gamma(1/\nu)y(1-y)}, \qquad 0 < y < 1.$$

For this distribution, the median has a simple closed form:

$$\mathrm{Med}(Y) = \frac{\exp(\mu)}{1+\exp(\mu)}.$$

This makes the location parameter especially interpretable after inverse-logit transformation.

2.2.5 Logistic Skew-Normal Type I Distribution

The logistic skew-normal type I distribution uses a skew-normal baseline distribution. If

$$z = \frac{\log\{y/(1-y)\} - \mu}{\sigma},$$

then the bounded density is

$$f(y;\mu,\sigma,\nu) = \frac{2}{\sigma y(1-y)} \phi(z)\Phi(\nu z),$$

where $\phi(\cdot)$ and $\Phi(\cdot)$ are the standard normal density and distribution function. Here μ controls location, σ controls dispersion, and $\nu \in \mathbb{R}$ controls skewness.

2.2.6 Logistic Skew-Normal Type II Distribution

The logistic skew-normal type II distribution is another JSBt model mentioned by Piccirilli, De Bastiani, and Jorge Luis Bazán (2026). It is based on a two-piece skew-normal baseline distribution. If

$$z = \frac{\log\{y/(1-y)\} - \mu}{\sigma},$$

then the bounded density can be written as

$$f(y;\mu,\sigma,\nu) = \frac{1}{y(1-y)}\frac{\nu}{\sigma(1+\nu^2)}\sqrt{\frac{2}{\pi}}\exp\left[-\frac{1}{2}\left\{\nu|z|\right\}^2\right], \qquad \log\left(\frac{y}{1-y}\right) < \mu,$$

and

$$f(y;\mu,\sigma,\nu) = \frac{1}{y(1-y)}\frac{\nu}{\sigma(1+\nu^2)}\sqrt{\frac{2}{\pi}}\exp\left[-\frac{1}{2}\left\{\frac{|z|}{\nu}\right\}^2\right], \qquad \log\left(\frac{y}{1-y}\right) \geq \mu,$$

with $\mu \in \mathbb{R}$, $\sigma > 0$, and $\nu > 0$. The value $\nu = 1$ gives the symmetric normal baseline, while values below or above one generate skewness by allowing different scales on the two sides of the location parameter.

In summary, the JSBt class is useful because it provides several bounded continuous distributions with different tail and skewness behavior. The models retain the baseline distribution parameters, which can then be modeled through covariates in a regression setting. The logistic, normal, and Student-t baselines mentioned by Rodrigues et al. (2020) are also members of the same JSBt family. They correspond, respectively, to logistic-logistic, logistic-normal, and logistic-t bounded distributions obtained by applying the same logit transformation to different real-line baseline distributions.

The following R code compares the probability density functions for the JSBt distributions when $\mu = 0$ and $\sigma = 1$. The LGU and LRG distributions have only these two parameters, while the remaining three-parameter distributions are shown for different values of the shape parameter ν.

Figure 1 shows that, for $\mu = 0$ and $\sigma = 1$, the JSBt families can have very different shapes even though they are generated through the same inverse-logit transformation. The LGU and LRG densities are mirror-like asymmetric cases: LGU places more density on smaller and middle response values before dropping near one, whereas LRG places more density on middle and larger response values after rising from zero. For LEG, larger values of ν concentrate the density strongly near the right boundary. In LPE, small ν produces a sharp central peak, while larger ν gives a flatter density with more mass away from the center. In LSN1, negative and positive values of ν move the skewness in opposite directions, while $\nu = 0$ gives the symmetric logistic-normal case centered around 0.5. LSN2 also controls skewness, but through a positive shape parameter: $\nu = 1$ gives the symmetric normal baseline, whereas values below or above one create different left and right tail scales.

The next two figures keep $\sigma = 1$ and the same values of ν, but change μ to show its effect on the location of the density.

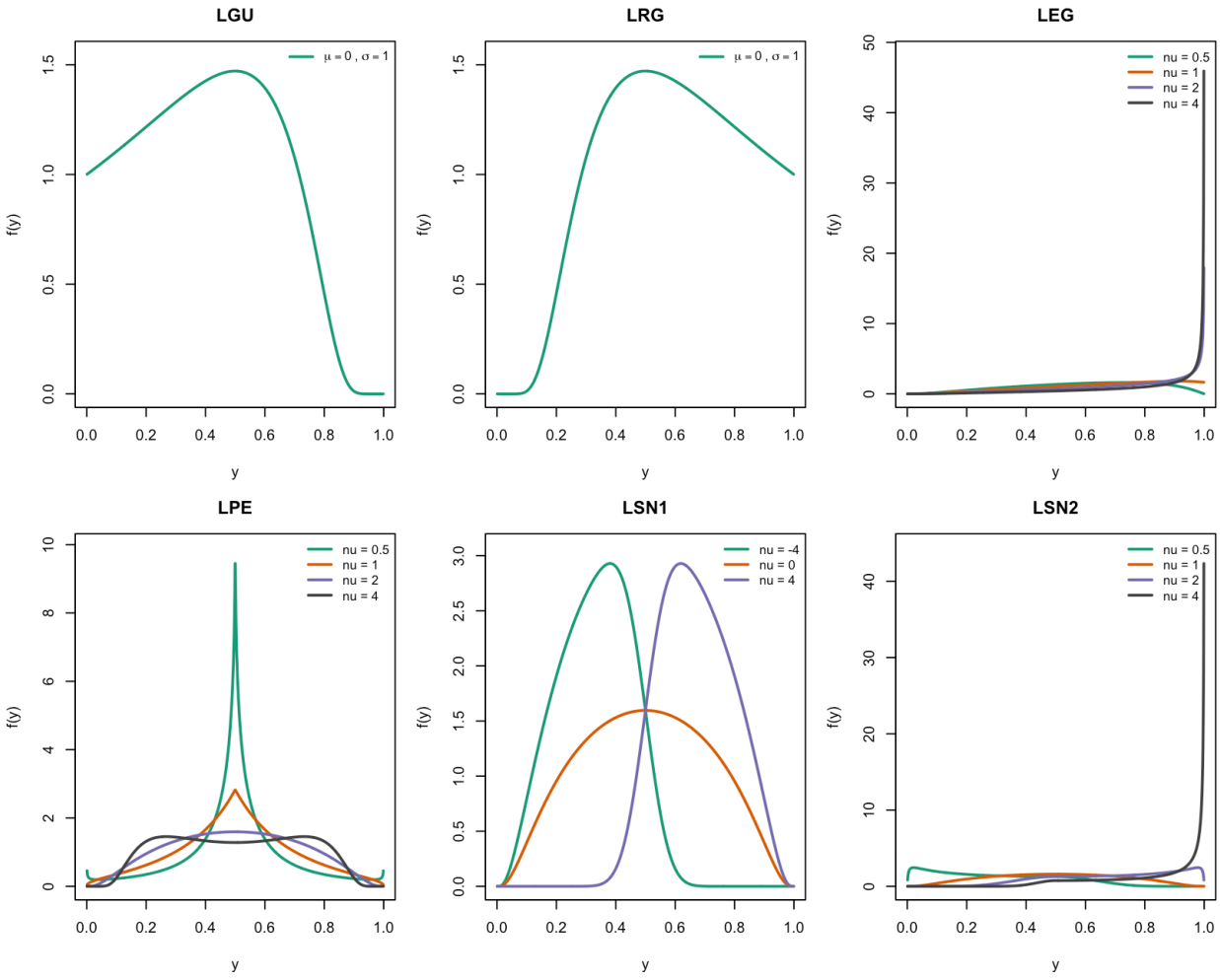


Figure 1: Probability density functions of the JSBt distributions for $\mu = 0$, $\sigma = 1$, and, for three-parameter models, selected values of ν.

```
plot_jsbt_comparison(mu_value = -1, sigma_value = 1)
```

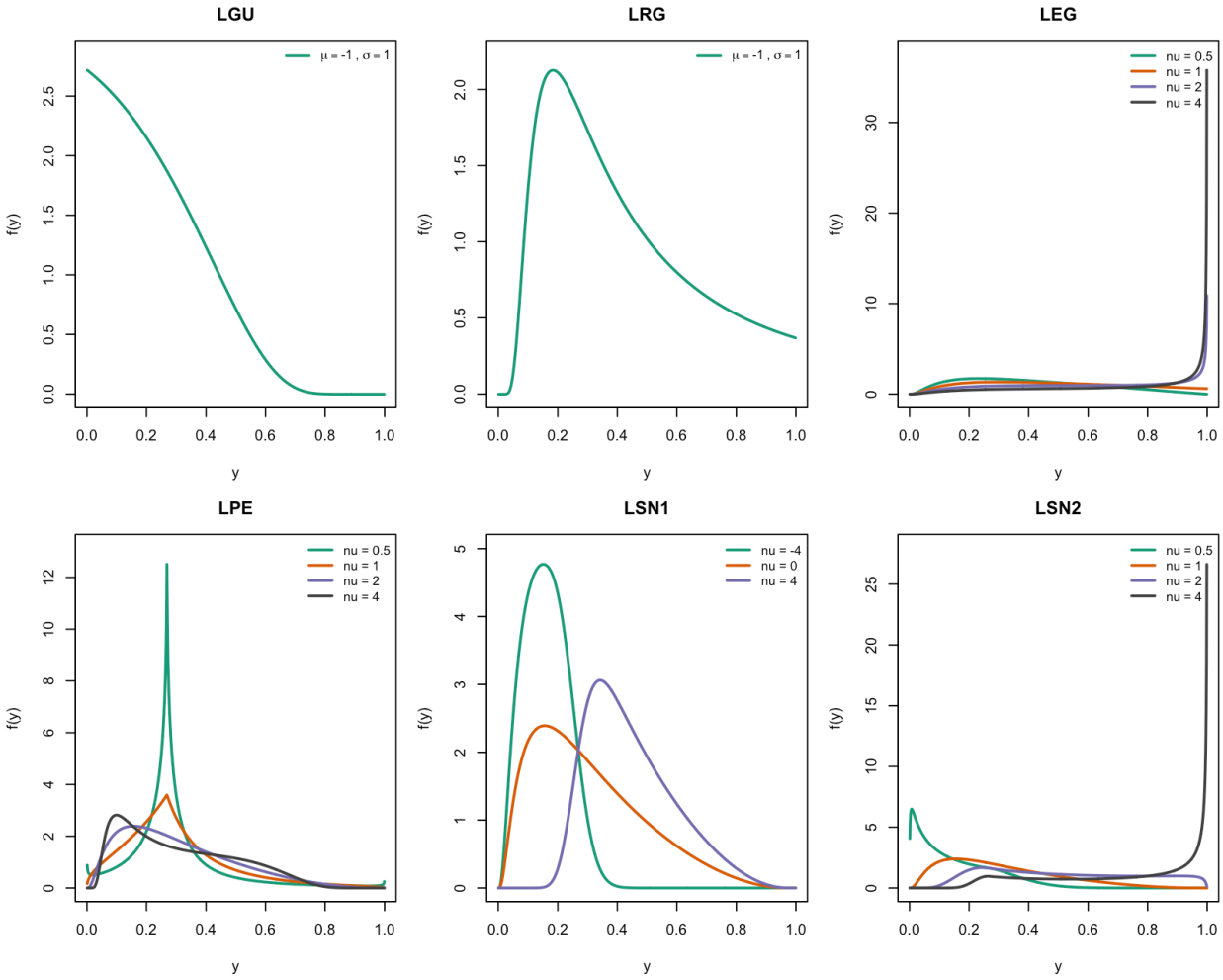


Figure 2: Probability density functions of the JSBt distributions for $\mu = -1$, $\sigma = 1$, and, for three-parameter models, selected values of ν.

Figure 2 shows the effect of decreasing the location parameter to $\mu = -1$. Since μ is defined on the logit scale, this shifts the baseline random variable W to the left and therefore moves the bounded response $Y = \exp(W)/\{1 + \exp(W)\}$ toward smaller values. The modes and high-density regions move toward the lower part of $(0, 1)$, although the amount of movement depends on the baseline family and on ν. The shape parameter still controls tail weight, boundary concentration, and skewness, but these shape effects now occur around a lower response level.

```
plot_jsbt_comparison(mu_value = 1, sigma_value = 1)
```

Figure 3 shows the corresponding rightward shift when $\mu = 1$. Increasing μ moves the distribution of W to the right, so the inverse-logit transformation assigns more density to larger values of Y. Compared with Figure 2, the density shapes are displaced toward the upper part of the unit interval. Thus, μ acts mainly as a location parameter on the logit scale: changing μ shifts the response distribution left or right, whereas σ controls dispersion and ν controls shape, tail behavior, boundary concentration, or skewness depending on the chosen JSBt family.

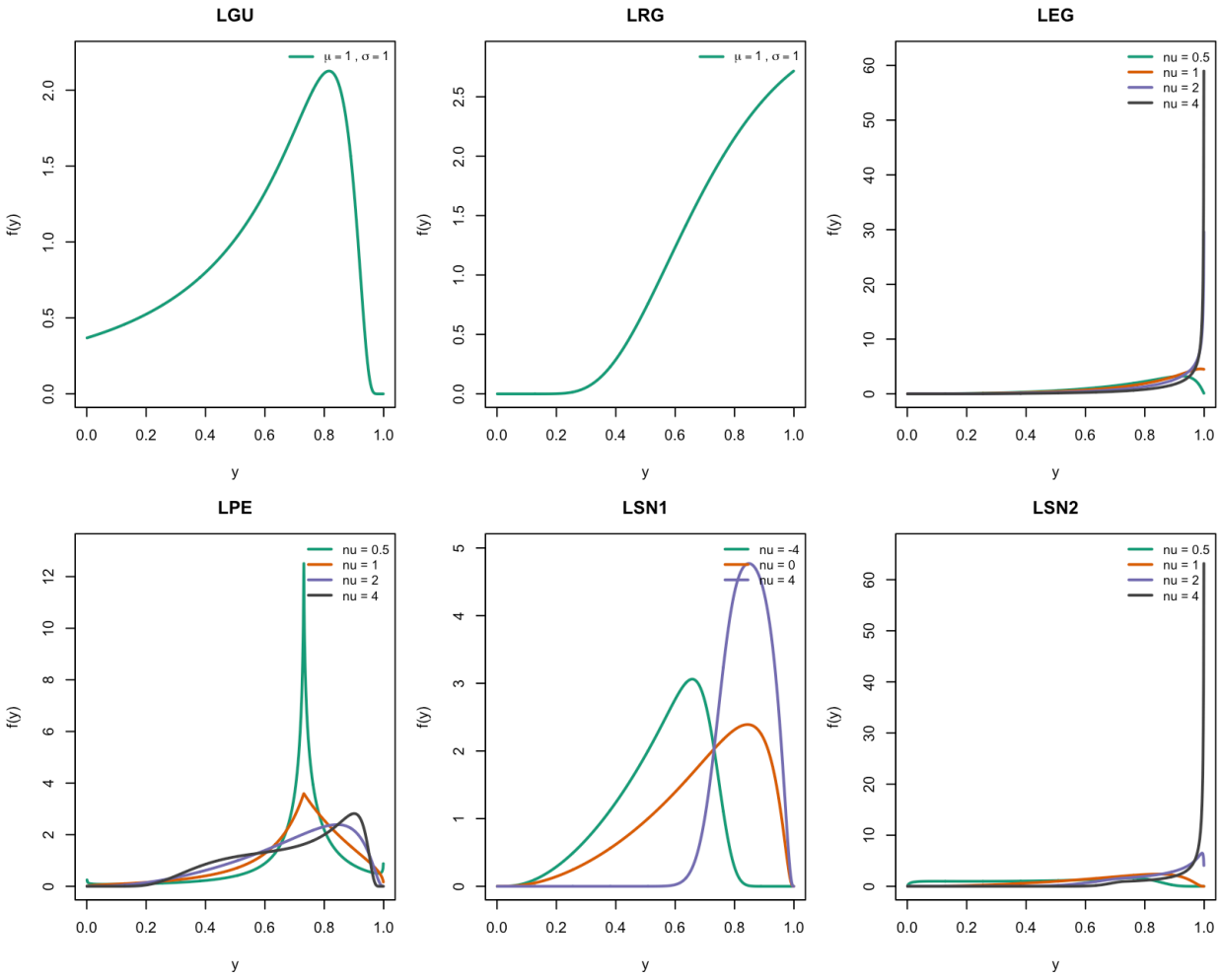


Figure 3: Probability density functions of the JSBt distributions for $\mu = 1$, $\sigma = 1$, and, for three-parameter models, selected values of ν.

Taken together, Figures 1–3 indicate that μ should be interpreted as a location parameter before the bounded transformation is applied. A smaller value of μ moves the density toward zero, and a larger value moves it toward one. The parameter σ is a scale parameter on the same logit scale. For fixed μ, increasing σ makes the underlying distribution of W more dispersed; after transformation to $(0, 1)$, this appears as a wider bounded density and, as shown in Table 1, a larger response-scale variance. Because the inverse-logit transformation is nonlinear, changes in σ can also affect the response-scale mean, especially for asymmetric distributions or when the density is close to the boundaries.

Table 1 compares the JSBt distributions considered in this notebook. The mean and variance values were computed numerically from the corresponding bounded densities with $\mu \in \{-1, 0, 1\}$ and $\sigma \in \{1, 2\}$. For three-parameter distributions, the additional shape parameter was fixed at a reference value: $\nu = 1$ for LEG, LPE, and LSN2, and $\nu = 0$ for LSN1.

Table 1: Mean and variance of the JSBt distributions for fixed reference values.

			$\mu = -1$		$\mu = 0$		$\mu = 1$	
Bounded distribution	Baseline distribution (parameters)	Reference values	$\mathbb{E}(Y)$	$\mathrm{Var}(Y)$	$\mathbb{E}(Y)$	$\mathrm{Var}(Y)$	$\mathbb{E}(Y)$	$\mathrm{Var}(Y)$
Logistic Gumbel	Gumbel (2) $\mu \in \mathbb{R}$, $\sigma > 0$	$\sigma = 1$	0.2283	0.0252	0.4037	0.0480	0.5964	0.0565
(LGU)		$\sigma = 2$	0.2467	0.0644	0.3786	0.0962	0.5180	0.1131
Logistic Reverse Gumbel	Reverse Gumbel (2) $\mu \in \mathbb{R}$, $\sigma > 0$	$\sigma = 1$	0.4036	0.0565	0.5963	0.0480	0.7717	0.0252
(LRG)		$\sigma = 2$	0.4820	0.1131	0.6214	0.0962	0.7533	0.0644
Logistic Exponential Gaussian	Exponential Gaussian (3) $\mu \in \mathbb{R}$, $\sigma > 0$, $\nu > 0$	$\sigma = 1$, $\nu = 1$	0.4889	0.0642	0.6708	0.0500	0.8190	0.0260
(LEG)		$\sigma = 2$, $\nu = 1$	0.4948	0.1075	0.6351	0.0967	0.7595	0.0709
Logistic Power Exponential	Power Exponential (3) $\mu \in \mathbb{R}$, $\sigma > 0$, $\nu > 0$	$\sigma = 1$, $\nu = 1$	0.2987	0.0299	0.5000	0.0371	0.7013	0.0299
(LPE)		$\sigma = 2$, $\nu = 1$	0.3373	0.0730	0.5000	0.0809	0.6627	0.0730
Logistic Skew-Normal Type I	Skew-normal type I (3) $\mu \in \mathbb{R}$, $\sigma > 0$, $\nu \in \mathbb{R}$	$\sigma = 1$, $\nu = 0$	0.3033	0.0334	0.5000	0.0434	0.6967	0.0334
(LSN1)		$\sigma = 2$, $\nu = 0$	0.3523	0.0877	0.5000	0.0986	0.6477	0.0877
Logistic Skew-Normal Type II	Skew-normal type II (3) $\mu \in \mathbb{R}$, $\sigma > 0$, $\nu > 0$	$\sigma = 1$, $\nu = 1$	0.3033	0.0334	0.5000	0.0434	0.6967	0.0334
(LSN2)		$\sigma = 2$, $\nu = 1$	0.3523	0.0877	0.5000	0.0986	0.6477	0.0877

The table also illustrates that $\mu = 0$ does not necessarily imply $\mathbb{E}(Y) = 0.5$. The location parameter is defined on the baseline logit scale, and the response-scale mean depends on the full baseline distribution, including its skewness and tail behavior. The additional rows with $\sigma = 2$ show the role of the scale parameter. Increasing σ generally increases $\mathrm{Var}(Y)$ and spreads the density over a wider part of the unit interval. Because the transformation from the real line to $(0, 1)$ is nonlinear, this increase in spread can also pull the response-scale mean toward the interior of the interval, especially when μ is far from zero.

2.2.7 Identification

For bounded continuous JSBt distributions, identification refers to whether different parameter values imply different probability density functions on $(0, 1)$. The transformation

$$Y = s(W) = \frac{\exp(W)}{1 + \exp(W)}$$

is one-to-one and strictly increasing, with inverse $Q_H(y) = \log\{y/(1-y)\}$. Therefore, identification of the bounded density is inherited from identification of the baseline density on the real line. If two parameter vectors θ_1 and θ_2 produce the same bounded density,

$$f(y;\theta_1) = f(y;\theta_2), \qquad 0 < y < 1,$$

then, after multiplying by $y(1-y)$ and writing $w = Q_H(y)$, the corresponding baseline densities satisfy

$$g(w;\theta_1) = g(w;\theta_2), \qquad -\infty < w < \infty.$$

Thus, provided the baseline family is identifiable, the induced JSBt family is identifiable.

The two-parameter LGU and LRG distributions are identified by their location and scale parameters, μ and σ. These parameters do not correspond directly to the response-scale mean and standard deviation of Y; instead, they describe the location and scale of the latent real-line variable W. The response-scale mean and variance are functions of the full baseline density and the inverse-logit transformation.

The three-parameter LEG, LPE, LSN1, and LSN2 distributions add a shape parameter ν. In these models, ν is identified through features of the full density beyond location and scale, such as skewness, kurtosis, tail behavior, and concentration near the boundaries of $(0, 1)$. In applications, theoretical identification does not guarantee stable estimation. When the sample has little information in the tails or near the boundaries, σ and ν may be weakly separated in practice because both can affect the apparent spread and shape of the fitted density.

2.2.8 Monotonicity

For bounded continuous JSBt distributions, monotonicity in μ is best understood as stochastic monotonicity, not as pointwise monotonicity of the density. Since μ is a location parameter on the logit scale, for fixed σ and fixed ν when present, increasing μ shifts the baseline distribution of W to the right. Because the inverse-logit transformation is increasing, the induced distribution of $Y = s(W)$ also shifts to the right.

Equivalently, for any fixed cutoff $0 < y < 1$,

$$\Pr(Y > y) = \Pr\{W > Q_H(y)\}$$

is nondecreasing as μ increases, when all other parameters are held fixed. Thus, larger values of μ place more probability on larger response values. This is the behavior shown in Figures 1–3: decreasing μ to -1 moves the densities toward zero, while increasing μ to 1 moves them toward one.

This monotonicity does not mean that the density value $f(y)$ increases at every point y when μ increases. Density curves can cross because probability mass is being redistributed across the unit interval. Also, if σ or ν changes at the same time as μ, the resulting densities need not be ordered by μ alone. Changes in σ alter dispersion, and changes in ν alter shape, tail behavior, boundary concentration, or skewness. Therefore, monotonicity in μ should be interpreted conditionally on the other distributional parameters being fixed.

2.2.9 Skewness and Kurtosis

Skewness and kurtosis describe aspects of the JSBt distributions that are not captured by the location and scale parameters alone. For $Y = s(W)$, the raw moments are

$$m_r = \mathbb{E}(Y^r) = \int_{-\infty}^{\infty} \{s(w)\}^r g(w;\theta)\, dw, \qquad r = 1, 2, 3, 4.$$

The standardized skewness and kurtosis are

$$\text{Skew}(Y) = \frac{\mathbb{E}\{Y - \mathbb{E}(Y)\}^3}{\{\text{Var}(Y)\}^{3/2}},$$

and

$$\text{Kurt}(Y) = \frac{\mathbb{E}\{Y - \mathbb{E}(Y)\}^4}{\{\text{Var}(Y)\}^2},$$

where kurtosis denotes the standardized fourth central moment, not excess kurtosis. These quantities are generally evaluated numerically because the inverse-logit transformation makes closed-form expressions unavailable for most JSBt families.

The six JSBt distributions differ in how skewness and kurtosis arise. LGU and LRG are complementary asymmetric distributions: the reversed Gumbel baseline changes the direction of skewness after transformation to $(0, 1)$. LEG introduces additional shape flexibility through $\nu > 0$, which can strongly affect boundary concentration and tail weight. LPE uses $\nu > 0$ to control tail thickness and peakedness; small values of ν can produce a sharp central peak, while larger values produce flatter densities. LSN1 uses $\nu \in \mathbb{R}$ as a skewness parameter: negative and positive values move the density in opposite directions, while $\nu = 0$ gives the symmetric logistic-normal case. LSN2 uses a positive skewness parameter; $\nu = 1$ gives the symmetric normal baseline and values different from one create two-piece asymmetry.

The figures and Table 1 show why skewness and kurtosis matter for bounded continuous modeling. Two distributions can have similar location or even similar mean values but very different tail behavior and concentration near zero or one. Consequently, the choice among LGU, LRG, LEG, LPE, LSN1, and LSN2 should be based not only on the fitted mean, but also on the shape of the full density and on diagnostics that reveal asymmetry, tail weight, and boundary behavior.

2.3 Regression Models for Bounded Continuous Responses

2.3.1 JSBt Regression Formulation of Rodrigues et al. (2020)

Rodrigues et al. (2020) illustrate how the JSBt construction can be used to formulate regression models for bounded continuous responses under the likelihood approach. For independent observations (z_i, y_i), $i = 1, \ldots, n$, with $0 < y_i < 1$, they consider

$$Y_i \sim \text{JSB}(\theta_i; (0, 1)), \qquad i = 1, \ldots, n,$$

where the bounded density is obtained by applying the logit quantile transformation to a baseline density on the real line. In the notation used in this notebook, if

$$w_i = \log\left(\frac{y_i}{1-y_i}\right),$$

then the density of Y_i can be written as

$$f(y_i;\theta_i) = g(w_i;\theta_i)\frac{1}{y_i(1-y_i)}, \qquad 0 < y_i < 1.$$

For two-parameter baseline distributions, the parameter vector is $\theta_i = (\mu_i, \sigma_i)$; for three-parameter distributions, it is $\theta_i = (\mu_i, \sigma_i, \nu_i)$. Rodrigues et al. (2020) specify a regression structure through link functions. In their application to the poverty data, they also consider logistic, normal, and Student-t baselines, which are additional JSBt cases generated by the same logit transformation; the Appendix gives the corresponding baseline and bounded densities. The covariate enters the location parameter, while the scale and shape parameters are constant:

$$g_\mu(\mu_i) = \beta_0 + \beta_1 z_i, \qquad g_\sigma(\sigma_i) = \delta_0, \qquad g_\nu(\nu_i) = \phi_0.$$

The link functions depend on the parameter space of each baseline distribution. For example, an identity link may be used for a real-line location parameter, while log links are used for positive scale and positive shape parameters. The log-likelihood for a sample $D = \{(z_i, y_i) : i = 1, \dots, n\}$ is

$$\ell(\beta_0, \beta_1, \delta_0, \phi_0 \mid D) = \sum_{i=1}^{n} \log g\left\{\log\left(\frac{y_i}{1-y_i}\right); \beta_0, \beta_1, \delta_0, \phi_0\right\} - \sum_{i=1}^{n} \log\{y_i(1-y_i)\}.$$

This formulation shows the central idea of the JSBt regression proposal: covariates are introduced through the parameters of the baseline real-line distribution, while the Jacobian term $\{y_i(1-y_i)\}^{-1}$ guarantees that the fitted density is defined on the bounded interval $(0, 1)$. Rodrigues et al. (2020) estimate these models by maximum likelihood using GAMLSS, compare alternative baseline distributions with information criteria such as AIC and SBC, and use standardized quantile residuals for model checking.

2.3.2 JSBt Regression Formulation of Piccirilli et al. (2026)

Piccirilli, De Bastiani, and Jorge Luis Bazán (2026) extend the JSBt construction to a flexible regression framework in which covariates may enter the location, scale, and shape parameters of the bounded distribution. A non-mixed regression version of their proposal can be written for independent bounded responses y_i, $i = 1, \dots, n$, with $0 < y_i < 1$, as

$$Y_i \sim \text{JSBt}(\mu_i, \sigma_i, \nu_i), \qquad i = 1, \dots, n,$$

with link functions

$$g_1(\mu_i) = x_{1i}^\top \beta, \qquad g_2(\sigma_i) = x_{2i}^\top \gamma, \qquad g_3(\nu_i) = x_{3i}^\top \delta.$$

The location parameter μ_i is defined on the real line because it is the location of the baseline distribution on the logit-transformed scale. Therefore, it should not be interpreted as the mean of Y_i. Instead, it controls the central tendency of the transformed response $\log\{Y_i/(1-Y_i)\}$. When direct interpretation on the response scale is desired, the inverse-logit transformation

$$\omega_i = \frac{\exp(\mu_i)}{1+\exp(\mu_i)}$$

can be used as a location index on $(0,1)$. For the logistic power exponential model, this value is the median of Y_i.

The scale parameter $\sigma_i > 0$ is commonly modeled with a log link. It controls residual dispersion after accounting for the location model. The shape parameter ν_i is included only for the three-parameter baseline distributions. Its link depends on the baseline distribution: for the logistic exponential Gaussian and logistic power exponential models, a log link is natural because $\nu > 0$; for logistic skew-normal type I, the identity link can be used because $\nu \in \mathbb{R}$; and for logistic skew-normal type II, a log link is natural because $\nu > 0$.

The log-likelihood contribution has the generic form

$$\ell(\Theta) = \sum_{i=1}^{n} \log g\left\{\log\left(\frac{y_i}{1-y_i}\right); \mu_i, \sigma_i, \nu_i\right\} - \sum_{i=1}^{n} \log\{y_i(1-y_i)\},$$

where $g(\cdot)$ is the density of the chosen baseline distribution and Θ collects all regression coefficients. The full Piccirilli, De Bastiani, and Jorge Luis Bazán (2026) paper also develops a mixed model for longitudinal or clustered bounded responses, but that mixed-model specification is not introduced here. Model comparison is usually based on AIC and BIC, while randomized quantile residuals are used to check the adequacy of the fitted distribution (Dunn and Smyth 1996). A practical GAMLSS example for bounded mixed regression is given in the second application of Piccirilli, De Bastiani, and Jorge Luis Bazán (2026); the accompanying data and analysis are available in the authors' repository.

2.4 Practical Estimation Using GAMLSS

This section shows how to fit JSBt regression models using GAMLSS. Two simulated bounded continuous regression datasets are considered. The first response is generated from the beta regression model implemented in GAMLSS, with covariates only in the mean parameter and a fixed dispersion parameter. The second response is generated from the Logistic-LSN1 model, again with covariates only in the location parameter and fixed remaining parameters. The fitted models include the beta benchmark and the JSBt bounded regression models discussed in Sections 2 and 3, including the Logistic-logistic, Logistic-normal, and Logistic-t cases.

GAMLSS estimates distributional regression models by maximum likelihood or penalized maximum likelihood, allowing each distributional parameter, such as location, scale, and shape, to have its own additive predictor (Rigby and Stasinopoulos 2005; Stasinopoulos and Rigby 2007). In practice, `gamlss()` fits these models by iterative backfitting algorithms for the distributional parameters. Thus, the same framework can fit a beta regression, a two-parameter JSBt regression, or a three-parameter JSBt regression by changing the distribution family and the formulas assigned to `mu`, `sigma`, and `nu` (Rigby et al. 2019).

The GAMLSS function `gen.Family()` can generate the logit-transformed families directly from baseline distributions available in `gamlss.dist`. The generated families use names such as `logitGU`, `logitRG`, `logitexGAUS`, `logitSN1`, `logitSN2`, `logitPE`, `logitLO`, `logitNO`, and `logitTF`. The function creates the density, distribution, quantile, random-generation, and fitting functions for a log or logit transformation of an existing continuous GAMLSS family; for example, `gen.Family("NO", "logit")` creates `dlogitNO()`, `plogitNO()`, `qlogitNO()`, `rlogitNO()`, and the `logitNO` family used inside `gamlss()` (Stasinopoulos and Rigby 2023).

The code in this section is organized as a reproducible workflow. The first chunk loads the required packages, the second chunk generates the logit-transformed JSBt families, the simulation chunk creates two controlled datasets, the fitting chunk applies the same candidate model set to both datasets, and the final chunks compare and diagnose the selected fits.

```
# Load the modeling functions, distributions, and plotting tools.
library(gamlss)
library(gamlss.dist)
library(ggplot2)
```

```
# Generate logit-transformed JSBt families from GAMLSS baselines.
gen.Family(family = "GU", type = "logit")
gen.Family(family = "RG", type = "logit")
gen.Family(family = "exGAUS", type = "logit")
gen.Family(family = "SN1", type = "logit")
gen.Family(family = "SN2", type = "logit")
gen.Family(family = "PE", type = "logit")
gen.Family(family = "LO", type = "logit")
gen.Family(family = "NO", type = "logit")
gen.Family(family = "TF", type = "logit")
```

After running the code above, it is possible to fit the different JSBt regression models available in GAMLSS. All models were included in Rodrigues et al. (2020), while the first six models are the focus of Piccirilli, De Bastiani, and Jorge Luis Bazán (2026).

2.4.1 Simulated Data for JSBt Regression Models

We simulate two bounded continuous responses using the same covariate values. We consider beta and Logistic-LSN1 regression settings. In the beta case, the mean parameter depends on the covariate through a logit link and the GAMLSS beta dispersion parameter is fixed. In the Logistic-LSN1 case, the same linear predictor enters the real-line LSN1 location parameter, while `sigma` and `nu` are fixed. The column `true_location` is the conditional mean for the beta data and the inverse-logit location index for the Logistic-LSN1 data.

```
# Use the same covariate values in both simulations.
set.seed(660)

n <- 500
x <- rnorm(n)

# Beta-generated case: x enters the conditional mean.
beta_data <- data.frame(Scenario = "Beta-generated", x = x)
beta_data$eta <- -0.40 + 1.20 * beta_data$x
beta_data$true_location <- plogis(beta_data$eta)
beta_data$sigma <- 0.20
beta_data$nu <- 0
beta_data$y <- rBE(
  n = n,
```

```r
  mu = beta_data$true_location,
  sigma = beta_data$sigma
)

# Logistic-LSN1-generated case: x enters the real-line location.
lsn1_data <- data.frame(Scenario = "Logistic-LSN1-generated", x = x)
lsn1_data$eta <- -0.40 + 1.20 * lsn1_data$x
lsn1_data$true_location <- plogis(lsn1_data$eta)
lsn1_data$sigma <- 0.45
lsn1_data$nu <- 5
set.seed(650)
lsn1_data$y <- rlogitSN1(
  n = n,
  mu = lsn1_data$eta,
  sigma = lsn1_data$sigma,
  nu = lsn1_data$nu
)

# Store both datasets in a list so the same fitting code can be reused.
simulations <- list(
  Beta = beta_data,
  LSN1 = lsn1_data
)

lapply(simulations, function(dat) summary(dat$y))
```

```
## $Beta
##      Min.   1st Qu.    Median      Mean   3rd Qu.      Max.
## 0.006354 0.224625 0.418256 0.425581 0.600160 0.975860
##
## $LSN1
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.03926 0.31050 0.48468 0.49325 0.67650 0.97966
```

```r
# Combine the two simulations only for graphical display.
sim_plot_data <- do.call(rbind, simulations)
mean_curve <- sim_plot_data[order(sim_plot_data$Scenario, sim_plot_data$x), ]

p1 <- ggplot(sim_plot_data, aes(x = y)) +
  geom_histogram(bins = 30, color = "black", fill = "grey80") +
  facet_wrap(~ Scenario) +
  theme_classic() +
  labs(x = "Simulated bounded response", y = "Count")

p2 <- ggplot(sim_plot_data, aes(x = x, y = y)) +
  geom_point(alpha = 0.35) +
  geom_line(
    data = mean_curve,
    aes(y = true_location),
    color = "red",
    linewidth = 1
  ) +
  facet_wrap(~ Scenario) +
  theme_classic() +
  labs(x = "Covariate x", y = "Simulated bounded response")

print(p1)
```

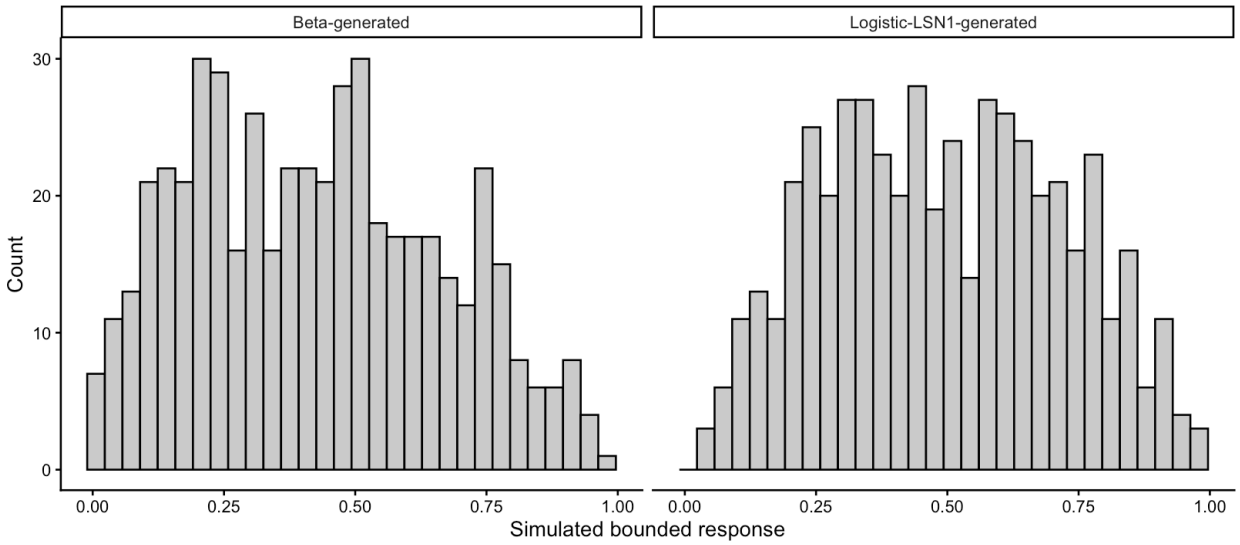


Figure 4: plot of chunk exploratory-plots

```
print(p2)
```

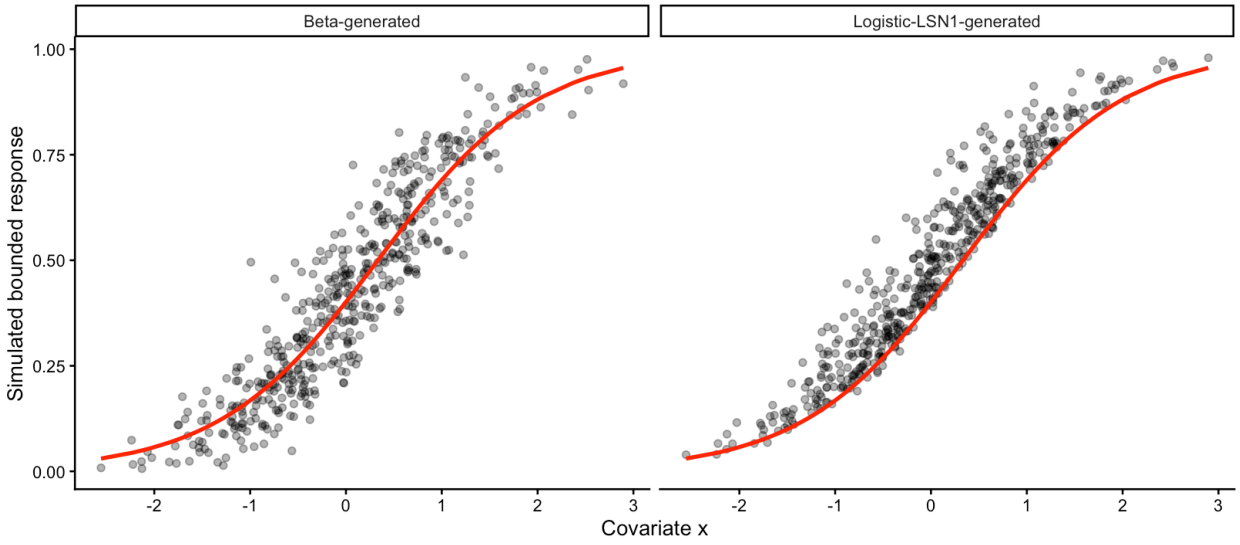


Figure 5: plot of chunk exploratory-plots

2.4.2 Fitting JSBt Bounded Regression Models

For each simulated dataset, all fitted models use the same predictor in the location or mean regression and keep the remaining parameters fixed. For the beta model, `mu` is the conditional mean of the response. For the JSBt models, the linear predictor is assigned to the real-line location parameter of the baseline distribution and is transformed back to the unit interval by the inverse-logit map.

```
ctrl <- gamlss.control(n.cyc = 100, trace = FALSE)

# Fit the beta benchmark and all JSBt candidates to one dataset.
```

```
fit_regression_models <- function(dat) {
  list(
    Beta = gamlss(
      y ~ x,
      sigma.formula = ~ 1,
      data = dat,
      family = BE,
      control = ctrl
    ),
    LGU = gamlss(
      y ~ x,
      sigma.formula = ~ 1,
      data = dat,
      family = logitGU,
      control = ctrl
    ),
    LRG = gamlss(
      y ~ x,
      sigma.formula = ~ 1,
      data = dat,
      family = logitRG,
      control = ctrl
    ),
    LLo = gamlss(
      y ~ x,
      sigma.formula = ~ 1,
      data = dat,
      family = logitLO,
      control = ctrl
    ),
    LN = gamlss(
      y ~ x,
      sigma.formula = ~ 1,
      data = dat,
      family = logitNO,
      control = ctrl
    ),
    LEG = gamlss(
      y ~ x,
      sigma.formula = ~ 1,
      nu.formula = ~ 1,
      data = dat,
      family = logitexGAUS,
      control = ctrl
    ),
    LPE = gamlss(
      y ~ x,
      sigma.formula = ~ 1,
      nu.formula = ~ 1,
      data = dat,
      family = logitPE,
      control = ctrl
    ),
    LSN1 = gamlss(
      y ~ x,
      sigma.formula = ~ 1,
      nu.formula = ~ 1,
      data = dat,
      family = logitSN1,
```

```
      control = ctrl
    ),
    LSN2 = gamlss(
      y ~ x,
      sigma.formula = ~ 1,
      nu.formula = ~ 1,
      data = dat,
      family = logitSN2,
      control = ctrl
    ),
    LT = gamlss(
      y ~ x,
      sigma.formula = ~ 1,
      nu.formula = ~ 1,
      data = dat,
      family = logitTF,
      control = ctrl
    )
  )
}

# Apply the same model set to the beta and Logistic-LSN1 simulations.
fits <- lapply(simulations, fit_regression_models)
```

2.4.3 Model Comparison

The following table compares the fitted beta and JSBt bounded regression models for both simulated datasets using AIC and BIC. Smaller values indicate a better trade-off between fit and complexity.

```
# Build one AIC/BIC table containing both simulation scenarios.
criteria <- do.call(
  rbind,
  lapply(names(fits), function(scenario_name) {
    model_list <- fits[[scenario_name]]
    data.frame(
      Scenario = scenario_name,
      Model = names(model_list),
      AIC = sapply(model_list, AIC),
      BIC = sapply(model_list, BIC),
      row.names = NULL
    )
  })
)

criteria <- criteria[order(criteria$Scenario, criteria$AIC), ]

# Select the best model in each simulation case by minimum AIC.
best_names <- sapply(names(fits), function(scenario_name) {
  scenario_criteria <- criteria[criteria$Scenario == scenario_name, ]
  scenario_criteria$Model[which.min(scenario_criteria$AIC)]
})

# Keep the selected fitted objects for diagnostics and plots.
best_models <- lapply(names(fits), function(scenario_name) {
  fits[[scenario_name]][[best_names[scenario_name]]]
})
names(best_models) <- names(fits)
```

```
criteria
```

```
##    Scenario Model        AIC        BIC
## 1      Beta  Beta -1089.6007 -1076.9569
## 4      Beta   LLo -1054.8953 -1042.2515
## 10     Beta    LT -1054.1932 -1037.3347
## 7      Beta   LPE -1049.5344 -1032.6760
## 9      Beta  LSN2 -1045.9920 -1029.1336
## 5      Beta    LN -1043.2603 -1030.6165
## 8      Beta  LSN1 -1041.2573 -1024.3989
## 6      Beta   LEG -1041.2402 -1024.3818
## 2      Beta   LGU  -984.0037  -971.3599
## 3      Beta   LRG  -888.5173  -875.8734
## 18     LSN1  LSN1 -1568.0935 -1551.2351
## 19     LSN1  LSN2 -1565.0561 -1548.1976
## 13     LSN1   LRG -1562.8701 -1550.2263
## 16     LSN1   LEG -1555.5870 -1538.7285
## 14     LSN1   LLo -1479.2904 -1466.6466
## 20     LSN1    LT -1477.4302 -1460.5718
## 17     LSN1   LPE -1472.2480 -1455.3896
## 15     LSN1    LN -1467.4929 -1454.8491
## 11     LSN1  Beta -1452.1055 -1439.4616
## 12     LSN1   LGU -1258.1553 -1245.5115
```

In the beta-generated case, the beta fit is the correctly specified benchmark. In the Logistic-LSN1-generated case, the LSN1 model is the correctly specified JSBt benchmark. The remaining fitted models are useful competitors: they show how different real-line baseline distributions can approximate the same bounded regression structure after the logistic transformation. In a real application, the preferred distribution should be chosen together with residual diagnostics and substantive interpretation, not by information criteria alone.

2.4.4 Residual Diagnostics

Quantile residuals of Dunn and Smyth (1996) are standard diagnostics in GAMLSS models. For a continuous response, they are computed as $r_i = \Phi^{-1}\{F(y_i; \widehat{\theta}_i)\}$; randomization is needed when the fitted distribution has discrete probability masses. For a continuous response, if the fitted model is adequate, the residuals should be approximately standard normal.

```
# Randomized quantile residuals should be approximately normal.
residual_list <- lapply(best_models, residuals)

par(mfrow = c(length(best_models), 2))
for (scenario_name in names(best_models)) {
  rq <- residual_list[[scenario_name]]
  best_name <- best_names[scenario_name]

  hist(
    rq,
    main = paste("Residuals:", scenario_name, "-", best_name),
    xlab = "Randomized quantile residuals",
    col = "grey80",
    border = "black"
```

```
  )
  qqnorm(rq, main = paste("Q-Q plot:", scenario_name, "-", best_name))
  qqline(rq, lwd = 2)
}
```

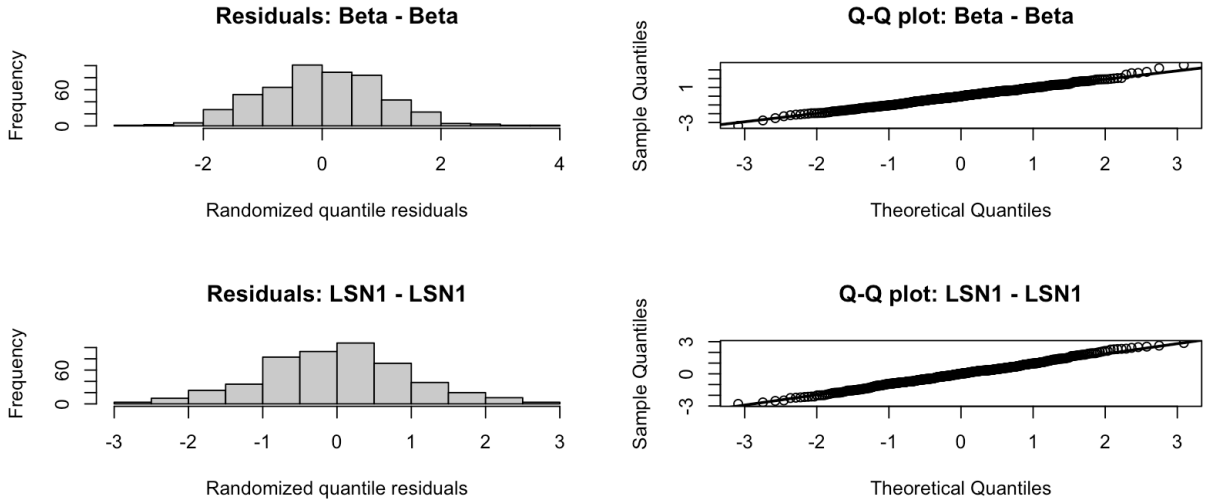


Figure 6: plot of chunk residual-diagnostics

```
par(mfrow = c(1, 1))

data.frame(
  Scenario = names(best_models),
  SelectedModel = best_names[names(best_models)],
  ShapiroPValue = sapply(residual_list, function(rq) shapiro.test(rq)$p.value),
  row.names = NULL
)
```

```
##   Scenario SelectedModel ShapiroPValue
## 1     Beta          Beta     0.7167631
## 2     LSN1          LSN1     0.5584834
```

2.4.5 Fitted Location on the Response Scale

For the beta regression model, the fitted `mu` parameter is already the conditional mean on the unit interval. For a selected JSBt model, the fitted location parameter is estimated on the real-line baseline scale, so the inverse-logit transformation gives a location index on the response scale. The plot is produced for the selected model in each simulation case.

```
# Put the selected fitted location on the response scale.
fitted_location_data <- do.call(
  rbind,
  lapply(names(best_models), function(scenario_name) {
    dat <- simulations[[scenario_name]]
    best_name <- best_names[scenario_name]
    best_model <- best_models[[scenario_name]]
    dat$SelectedModel <- best_name
    dat$mu_hat <- fitted(best_model, what = "mu")
    dat$location_hat <- if (best_name == "Beta") {
```

```
      dat$mu_hat
    } else {
      plogis(dat$mu_hat)
    }
    dat
  })
)

plot_data <- fitted_location_data[
  order(fitted_location_data$Scenario, fitted_location_data$x),
]

ggplot(fitted_location_data, aes(x = x, y = y)) +
  geom_point(alpha = 0.35) +
  geom_line(
    data = plot_data,
    aes(y = true_location),
    linetype = 2,
    linewidth = 1
  ) +
  geom_line(
    data = plot_data,
    aes(y = location_hat),
    color = "red",
    linewidth = 1
  ) +
  facet_wrap(~ Scenario) +
  theme_classic() +
  labs(
    x = "Covariate x",
    y = "Response, true mean, and fitted location",
    title = "Fitted response-scale location by simulation case"
  )
```

The dashed curve is the true beta regression mean in the beta-generated case and the true response-scale location index in the Logistic-LSN1-generated case. The red curve is the fitted mean when the selected model is the beta regression model, or the fitted JSBt location index transformed to the response scale otherwise. For asymmetric JSBt families such as LSN1, this response-scale location is not necessarily equal to the conditional mean of Y, but it remains useful for interpreting how the covariate moves the center of the bounded distribution.

This completes the simulated workflow: generate bounded continuous data from a beta mean regression and from a Logistic-LSN1 location regression with fixed remaining parameters, construct the JSBt families, fit the beta and JSBt bounded regression models under the same mean-only covariate structure, compare them with AIC and BIC, and assess the selected models using randomized quantile residuals.

2.5 Exercises

1. Fit the JSBt regression models to the data used in the second application of Piccirilli, De Bastiani, and Jorge Luis Bazán (2026), namely the mortality dataset available at https://github.com/Giovannipcl/Bronchial-and-lung-cancer-mortality-rate-in-Brazil. Use the general workflow presented in Section 4: define the candidate JSBt families, specify the regression structure for the location parameter, fit all candidate models, and compare them using AIC,

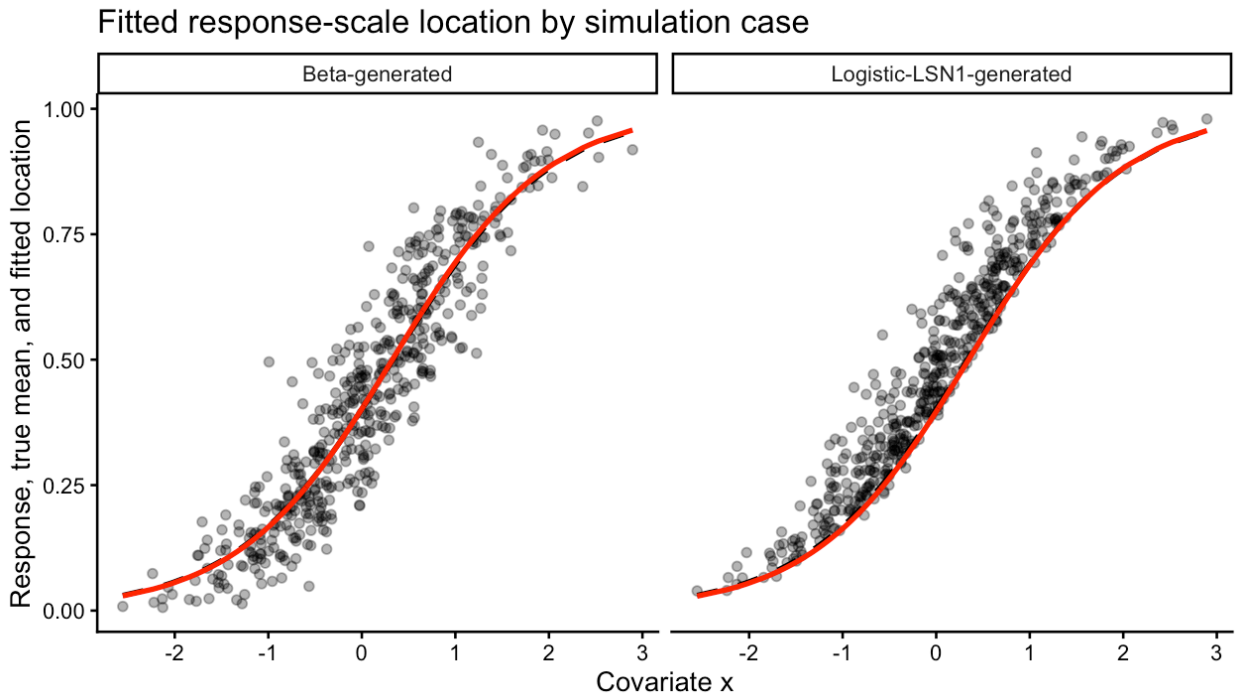


Figure 7: plot of chunk fitted-location

BIC, and randomized quantile residuals. Choose the best model according to the joint evidence from information criteria and diagnostics. Then interpret the estimated regression coefficients, making clear whether each coefficient acts on the real-line baseline location scale or, after inverse-logit transformation, on the bounded response scale.

2. Simulate a dataset from the Logistic-logistic regression model and compare the fitted results obtained with the available R implementations. At a minimum, fit the model using the JSBt construction in GAMLSS and compare it with the implementations available in `llbayesireg` and `unitquantreg`, when these packages are installed. Take care with the different parameterizations: the Logistic-logistic distribution is also called the L-Logistic distribution in the literature, and the meaning of location, precision, scale, or shape parameters may differ across packages. The comparison should therefore report not only the estimated coefficients and fit criteria, but also the parameterization used by each package and any transformations needed to put the estimates on a common interpretive scale.

3. Use one bounded response dataset available in the `betareg` package, such as `GasolineYield`, to compare classical beta regression with the JSBt regression models considered in this paper (Cribari-Neto and Zeileis 2010). Fit the beta model using `betareg` and fit the JSBt candidate models using GAMLSS after generating the corresponding logit-transformed families. Compare AIC, BIC, fitted response-scale location curves, and randomized quantile residuals. Discuss whether the best JSBt model gives a visibly different interpretation of covariate effects from the beta regression model, especially near the lower or upper boundary of the response.

4. Use a proportional response dataset available in the `simplexreg` package, such as `sdac`, to compare simplex regression with the JSBt regression models (Zhang et al. 2016). Fit the

simplex model with `simplexreg` using an appropriate mean link and fit the JSBt models with the same covariates in the location parameter. Compare the selected models using fit criteria and residual diagnostics, and explain how the interpretation of dispersion differs between the simplex model and the JSBt models. Pay special attention to parameterization, because the simplex model is formulated directly on the mean and dispersion of a simplex distribution, whereas the JSBt models are generated by transforming a real-line baseline distribution to the unit interval.

5. Implement a Bayesian approach for the models studied here.

2.6 Conclusions

Bounded continuous responses require models that respect the support of the data and can represent features that are not easily captured by ordinary linear models. The JSBt construction provides a flexible way to generate distributions on the unit interval by combining a real-line baseline distribution with the logistic quantile transformation proposed by Rodrigues et al. (2020). This approach gives a unified family of bounded distributions whose parameters retain the interpretation of the baseline distribution: location, scale, and, when available, shape.

The regression framework motivated by Piccirilli, De Bastiani, and Jorge Luis Bazán (2026) extends this idea by allowing covariates to enter the location, scale, and shape parameters. In this way, the model can describe changes not only in central tendency but also in residual dispersion, asymmetry, and tail behavior. Their paper also develops a mixed-model version for longitudinal and clustered data, but the theoretical formulation in Section 3 focuses on the non-mixed regression case.

The simulated GAMLSS example illustrates the practical workflow: define the logit-transformed JSBt families, simulate bounded continuous data, fit competing bounded regression models, compare them using AIC and BIC, and evaluate the selected model through randomized quantile residuals. In applied work, the choice among LGU, LRG, LEG, LPE, LSN1, LSN2, LLo, LN, and LT should be guided by both goodness-of-fit criteria and substantive interpretation, because each baseline distribution induces different behavior near the boundaries and in the tails of the bounded response distribution.

2.7 Appendix: Additional JSBt Baseline Distributions

Rodrigues et al. (2020) also discuss JSBt models generated from logistic, normal, and Student-t baseline distributions. These cases are obtained with the same transformation used in Section 2:

$$w = \log\left(\frac{y}{1-y}\right), \qquad f(y;\theta) = g(w;\theta)\frac{1}{y(1-y)}, \qquad 0 < y < 1.$$

The parameterization used below follows the baseline distributions as implemented in GAMLSS, where μ and σ are location and scale parameters for the real-line variable $W = \log\{Y/(1-Y)\}$. This differs from the parameterizations used in the corresponding papers. Lemonte and Bazán (2016) write the generalized Johnson-S_B distribution using γ and δ through the transformed variable $\gamma + \delta \log\{y/(1-y)\}$. For symmetric baselines, this corresponds to the GAMLSS-style location-scale form through $\mu = -\gamma/\delta$ and $\sigma = 1/\delta$. Similarly, Paz et al. (2019) parameterize the L-Logistic distribution by its median $m \in (0,1)$ and dispersion parameter $b > 0$. In the GAMLSS-style logistic

baseline form used here, the correspondence is $m = \exp(\mu)/\{1+\exp(\mu)\}$ and $b = 1/\sigma$. Thus, larger b in Paz et al. (2019) means smaller dispersion, whereas larger σ in the baseline-parameterized JSBt representation means larger dispersion on the logit scale.

The estimation strategies also differ across these works. Lemonte and Bazán (2016) discuss maximum likelihood and least-squares estimation for the generalized Johnson-S_B class, and their regression application is implemented in the R program using nonlinear optimization, specifically the BFGS quasi-Newton algorithm. In contrast, Paz et al. (2019) develop the L-Logistic model mainly under a Bayesian framework, using Markov chain Monte Carlo estimation with a hybrid Metropolis-Hastings and Gibbs sampling algorithm implemented in R. The GAMLSS-style representation used in this appendix is therefore a common JSBt notation for comparing densities, but it should not be confused with the estimation parameterizations used in those papers.

In the literature, the logistic-logistic distribution is usually named the L-Logistic distribution, following Tadikamalla and Johnson (1982) and Paz et al. (2019). This model can be fitted with the R package `llbayesireg`, which implements Bayesian L-Logistic regression (Fonsêca and Paz 2019). It is also available in `unitquantreg` as the unit-Logistic family for parametric quantile regression of bounded data (Menezes and Mazucheli 2021; Mazucheli et al. 2022). The logistic-normal distribution corresponds to the classical Johnson S_B distribution introduced by Johnson (1949) and generalized by Lemonte and Bazán (2016). The Johnson S_B model can also be fitted with `unitquantreg`, which includes the `johnsonsb` family and allows regression structures for bounded quantiles and shape parameters (Menezes and Mazucheli 2021; Mazucheli et al. 2022).

For the logistic baseline distribution,

$$g(w;\mu,\sigma) = \frac{\exp\{(w-\mu)/\sigma\}}{\sigma\left[1+\exp\{(w-\mu)/\sigma\}\right]^2}, \qquad \mu \in \mathbb{R},\ \sigma > 0.$$

The corresponding logistic-logistic bounded density is

$$f(y;\mu,\sigma) = \frac{\exp\left[\{\log(y/(1-y)) - \mu\}/\sigma\right]}{\sigma y(1-y)\left(1+\exp\left[\{\log(y/(1-y)) - \mu\}/\sigma\right]\right)^2}.$$

For the normal baseline distribution,

$$g(w;\mu,\sigma) = \frac{1}{\sigma\sqrt{2\pi}}\exp\left[-\frac{1}{2}\left(\frac{w-\mu}{\sigma}\right)^2\right], \qquad \mu \in \mathbb{R},\ \sigma > 0.$$

The corresponding logistic-normal bounded density is

$$f(y;\mu,\sigma) = \frac{1}{\sigma\sqrt{2\pi}y(1-y)}\exp\left[-\frac{1}{2}\left(\frac{\log(y/(1-y)) - \mu}{\sigma}\right)^2\right].$$

For the Student-t baseline distribution with ν degrees of freedom,

$$g(w;\mu,\sigma,\nu) = \frac{\Gamma\{(\nu+1)/2\}}{\sigma\sqrt{\nu\pi}\Gamma(\nu/2)}\left[1+\frac{1}{\nu}\left(\frac{w-\mu}{\sigma}\right)^2\right]^{-(\nu+1)/2}, \qquad \mu \in \mathbb{R},\ \sigma > 0,\ \nu > 0.$$

The corresponding logistic-t bounded density is

$$f(y;\mu,\sigma,\nu) = \frac{\Gamma\{(\nu+1)/2\}}{\sigma\sqrt{\nu\pi}\Gamma(\nu/2)y(1-y)}\left[1+\frac{1}{\nu}\left(\frac{\log(y/(1-y))-\mu}{\sigma}\right)^2\right]^{-(\nu+1)/2}.$$

Table A1 reports numerical mean and variance values for these additional JSBt distributions. For the logistic-t case, the reference value is $\nu = 5$.

Table A1: Mean and variance of additional JSBt distributions for fixed reference values.

			$\mu=-1$		$\mu=0$		$\mu=1$	
Bounded distribution	Baseline distribution (parameters)	Reference values	$\mathbb{E}(Y)$	$\mathrm{Var}(Y)$	$\mathbb{E}(Y)$	$\mathrm{Var}(Y)$	$\mathbb{E}(Y)$	$\mathrm{Var}(Y)$
Logistic-Logistic (LLo)	Logistic (2) $\mu\in\mathbb{R}$, $\sigma>0$	$\sigma=1$	0.3387	0.0730	0.5000	0.0833	0.6613	0.0730
Logistic-Logistic (LLo)	Logistic (2) $\mu\in\mathbb{R}$, $\sigma>0$	$\sigma=2$	0.3942	0.1360	0.5000	0.1427	0.6058	0.1360
Logistic-Normal (LN)	Normal (2) $\mu\in\mathbb{R}$, $\sigma>0$	$\sigma=1$	0.3033	0.0334	0.5000	0.0434	0.6967	0.0334
Logistic-Normal (LN)	Normal (2) $\mu\in\mathbb{R}$, $\sigma>0$	$\sigma=2$	0.3523	0.0877	0.5000	0.0986	0.6477	0.0877
Logistic-t (LT)	Student-t (3) $\mu\in\mathbb{R}$, $\sigma>0$, $\nu>0$	$\sigma=1$, $\nu=5$	0.3123	0.0440	0.5000	0.0534	0.6877	0.0440
Logistic-t (LT)	Student-t (3) $\mu\in\mathbb{R}$, $\sigma>0$, $\nu>0$	$\sigma=2$, $\nu=5$	0.3614	0.0990	0.5000	0.1080	0.6386	0.0990

The next figures use the same format as Figures 1–3. The logistic and normal baseline cases have only location and scale parameters, while the logistic-t case is shown for different values of ν.

```
plot_appendix_jsbt_comparison(mu_value = 0, sigma_value = 1)
```

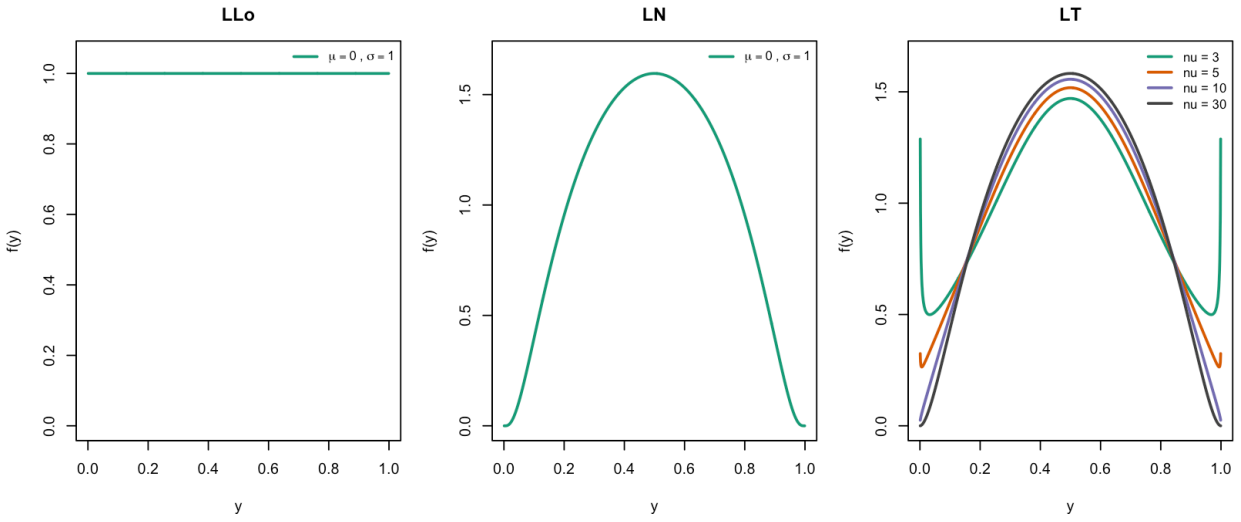


Figure 8: Probability density functions of additional JSBt distributions for $\mu = 0$, $\sigma = 1$, and, for logistic-t, selected values of ν.

For $\mu = 0$, the logistic-logistic and logistic-normal cases are symmetric around 0.5 on the response scale. The logistic-t case also centers around 0.5, but smaller values of ν give heavier tails on the logit scale and therefore more density near the boundaries. In the LLo panel, the density is flat at one because when $W \sim \mathrm{Logistic}(0,1)$ and $Y = \exp(W)/\{1+\exp(W)\}$, the transformed response is uniform on $(0,1)$. Thus, for $\mu = 0$ and $\sigma = 1$, the LLo density is $f(y) = 1$ for $0 < y < 1$.

```
plot_appendix_jsbt_comparison(mu_value = -1, sigma_value = 1)
```

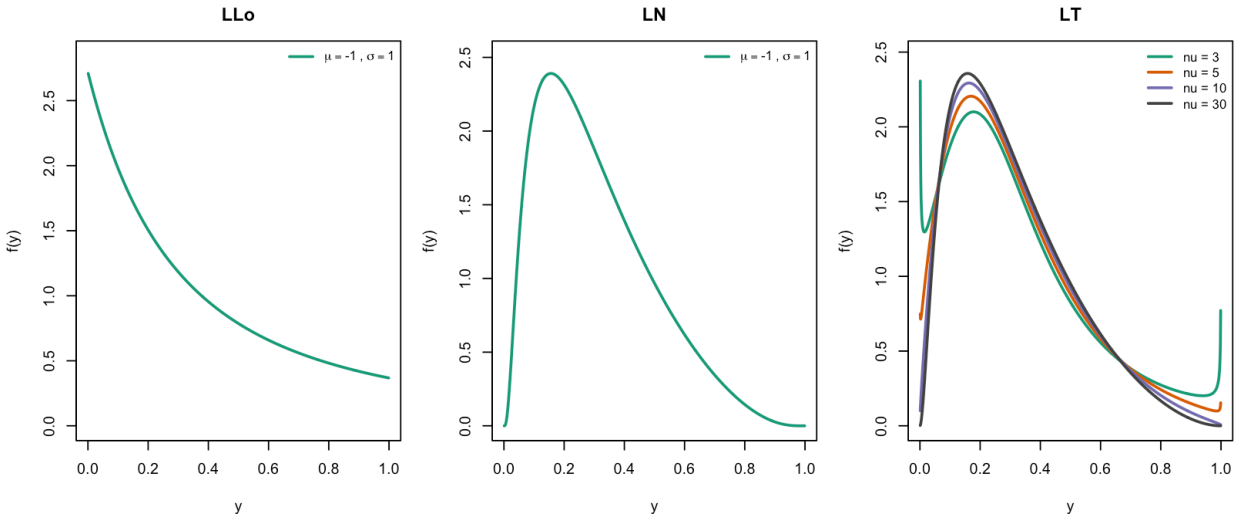


Figure 9: Probability density functions of additional JSBt distributions for $\mu = -1$, $\sigma = 1$, and, for logistic-t, selected values of ν.

For $\mu = -1$, the densities shift toward smaller values of Y. This is the same location effect observed in the main JSBt families: changing μ moves the distribution on the logit scale, and the monotone inverse-logit transformation carries that shift to the bounded scale.

```
plot_appendix_jsbt_comparison(mu_value = 1, sigma_value = 1)
```

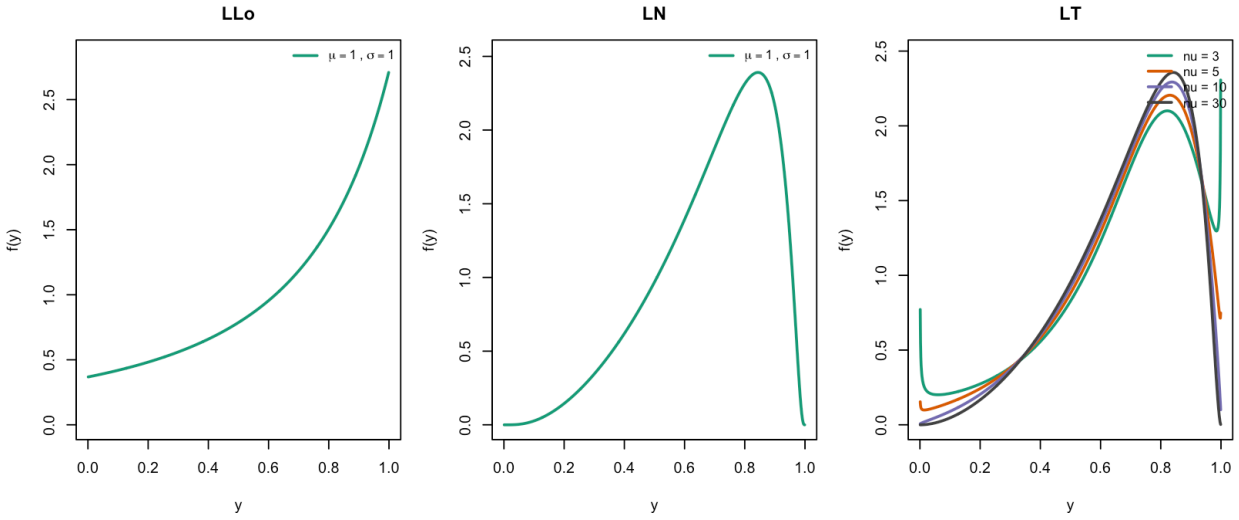


Figure 10: Probability density functions of additional JSBt distributions for $\mu = 1$, $\sigma = 1$, and, for logistic-t, selected values of ν.

For $\mu = 1$, the densities shift toward larger values of Y. The comparison with $\mu = -1$ confirms that μ controls location, while σ controls spread and the Student-t degrees of freedom ν controls tail thickness.

Chapter 3

Bounded Count Regression

3.1 Introduction

Bounded count data are observed in many applied problems in which the response is a number of events, successes, symptoms, correct answers, or affirmative responses out of a fixed maximum. Typical examples include health scores with an upper limit, educational item scores, numbers of positive tests, and grouped binary outcomes. In these situations, the support of the response is finite, $Y \in \{0, 1, \ldots, m\}$, and the statistical model must respect both the discreteness and the upper bound.

The binomial distribution is the natural starting point for bounded counts because it represents the number of successes in m Bernoulli trials. However, empirical bounded count data often show more variation than the binomial model allows, as well as asymmetry, heavy tails, or excess probability near the boundary values 0 and m. The beta-binomial distribution addresses overdispersion by allowing the success probability to vary across observational units. The beta-2-binomial distribution extends this idea further by introducing an additional shape parameter that can improve the representation of tail and boundary behavior (Bayes et al. 2024).

This notebook summarizes the binomial, beta-binomial (BB), and beta-2-binomial (B2B) distributions for bounded counts using a common mean-dispersion notation based on μ, δ, and ν. It then discusses identification, monotonicity, skewness, kurtosis, and regression formulations. The practical section illustrates estimation with GAMLSS (Rigby and Stasinopoulos 2005; Stasinopoulos and Rigby 2007), `gamlss.dist` (Stasinopoulos and Rigby 2023), `gamlss.data` (Stasinopoulos and Rigby 2025), `numDeriv` (Gilbert and Varadhan 2019), and the `RobustDiscrete` package (Bayes et al. 2026), including simulated examples and an applied B2B example from the package repository.

3.2 Bounded Count Distributions

Bounded count responses arise when the observed outcome is the number of successes, events, or correct responses out of a fixed maximum number of opportunities. In this setting, a response variable Y has support $\{0, 1, \ldots, m\}$, where m is known. The binomial distribution is the basic model for such data, while the beta-binomial and beta-2-binomial distributions extend it by allowing extra-binomial variation and more flexible distributional shapes (Johnson et al. 2005; Bayes et al. 2024).

3.2.1 Binomial Distribution

Let Y be a bounded count variable with support $Y \in \{0, 1, \dots, m\}$. The binomial distribution assumes that Y is the number of successes in m conditionally independent Bernoulli trials with common success probability μ, written

$$Y \sim \text{Binomial}(m, \mu).$$

Its probability mass function is

$$\Pr(Y = y \mid m, \mu) = \binom{m}{y} \mu^y (1-\mu)^{m-y}, \qquad y = 0, 1, \dots, m,$$

where $0 < \mu < 1$. The parameter m is the maximum possible count or number of trials, and μ is the probability of success on each trial.

The mean and variance are

$$\mathbb{E}(Y) = m\mu, \qquad \text{Var}(Y) = m\mu(1-\mu).$$

Thus, in the binomial model, the variance is fully determined by m and μ. This makes the model parsimonious, but it can be too restrictive when the data show overdispersion or more responses near the boundaries than the binomial distribution can accommodate (Johnson et al. 2005).

3.2.2 Beta-Binomial Distribution

Let Y be a count variable with support $Y \in \{0, 1, \dots, m\}$. Then Y follows a beta-binomial distribution with maximum count m and positive shape parameters $\alpha > 0$ and $\beta > 0$, written

$$Y \sim \text{BetaBinomial}(m, \alpha, \beta),$$

if its probability mass function is

$$\Pr(Y = y \mid m, \alpha, \beta) = \binom{m}{y} \frac{B(y+\alpha, m-y+\beta)}{B(\alpha, \beta)}, \qquad y = 0, 1, \dots, m,$$

where

$$B(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}$$

is the beta function. This parameterization is named as the traditional Beta Binomial parameterization.

The beta-binomial distribution can be interpreted as a binomial model in which the success probability is itself random:

$$Y \mid \pi \sim \text{Binomial}(m, \pi), \qquad \pi \sim \text{Beta}(\alpha, \beta).$$

Thus, m is the number of trials or the maximum possible count, π is the latent success probability, and α and β determine the distribution of π. Larger values of α relative to β imply larger expected success probabilities, while smaller values of $\alpha + \beta$ imply more heterogeneity in π and therefore more dispersion in Y.

The mean and variance are

$$\mathbb{E}(Y) = m\frac{\alpha}{\alpha+\beta},$$

and

$$\mathrm{Var}(Y) = \frac{m\alpha\beta(\alpha+\beta+m)}{(\alpha+\beta)^2(\alpha+\beta+1)}.$$

Equivalently, if

$$\mu = \frac{\alpha}{\alpha+\beta}, \qquad \delta = \frac{1}{\alpha+\beta+1},$$

then

$$\mathbb{E}(Y) = m\mu, \qquad \mathrm{Var}(Y) = m\mu(1-\mu)\{1+(m-1)\delta\}.$$

In this representation, μ is the marginal mean success probability and δ is an intraclass-correlation or overdispersion parameter. When $\delta = 0$, the variance reduces to the binomial variance $m\mu(1-\mu)$; when $\delta > 0$, the beta-binomial distribution allows extra-binomial variation.

Solving for the original shape parameters gives

$$\alpha = \mu\frac{1-\delta}{\delta}, \qquad \beta = (1-\mu)\frac{1-\delta}{\delta}.$$

Therefore, the beta-binomial distribution can be written directly as a function of μ and δ:

$$Y \sim \mathrm{BetaBinomial}\left(m,\ \mu, \delta\right),$$

or equivalently,

$$\Pr(Y = y \mid m, \mu, \delta) = \binom{m}{y} \frac{B\left(y + \mu\frac{1-\delta}{\delta}, m - y + (1-\mu)\frac{1-\delta}{\delta}\right)}{B\left(\mu\frac{1-\delta}{\delta}, (1-\mu)\frac{1-\delta}{\delta}\right)}, \qquad y = 0, 1, \ldots, m.$$

This new parameterization is named here as the mean-dispersion parameterization.

3.2.3 Beta-2-Binomial Distribution

Bayes et al. (2024) proposed the beta-2-binomial (B2B) distribution for bounded count data as a flexible extension of the beta-binomial distribution. We write

$$Y \sim \mathrm{B2B}(m, \mu, \delta, \nu),$$

where m is the maximum possible count, μ is the marginal mean success probability, δ is an overdispersion parameter, and ν is an additional shape parameter.

The B2B distribution can be defined hierarchically as

$$Y \mid W = w \sim \mathrm{BetaBinomial}(m, \mu, w), \qquad W \sim \mathrm{Beta}(\delta, \nu),$$

where the beta distribution is written in the same mean-dispersion parameterization used above. Therefore, its probability mass function is

$$\mathrm{b2b}(y \mid m, \mu, \delta, \nu) = \int_0^1 \mathrm{bb}(y \mid m, \mu, w)\, \mathrm{be}(w \mid \delta, \nu)\, dw, \qquad y = 0, 1, \ldots, m,$$

where

$$\mathrm{bb}(y \mid m, \mu, w) = \binom{m}{y} \frac{B\left(y + \mu\frac{1-w}{w}, m - y + (1-\mu)\frac{1-w}{w}\right)}{B\left(\mu\frac{1-w}{w}, (1-\mu)\frac{1-w}{w}\right)},$$

and

$$\text{be}(w \mid \delta, \nu) = \frac{w^{\delta(1-\nu)/\nu-1}(1-w)^{(1-\delta)(1-\nu)/\nu-1}}{B\left(\delta\frac{1-\nu}{\nu}, (1-\delta)\frac{1-\nu}{\nu}\right)}, \qquad 0 < w < 1.$$

This probability function generally requires numerical integration (Bayes et al. 2024).

The B2B distribution is constructed to preserve the same mean and variance structure as the beta-binomial model:

$$\mathbb{E}(Y) = m\mu,$$

and

$$\text{Var}(Y) = m\mu(1-\mu)\{1 + (m-1)\delta\}.$$

Thus, δ controls the inflation of the binomial variance, as in the beta-binomial model, while ν provides additional flexibility in the shape of the distribution. This extra shape parameter can help accommodate skewness, kurtosis, outlying counts, or excess observations at the boundaries 0 and m (Bayes et al. 2024).

In summary, the binomial distribution is the most restrictive of the three models because its variance is determined only by m and μ. The beta-binomial distribution preserves the same mean but introduces the dispersion parameter δ, inflating the binomial variance and allowing positive dependence among the underlying Bernoulli trials. The B2B distribution preserves the same mean and variance form as the beta-binomial distribution but adds the shape parameter ν, which controls tail behavior and the amount of probability assigned to extreme or outlying bounded counts. Thus, the sequence Binomial–BB–B2B moves from a parsimonious bounded-count model to increasingly flexible models for overdispersion and distributional shape.

Figure 1 illustrates the effect of replacing the binomial distribution by the BB and B2B distributions. The four panels vary the mean parameter μ, while the overdispersion parameter is fixed at $\delta = 0.3$ for the BB and B2B distributions. For the B2B distribution, three values of the additional shape parameter, $\nu = 0.2, 0.5, 0.8$, are shown. This comparison highlights that the binomial distribution is concentrated around its mean, the BB distribution spreads probability more broadly because of overdispersion, and the B2B distribution can further modify the tails and the probability of extreme bounded counts.

```
## quartz_off_screen
##                 2
```

In Figure 1, all distributions in each panel share the same value of μ, so differences among the curves are due to the dispersion and shape parameters. The BB curve is more dispersed than the binomial curve because $\delta = 0.3$ inflates the binomial variance. The B2B curves preserve the same mean and variance form as the BB model, but changes in ν alter the shape of the probability mass function, especially near the tails and boundaries. This behavior motivates the use of the B2B distribution when bounded count data show both overdispersion and unusual tail or boundary behavior.

3.2.4 Identification

For bounded count distributions, identification refers to whether different parameter values imply different probability mass functions on $\{0, 1, \ldots, m\}$. The maximum count m is assumed to be known. The binomial model has a single free parameter, μ, and is identified by its mean:

$$\mu = \frac{\mathbb{E}(Y)}{m}.$$

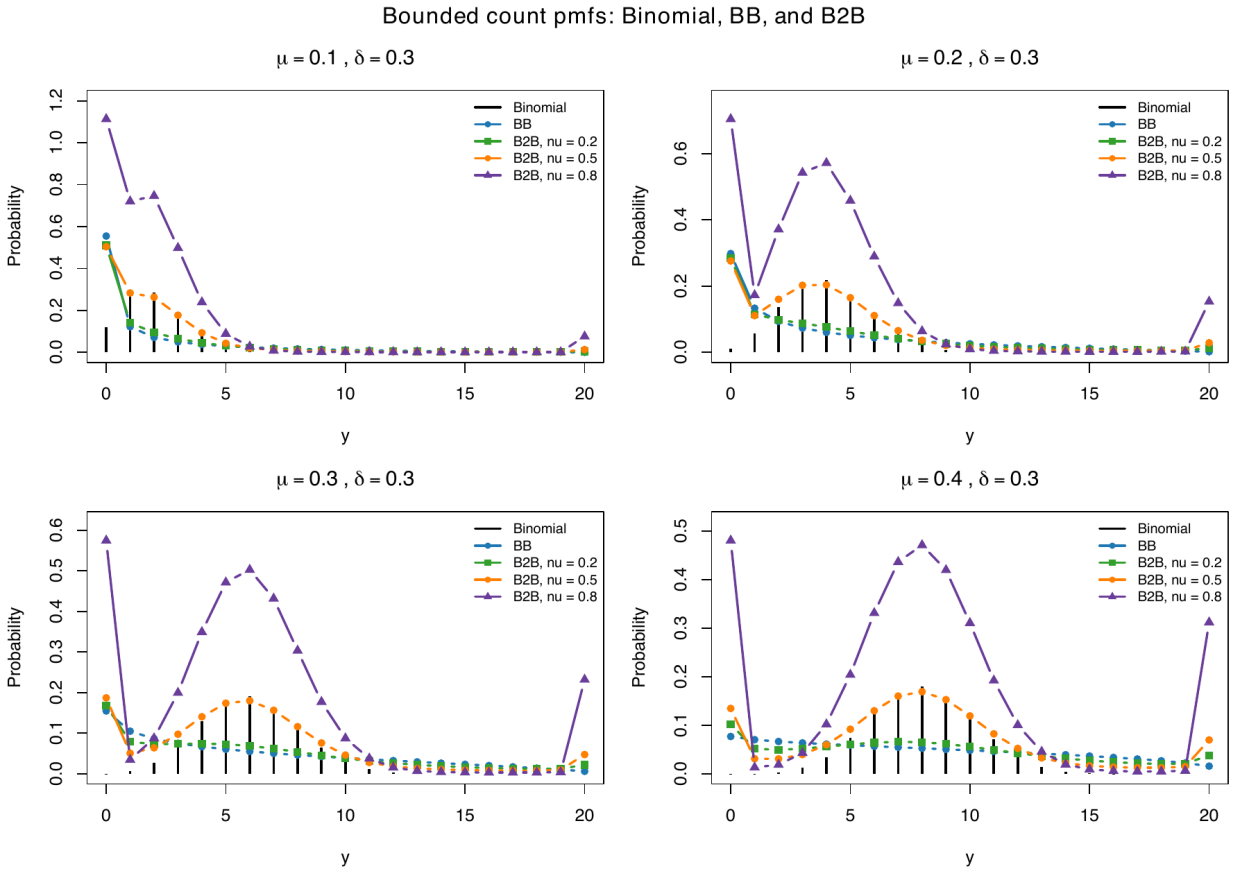


Figure 1: Probability mass functions of the binomial, beta-binomial (BB), and beta-2-binomial (B2B) distributions for $m = 20$, $\delta = 0.3$, $\mu \in \{0.1, 0.2, 0.3, 0.4\}$, and $\nu \in \{0.2, 0.5, 0.8\}$. The figure follows the structure of Figure 2 in Bayes et al. (2024).

Thus, for fixed m, two binomial distributions with different values of μ produce different probability mass functions.

The BB distribution has two free parameters, μ and δ. The mean identifies μ, while the variance identifies δ whenever $0 < \mu < 1$ and $m > 1$:

$$\mu = \frac{\mathbb{E}(Y)}{m}, \qquad \delta = \frac{\mathrm{Var}(Y)/\{m\mu(1-\mu)\} - 1}{m-1}.$$

Consequently, the BB distribution is identified by the first two moments under the usual parameter restrictions $0 < \mu < 1$ and $0 < \delta < 1$. The boundary case $\delta \to 0$ corresponds to the binomial model, so a BB model with $\delta = 0$ is observationally equivalent to a binomial model with the same m and μ.

The B2B distribution has three free parameters: μ, δ, and ν. Its mean and variance have the same form as in the BB distribution, so the first two moments identify μ and δ but not ν. The additional shape parameter ν is identified through higher-order features of the probability mass function, such as skewness, kurtosis, tail behavior, and the amount of mass near the boundaries. In particular, as $\nu \to 0$, the B2B distribution reduces to the BB distribution with the same m, μ, and δ (Bayes et al. 2024); therefore, this limiting case is not distinguishable from the BB model. Away from this boundary, identification of ν depends on information in the full bounded-count distribution beyond the mean and variance.

In applications, theoretical identification does not guarantee stable estimation. The extra parameters δ and ν require enough information about dispersion, tails, and boundary counts. When m is small, sample sizes are limited, or there are few observations near 0 or m, the BB and B2B models may be weakly identified in practice. In such cases, comparison with the simpler binomial and BB models, together with diagnostics for overdispersion and extreme counts, is important.

3.2.5 Monotonicity

For bounded count distributions, monotonicity can be understood as stochastic monotonicity in the mean parameter μ: larger values of μ should shift the distribution toward larger counts. Equivalently, for fixed m and for any cutoff $c \in \{0, \dots, m\}$, the upper-tail probability

$$\Pr(Y \geq c)$$

should be nondecreasing as μ increases.

The binomial distribution satisfies this property directly. If $Y \sim \mathrm{Binomial}(m, \mu)$, then increasing μ increases the probability of success in each Bernoulli trial. Therefore, for fixed m, the binomial distribution is stochastically increasing in μ. In particular, larger values of μ imply larger values of $\mathbb{E}(Y) = m\mu$ and shift probability mass from lower counts toward higher counts.

The BB distribution also preserves monotonicity in μ for each fixed value of δ. In the hierarchical representation,

$$Y \mid \pi \sim \mathrm{Binomial}(m, \pi), \qquad \pi \sim \mathrm{Beta}(\mu, \delta),$$

where the beta distribution is written in the mean-dispersion parameterization. For fixed δ, increasing μ shifts the mixing distribution of π toward larger success probabilities, and the conditional binomial distribution is itself monotone in π. Hence, the marginal BB distribution is stochastically increasing in μ. The parameter δ does not reverse this monotonicity; instead, it controls dispersion. As δ increases, probability mass is spread more strongly toward the tails, but the distribution is

not generally monotone in δ at every count value because increasing dispersion can increase tail probabilities while decreasing probabilities near the center.

This monotonicity statement is therefore conditional on fixing δ. If δ changes jointly with μ, the distributions are not necessarily ordered by μ alone. For two BB distributions,

$$Y_1 \sim \text{BetaBinomial}(m, \mu_1, \delta_1), \qquad Y_2 \sim \text{BetaBinomial}(m, \mu_2, \delta_2),$$

the condition $\mu_2 > \mu_1$ does not by itself imply

$$\Pr(Y_2 \geq c) \geq \Pr(Y_1 \geq c) \qquad \text{for all } c.$$

A larger value of δ_2 may increase the probability of both very low and very high counts, while decreasing probability near the center. As a result, the probability mass functions or distribution functions can cross. Hence, μ remains the mean-success parameter, but stochastic monotonicity in μ is guaranteed only when the dispersion parameter δ is held fixed.

The B2B distribution similarly preserves monotonicity in μ for fixed values of δ and ν. Its hierarchical form can be written as

$$Y \mid W = w \sim \text{BetaBinomial}(m, \mu, w), \qquad W \sim \text{Beta}(\delta, \nu).$$

For every fixed w, the conditional BB distribution is stochastically increasing in μ. Averaging these monotone conditional distributions over the mixing distribution of W preserves monotonicity. Thus, for fixed δ and ν, larger values of μ shift the B2B distribution toward larger counts. The parameters δ and ν affect dispersion, tails, and boundary behavior: δ controls the variance inflation, while ν controls additional shape flexibility. As with the BB model, changes in δ or ν should be interpreted as changes in spread and shape, not as simple monotone shifts of the distribution over counts.

If δ and ν are not fixed, monotonicity in μ is again not guaranteed by μ alone. For two B2B distributions,

$$Y_1 \sim \text{B2B}(m, \mu_1, \delta_1, \nu_1), \qquad Y_2 \sim \text{B2B}(m, \mu_2, \delta_2, \nu_2),$$

the inequality $\mu_2 > \mu_1$ does not necessarily imply stochastic ordering of Y_2 over Y_1. Changes in δ alter the variance inflation, while changes in ν alter tail and boundary behavior. These shape changes can counteract the rightward shift induced by a larger μ, especially for bounded counts where extra mass can be assigned simultaneously to the lower and upper boundaries. Therefore, the B2B distribution is monotone in μ only conditionally on fixed δ and ν; when all three parameters vary, comparisons must be made using the full probability mass function or tail probabilities rather than μ alone.

3.2.6 Skewness and Kurtosis

This subsection collects the skewness and kurtosis coefficients for the binomial, BB, and B2B distributions. Throughout, kurtosis denotes the usual standardized fourth central moment, not excess kurtosis. Let

$$q = \mu(1-\mu), \qquad A_x = 1 + (m-1)x.$$

For $Y \sim \text{Binomial}(m, \mu)$,

$$\text{Skew}_{\text{Bin}}(m, \mu) = \frac{1-2\mu}{\sqrt{mq}},$$

and

$$\text{Kurt}_{\text{Bin}}(m,\mu) = 3 + \frac{1-6q}{mq}.$$

For $Y \sim \text{BetaBinomial}(m,\mu,\delta)$,

$$\text{Skew}_{\text{BB}}(m,\mu,\delta) = \frac{(1-2\mu)\{1+(2m-1)\delta\}}{(1+\delta)\sqrt{mqA_\delta}},$$

and

$$\text{Kurt}_{\text{BB}}(m,\mu,\delta) = \frac{k_0 + k_1\delta + k_2\delta^2 + k_3\delta^3}{(1+\delta)(1+2\delta)mqA_\delta^2},$$

where

$$\begin{aligned}
k_0 &= 3q(m-2)+1,\\
k_1 &= 3q(2m^2-11m+6)+7m-4,\\
k_2 &= 3q(m^3-16m^2+19m-6)+12m^2-15m+5,\\
k_3 &= 3q(m-1)(-7m^2+7m-2)+2(m-1)(3m^2-3m+1).
\end{aligned}$$

Let

$$a_k = \mathbb{E}\left(\frac{W^k}{1+W}\right), \qquad k=0,1,2,$$

and

$$d_k = \mathbb{E}\left\{\frac{W^k}{(1+W)(1+2W)}\right\}, \qquad k=0,1,2,3,$$

where $W \sim \text{Beta}(\delta,\nu)$.

For $Y \sim \text{B2B}(m,\mu,\delta,\nu)$,

$$\text{Skew}_{\text{B2B}}(m,\mu,\delta,\nu) = \frac{(1-2\mu)\{a_0+(3m-2)a_1+(2m^2-3m+1)a_2\}}{\sqrt{mqA_\delta^3}},$$

and

$$\text{Kurt}_{\text{B2B}}(m,\mu,\delta,\nu) = \frac{k_0d_0+k_1d_1+k_2d_2+k_3d_3}{mqA_\delta^2}.$$

In the following we include relations between the skewness and kurtosis of the distributions introduced.

Theorem 1 (Skewness relationships for the BB and B2B distributions). *Let $q=\mu(1-\mu)$ and $A_x = 1+(m-1)x$.*

(a) If $Y \sim \text{BetaBinomial}(m,\mu,\delta)$, then

$$\text{Skew}_{\text{BB}}(m,\mu,\delta) = K(m,\delta) \times \text{Skew}_{\text{Bin}}(m,\mu),$$

where $K(m,\delta) = \left[\frac{1+(2m-1)\delta}{(1+\delta)\sqrt{A_\delta}}\right]$.

*(b) If $Y \sim \text{B2B}(m,\mu,\delta,\nu)$ and $W \sim \text{Beta}(\delta,\nu)$, then

$$\text{Skew}_{\text{B2B}}(m,\mu,\delta,\nu) = \frac{\mathbb{E}\left[\text{Skew}_{\text{BB}}(m,\mu,W)A_W^{3/2}\right]}{A_\delta^{3/2}}.$$

In the limiting special case $\delta \to 0$, $A_\delta \to 1$ and $K(m, \delta) \to 1$. Therefore,

$$\text{Skew}_{\text{BB}}(m, \mu, 0) = \text{Skew}_{\text{Bin}}(m, \mu),$$

which is the skewness of the binomial distribution. In the limiting special case $\nu \to 0$, the mixing variable satisfies $W \to \delta$ in distribution, and

$$\text{Skew}_{\text{B2B}}(m, \mu, \delta, 0) = \text{Skew}_{\text{BB}}(m, \mu, \delta).$$

If additionally $\delta \to 0$, this reduces to $\text{Skew}_{\text{Bin}}(m, \mu)$.*

Theorem 2 (Kurtosis relationships for the BB and B2B distributions). *Let $q = \mu(1-\mu)$ and $A_x = 1 + (m-1)x$.*

*(a) If $Y \sim \text{BetaBinomial}(m, \mu, \delta)$, then

$$\text{Kurt}_{\text{BB}}(m, \mu, \delta) = \frac{mq\ \text{Kurt}_{\text{Bin}}(m, \mu) + k_1\delta + k_2\delta^2 + k_3\delta^3}{(1+\delta)(1+2\delta)mqA_\delta^2}.$$

*(b) If $Y \sim \text{B2B}(m, \mu, \delta, \nu)$ and $W \sim \text{Beta}(\delta, \nu)$, then

$$\text{Kurt}_{\text{B2B}}(m, \mu, \delta, \nu) = \frac{\mathbb{E}\left[\text{Kurt}_{\text{BB}}(m, \mu, W)A_W^2\right]}{A_\delta^2}.$$

In the limiting special case $\delta \to 0$ ($A_\delta \to 1$), all terms involving δ vanish. Hence,

$$\text{Kurt}_{\text{BB}}(m, \mu, 0) = \text{Kurt}_{\text{Bin}}(m, \mu),$$

which is the kurtosis of the binomial distribution. In the limiting special case $\nu \to 0$, $W \to \delta$ in distribution, and

$$\text{Kurt}_{\text{B2B}}(m, \mu, \delta, 0) = \text{Kurt}_{\text{BB}}(m, \mu, \delta).$$

If additionally $\delta \to 0$, this reduces to $\text{Kurt}_{\text{Bin}}(m, \mu)$.*

These expressions for the skewness and kurtosis show how the three distributions differ beyond their mean and variance. The binomial skewness depends only on μ and m: it is positive when $\mu < 0.5$, negative when $\mu > 0.5$, and equal to zero when $\mu = 0.5$. Its kurtosis is also fixed once m and μ are specified. The BB distribution keeps the same sign of skewness as the binomial model, because the factor multiplying $\text{Skew}_{\text{Bin}}(m, \mu)$ is positive, but δ changes the magnitude of skewness and kurtosis. Thus, overdispersion affects not only the variance but also the asymmetry and tail weight of the bounded count distribution. The B2B distribution adds a further layer of flexibility: its skewness and kurtosis are weighted averages of the corresponding BB quantities over the mixing distribution of W. Consequently, ν can change the tail behavior and the concentration near extreme counts without changing the mean and variance form. In this sense, the moment-shape relationships parallel the distributional hierarchy: the binomial model fixes dispersion and shape, the BB model adds overdispersion through δ, and the B2B model adds shape flexibility through ν.

3.3 Regression Models for Bounded Count Responses

Let $Y_i \in \{0, 1, \ldots, m_i\}$ denote a bounded count response for observation i, where m_i is known. Regression models for bounded count data are obtained by allowing one or more distributional parameters to depend on covariates. In all three models below, the mean parameter is linked to covariates through

$$g_\mu(\mu_i) = \mathbf{x}_i^\top \beta,$$

where $g_\mu(\cdot)$ is commonly the logit or probit link, so that $0 < \mu_i < 1$.

3.3.1 Binomial Regression Model

The binomial regression model is

$$Y_i \mid \mu_i \sim \text{Binomial}(m_i, \mu_i), \qquad g_\mu(\mu_i) = \mathbf{x}_i^\top \beta.$$

The conditional mean and variance are

$$\mathbb{E}(Y_i \mid \mathbf{x}_i) = m_i \mu_i, \qquad \text{Var}(Y_i \mid \mathbf{x}_i) = m_i \mu_i (1 - \mu_i).$$

This model is appropriate when the binomial variance is adequate and the within-count trials can be treated as conditionally independent with a common success probability.

3.3.2 BB Regression Model

The BB regression model extends binomial regression by adding an overdispersion parameter:

$$Y_i \mid \mu_i, \delta_i \sim \text{BetaBinomial}(m_i, \mu_i, \delta_i).$$

The mean parameter can depend on covariates through

$$g_\mu(\mu_i) = \mathbf{x}_i^\top \beta,$$

and the dispersion parameter can also depend on covariates through

$$g_\delta(\delta_i) = \mathbf{z}_i^\top \alpha,$$

where $g_\delta(\cdot)$ is typically a logit link, ensuring $0 < \delta_i < 1$. Thus, it is possible to include covariates in both μ_i and δ_i. The conditional mean and variance are

$$\mathbb{E}(Y_i \mid \mathbf{x}_i) = m_i \mu_i,$$

and

$$\text{Var}(Y_i \mid \mathbf{x}_i, \mathbf{z}_i) = m_i \mu_i (1 - \mu_i)\{1 + (m_i - 1)\delta_i\}.$$

Covariates in μ_i explain changes in the expected proportion of successes, whereas covariates in δ_i explain changes in overdispersion.

3.3.3 B2B Regression Model

The B2B regression model extends the BB regression model by adding a shape parameter:

$$Y_i \mid \mu_i, \delta_i, \nu_i \sim \text{B2B}(m_i, \mu_i, \delta_i, \nu_i).$$

Following Bayes et al. (2024), covariates can be introduced in the mean, dispersion, and shape parameters:

$$g_\mu(\mu_i) = \mathbf{x}_i^\top \beta, \qquad g_\delta(\delta_i) = \mathbf{z}_i^\top \alpha, \qquad g_\nu(\nu_i) = \mathbf{u}_i^\top \gamma,$$

with links such as the logit link to enforce $0 < \mu_i < 1$, $0 < \delta_i < 1$, and $0 < \nu_i < 1$. The conditional mean and variance are

$$\mathbb{E}(Y_i \mid \mathbf{x}_i) = m_i \mu_i,$$

and

$$\text{Var}(Y_i \mid \mathbf{x}_i, \mathbf{z}_i) = m_i \mu_i (1 - \mu_i)\{1 + (m_i - 1)\delta_i\}.$$

The parameter ν_i does not change this mean and variance form; instead, it controls additional shape features such as tail behavior, boundary mass, skewness, and kurtosis.

3.3.4 Likelihood Functions

Assume that $Y_1, \dots, Y_n$ are conditionally independent bounded count responses, with observed values $y_i \in \{0, 1, \dots, m_i\}$. The likelihood of a bounded count regression model is obtained by multiplying the probability mass functions evaluated at the observation-specific parameters.

For the binomial regression model, let

$$\mu_i = g_\mu^{-1}(\mathbf{x}_i^\top \beta).$$

The likelihood for β is

$$L_{\text{Bin}}(\beta) = \prod_{i=1}^{n} \binom{m_i}{y_i} \mu_i^{y_i} (1-\mu_i)^{m_i - y_i},$$

and the log-likelihood is

$$\ell_{\text{Bin}}(\beta) = \sum_{i=1}^{n} \left[\log \binom{m_i}{y_i} + y_i \log(\mu_i) + (m_i - y_i) \log(1-\mu_i) \right].$$

For the BB regression model, let

$$\mu_i = g_\mu^{-1}(\mathbf{x}_i^\top \beta), \qquad \delta_i = g_\delta^{-1}(\mathbf{z}_i^\top \alpha).$$

Define

$$\alpha_i = \mu_i \frac{1-\delta_i}{\delta_i}, \qquad \beta_i = (1-\mu_i)\frac{1-\delta_i}{\delta_i}.$$

Then the likelihood for (β, α) is

$$L_{\text{BB}}(\beta, \alpha) = \prod_{i=1}^{n} \binom{m_i}{y_i} \frac{B(y_i + \alpha_i, m_i - y_i + \beta_i)}{B(\alpha_i, \beta_i)}.$$

Equivalently, the log-likelihood is

$$\ell_{\text{BB}}(\beta, \alpha) = \sum_{i=1}^{n} \left[\log \binom{m_i}{y_i} + \log B(y_i + \alpha_i, m_i - y_i + \beta_i) - \log B(\alpha_i, \beta_i) \right].$$

For the B2B regression model, let

$$\mu_i = g_\mu^{-1}(\mathbf{x}_i^\top \beta), \qquad \delta_i = g_\delta^{-1}(\mathbf{z}_i^\top \alpha), \qquad \nu_i = g_\nu^{-1}(\mathbf{u}_i^\top \gamma).$$

Using the B2B probability mass function defined in the previous section, the likelihood for (β, α, γ) is

$$L_{\text{B2B}}(\beta, \alpha, \gamma) = \prod_{i=1}^{n} \text{b2b}(y_i \mid m_i, \mu_i, \delta_i, \nu_i),$$

where

$$\text{b2b}(y_i \mid m_i, \mu_i, \delta_i, \nu_i) = \int_0^1 \text{bb}(y_i \mid m_i, \mu_i, w)\, \text{be}(w \mid \delta_i, \nu_i)\, dw.$$

The corresponding log-likelihood is

$$\ell_{\text{B2B}}(\beta, \alpha, \gamma) = \sum_{i=1}^{n} \log \left\{ \text{b2b}(y_i \mid m_i, \mu_i, \delta_i, \nu_i) \right\}.$$

Because the B2B probability mass function involves an integral, maximum likelihood estimation generally requires numerical integration or a dedicated implementation of the B2B family (Bayes et al. 2024).

3.3.5 Monotonicity, Covariates, and Model Establishment

Monotonicity of the distribution in μ is a desirable interpretability property for a regression model, but it is not by itself the only condition needed to propose a regression model. It ensures that, when other parameters are fixed, increasing the linear predictor for μ_i produces a distribution that shifts toward larger counts. This makes β interpretable as governing the expected proportion of successes. However, a regression model also requires valid parameter spaces, identifiability, appropriate link functions, and a conditional distribution that represents the variation in the data.

It is possible to include covariates in both μ_i and δ_i for the BB model, and in μ_i, δ_i, and ν_i for the B2B model. When covariates are included in δ_i or ν_i, the model is no longer only a mean regression model: it becomes a distributional regression model. In this case, covariates may affect the mean, the variance inflation, and the shape of the response distribution.

These regression models are well established under the following conditions. First, the link functions must map the real line into the correct parameter spaces, such as $(0, 1)$ for μ_i, δ_i, and ν_i. Second, the parameters must be identifiable from the data. In particular, δ_i requires information about overdispersion beyond the binomial variance, and ν_i requires information about tail behavior or boundary counts beyond the mean and variance. Third, the covariate structures for μ_i, δ_i, and ν_i should not be unnecessarily complex or collinear, because weak information about dispersion and shape can lead to unstable estimation. Finally, monotonicity in μ_i should be interpreted conditionally on fixed dispersion and shape parameters. If δ_i or ν_i change together with μ_i, the full distributions may not be ordered by μ_i alone, even though $\mathbb{E}(Y_i) = m_i\mu_i$ remains increasing in μ_i.

3.4 Practical Estimation Using GAMLSS

The generalized additive models for location, scale, and shape framework (GAMLSS) provides a practical way to estimate bounded count regression models whose parameters may depend on covariates (Rigby and Stasinopoulos 2005; Stasinopoulos and Rigby 2007). Instead of modeling only the conditional mean, GAMLSS allows different distributional parameters to have their own regression equations. For bounded count data, this is useful because μ_i can describe the expected proportion of successes, while dispersion or shape parameters can describe departures from the binomial variance.

In the `gamlss` implementation, the response for binomial-type models is usually provided as a two-column matrix containing the number of successes and the number of failures. Thus, for a response $Y_i \in \{0, \dots, m_i\}$, the response matrix is

$$(Y_i,\ m_i - Y_i)\,.$$

The package `gamlss.dist` includes the binomial family `BI()` and the beta-binomial family `BB()` (Stasinopoulos and Rigby 2023). The beta-binomial family in `gamlss.dist` uses a dispersion parameter `sigma`. Its relation with the notation used in this paper is

$$\delta = \frac{\sigma}{1+\sigma}, \qquad \sigma = \frac{\delta}{1-\delta}.$$

Therefore, fitted values of `sigma` can be translated to the overdispersion parameter δ using this transformation.

3.4.1 Installing and Loading the Package

The packages can be installed from CRAN and loaded as follows (Stasinopoulos and Rigby 2007; Stasinopoulos and Rigby 2023).

```
install.packages(c("gamlss", "gamlss.dist"))

library(gamlss)
library(gamlss.dist)
```

The diagnostics below use normalized randomized quantile residuals (Dunn and Smyth 1996). For a discrete response, these residuals randomize within the fitted cumulative probability interval for each observed count and transform the result to a standard normal quantile.

```
compare_models <- function(fit_bi, fit_bb, n) {
  tab <- data.frame(
    model = c("Binomial", "Beta-binomial"),
    AIC = c(AIC(fit_bi), AIC(fit_bb)),
    BIC = c(GAIC(fit_bi, k = log(n)), GAIC(fit_bb, k = log(n)))
  )
  tab$AIC <- round(tab$AIC, 2)
  tab$BIC <- round(tab$BIC, 2)
  tab
}

residual_summary <- function(fit) {
  r <- residuals(fit)
  c(
    mean = mean(r),
    sd = sd(r),
    min = min(r),
    median = median(r),
    max = max(r)
  )
}

plot_residual_diagnostics <- function(fit_bi, fit_bb, main_label) {
  old_par <- par(mfrow = c(2, 2), mar = c(4, 4, 3, 1))
  on.exit(par(old_par))

  plot(fitted(fit_bi), residuals(fit_bi),
       xlab = "Fitted values", ylab = "Normalized quantile residuals",
       main = paste(main_label, "- BI"))
  abline(h = 0, col = 2, lty = 2)
  qqnorm(residuals(fit_bi), main = "Normal Q-Q: BI")
  qqline(residuals(fit_bi), col = 2)

  plot(fitted(fit_bb), residuals(fit_bb),
       xlab = "Fitted values", ylab = "Normalized quantile residuals",
       main = paste(main_label, "- BB"))
  abline(h = 0, col = 2, lty = 2)
  qqnorm(residuals(fit_bb), main = "Normal Q-Q: BB")
  qqline(residuals(fit_bb), col = 2)
}

compare_three_models <- function(fit_bi, fit_bb, fit_b2b, n) {
  tab <- data.frame(
    model = c("Binomial", "Beta-binomial", "B2B"),
    AIC = c(AIC(fit_bi), AIC(fit_bb), AIC(fit_b2b)),
```

```
    BIC = c(
      GAIC(fit_bi, k = log(n)),
      GAIC(fit_bb, k = log(n)),
      GAIC(fit_b2b, k = log(n))
    )
  )
  tab$AIC <- round(tab$AIC, 2)
  tab$BIC <- round(tab$BIC, 2)
  tab
}

plot_residual_diagnostics_three <- function(fit_bi, fit_bb, fit_b2b, main_label) {
  old_par <- par(mfrow = c(3, 2), mar = c(4, 4, 3, 1))
  on.exit(par(old_par))

  fits <- list(BI = fit_bi, BB = fit_bb, B2B = fit_b2b)
  for (model_name in names(fits)) {
    fit <- fits[[model_name]]
    plot(fitted(fit), residuals(fit),
         xlab = "Fitted values", ylab = "Normalized quantile residuals",
         main = paste(main_label, "-", model_name))
    abline(h = 0, col = 2, lty = 2)
    qqnorm(residuals(fit), main = paste("Normal Q-Q:", model_name))
    qqline(residuals(fit), col = 2)
  }
}
```

The installation command only needs to be run once. In a reproducible analysis, it is common to keep the installation line commented and load the packages at the beginning of the script or notebook.

3.4.2 Example: Simulated Binomial Regression

The following example simulates bounded count data from a binomial regression model with $m = 20$ trials and a logit link for μ_i:

$$\text{logit}(\mu_i) = -0.4 + 1.1x_i.$$

```
set.seed(2026)
n <- 400
m <- 20
x <- runif(n, -1.5, 1.5)

eta <- -0.4 + 1.1 * x
mu <- plogis(eta)
y <- rBI(n, bd = m, mu = mu)

dat_bin <- data.frame(
  y = y,
  failures = m - y,
  x = x
)

fit_bin_BI <- gamlss(
  cbind(y, failures) ~ x,
  family = BI,
  data = dat_bin,
  trace = FALSE
)
```

```
fit_bin_BB <- gamlss(
  cbind(y, failures) ~ x,
  sigma.formula = ~ 1,
  family = BB,
  data = dat_bin,
  trace = FALSE
)

fit_bin_BI
```

```
##
## Family:  c("BI", "Binomial")
## Fitting method: RS()
##
## Call:  gamlss(formula = cbind(y, failures) ~ x, family = BI,
##     data = dat_bin, trace = FALSE)
##
## Mu Coefficients:
## (Intercept)            x
##      -0.412        1.143
##
##  Degrees of Freedom for the fit: 2 Residual Deg. of Freedom   398
## Global Deviance:     1623.15
##             AIC:     1627.15
##             SBC:     1635.14
```

```
summary(fit_bin_BI)
```

```
## ******************************************************************
## Family:  c("BI", "Binomial")
##
## Call:  gamlss(formula = cbind(y, failures) ~ x, family = BI, data = dat_bin,
##     trace = FALSE)
##
## Fitting method: RS()
##
## ------------------------------------------------------------------
## Mu link function:  logit
## Mu Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.41204    0.02543  -16.20   <2e-16 ***
## x            1.14293    0.03190   35.83   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## ------------------------------------------------------------------
## No. of observations in the fit:  400
## Degrees of Freedom for the fit:  2
##       Residual Deg. of Freedom:  398
##                       at cycle:  2
```

```
##
## Global Deviance:     1623.154
##             AIC:     1627.154
##             SBC:     1635.137
## *******************************************************************
```

```
fit_bin_BB
```

```
##
## Family:  c("BB", "Beta Binomial")
## Fitting method: RS()
##
## Call:  gamlss(formula = cbind(y, failures) ~ x, sigma.formula = ~1,
##     family = BB, data = dat_bin, trace = FALSE)
##
## Mu Coefficients:
## (Intercept)            x
##     -0.3652       1.1237
## Sigma Coefficients:
## (Intercept)
##      -36.01
##
##  Degrees of Freedom for the fit: 3 Residual Deg. of Freedom   397
## Global Deviance:     1623.15
##             AIC:     1629.15
##             SBC:     1641.13
```

```
summary(fit_bin_BB)
```

```
## *******************************************************************
## Family:  c("BB", "Beta Binomial")
##
## Call:  gamlss(formula = cbind(y, failures) ~ x, sigma.formula = ~1,
##     family = BB, data = dat_bin, trace = FALSE)
##
## Fitting method: RS()
##
## ------------------------------------------------------------------
## Mu link function:  logit
## Mu Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.36519    0.02513  -14.53   <2e-16 ***
## x            1.12368    0.03131   35.89   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## ------------------------------------------------------------------
## Sigma link function:  log
## Sigma Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept) -36.010068   0.001668  -21592   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## ------------------------------------------------------------------
## No. of observations in the fit:  400
## Degrees of Freedom for the fit:  3
##       Residual Deg. of Freedom:  397
##                       at cycle:  6
##
## Global Deviance:     1623.155
##             AIC:     1629.155
##             SBC:     1641.129
## ******************************************************************
```

```
coef(fit_bin_BI, what = "mu")
```

```
## (Intercept)           x
##  -0.4120373   1.1429346
```

```
coef(fit_bin_BB, what = "mu")
```

```
## (Intercept)           x
##   -0.365188    1.123675
```

```
sigma_hat_bin <- exp(coef(fit_bin_BB, what = "sigma"))
delta_hat_bin <- sigma_hat_bin / (1 + sigma_hat_bin)
delta_hat_bin
```

```
##  (Intercept)
## 2.296288e-16
```

```
compare_models(fit_bin_BI, fit_bin_BB, n)
```

```
##           model     AIC     BIC
## 1      Binomial 1627.15 1635.14
## 2 Beta-binomial 1629.15 1641.13
```

```
deviance(fit_bin_BI)
```

```
## [1] 1623.154
```

```
deviance(fit_bin_BB)
```

```
## [1] 1623.155
```

```
round(rbind(
  Binomial = residual_summary(fit_bin_BI),
  `Beta-binomial` = residual_summary(fit_bin_BB)
), 3)
```

```
##                  mean    sd    min median   max
## Binomial        0.013 0.957 -2.746  0.028 2.810
## Beta-binomial -0.003 0.953 -2.885 -0.010 2.607
```

```
plot_residual_diagnostics(fit_bin_BI, fit_bin_BB, "Binomial data")
```

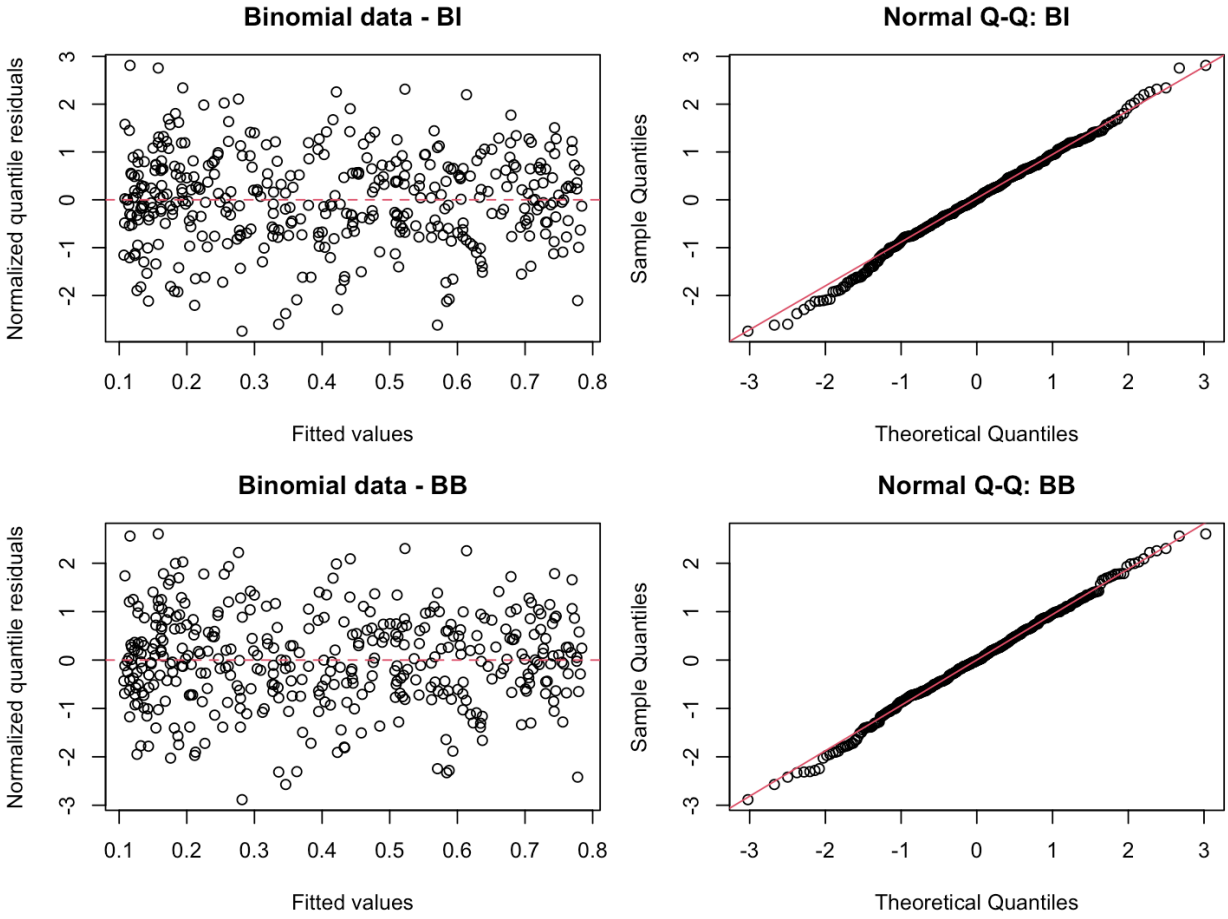


Figure 2: Residual diagnostics for the binomial and beta-binomial fits to data simulated from the binomial regression model.

For this simulated data set, the binomial model recovers the true regression coefficients well: the fitted intercept is approximately -0.412 and the fitted slope is approximately 1.143, close to the generating values -0.4 and 1.1. The beta-binomial model estimates an overdispersion parameter essentially equal to zero ($\hat{\delta} \approx 0$), so it collapses to the binomial special case. The binomial model has the smaller AIC and BIC, while both residual summaries have mean close to zero and standard deviation close to one. Therefore, the diagnostic evidence supports choosing the simpler binomial regression model.

3.4.3 Example: Simulated Beta-Binomial Regression

The next example simulates data from a beta-binomial regression model with the same mean structure and constant overdispersion $\delta = 0.30$. Since `BB()` uses `sigma`, the simulation uses

$$\sigma = \frac{0.30}{1 - 0.30}.$$

```
set.seed(2027)
n <- 400
m <- 20
x <- runif(n, -1.5, 1.5)

eta <- -0.4 + 1.1 * x
mu <- plogis(eta)
delta <- 0.30
sigma <- delta / (1 - delta)

y <- rBB(n, bd = m, mu = mu, sigma = sigma)

dat_bb <- data.frame(
  y = y,
  failures = m - y,
  x = x
)

fit_bb_BI <- gamlss(
  cbind(y, failures) ~ x,
  family = BI,
  data = dat_bb,
  trace = FALSE
)

fit_bb_BB <- gamlss(
  cbind(y, failures) ~ x,
  sigma.formula = ~ 1,
  family = BB,
  data = dat_bb,
  trace = FALSE
)

fit_bb_BI
```

```
##
## Family:  c("BI", "Binomial")
## Fitting method: RS()
##
## Call:  gamlss(formula = cbind(y, failures) ~ x, family = BI,
```

```
##     data = dat_bb, trace = FALSE)
##
## Mu Coefficients:
## (Intercept)            x
##     -0.3459       1.0872
##
##  Degrees of Freedom for the fit: 2 Residual Deg. of Freedom   398
## Global Deviance:     4013.31
##             AIC:     4017.31
##             SBC:     4025.3
```

```
summary(fit_bb_BI)
```

```
## ******************************************************************
## Family:  c("BI", "Binomial")
##
## Call:  gamlss(formula = cbind(y, failures) ~ x, family = BI, data = dat_bb,
##     trace = FALSE)
##
## Fitting method: RS()
##
## ------------------------------------------------------------------
## Mu link function:  logit
## Mu Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.34591    0.02491  -13.89   <2e-16 ***
## x            1.08719    0.03183   34.16   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## ------------------------------------------------------------------
## No. of observations in the fit:  400
## Degrees of Freedom for the fit:  2
##       Residual Deg. of Freedom:  398
##                       at cycle:  2
##
## Global Deviance:     4013.314
##             AIC:     4017.314
##             SBC:     4025.297
## ******************************************************************
```

```
fit_bb_BB
```

```
##
## Family:  c("BB", "Beta Binomial")
## Fitting method: RS()
##
## Call:  gamlss(formula = cbind(y, failures) ~ x, sigma.formula = ~1,
##     family = BB, data = dat_bb, trace = FALSE)
##
```

```
## Mu Coefficients:
## (Intercept)            x
##     -0.3568       1.0779
## Sigma Coefficients:
## (Intercept)
##     -0.7645
##
##  Degrees of Freedom for the fit: 3 Residual Deg. of Freedom   397
## Global Deviance:     2185.89
##             AIC:     2191.89
##             SBC:     2203.87
```

```
summary(fit_bb_BB)
```

```
## ******************************************************************
## Family:  c("BB", "Beta Binomial")
##
## Call:  gamlss(formula = cbind(y, failures) ~ x, sigma.formula = ~1,
##     family = BB, data = dat_bb, trace = FALSE)
##
## Fitting method: RS()
##
## ------------------------------------------------------------------
## Mu link function:  logit
## Mu Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.35682    0.06224  -5.733 1.95e-08 ***
## x            1.07790    0.07887  13.668  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## ------------------------------------------------------------------
## Sigma link function:  log
## Sigma Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.76448    0.07332  -10.43   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## ------------------------------------------------------------------
## No. of observations in the fit:  400
## Degrees of Freedom for the fit:  3
##       Residual Deg. of Freedom:  397
##                       at cycle:  4
##
## Global Deviance:     2185.892
##             AIC:     2191.892
##             SBC:     2203.866
## ******************************************************************
```

```r
coef(fit_bb_BI, what = "mu")
```

```
## (Intercept)           x
##   -0.345905    1.087190
```

```r
coef(fit_bb_BB, what = "mu")
```

```
## (Intercept)           x
##  -0.3568165   1.0778971
```

```r
sigma_hat <- exp(coef(fit_bb_BB, what = "sigma"))
delta_hat <- sigma_hat / (1 + sigma_hat)
delta_hat
```

```
## (Intercept)
##   0.3176744
```

```r
compare_models(fit_bb_BI, fit_bb_BB, n)
```

```
##            model     AIC     BIC
## 1      Binomial 4017.31 4025.30
## 2 Beta-binomial 2191.89 2203.87
```

```r
deviance(fit_bb_BI)
```

```
## [1] 4013.314
```

```r
deviance(fit_bb_BB)
```

```
## [1] 2185.892
```

```r
round(rbind(
  Binomial = residual_summary(fit_bb_BI),
  `Beta-binomial` = residual_summary(fit_bb_BB)
), 3)
```

```
##                  mean    sd    min median   max
## Binomial       -0.056 2.631 -7.308 -0.142 6.919
## Beta-binomial   0.013 0.990 -3.495  0.012 3.084
```

```r
plot_residual_diagnostics(fit_bb_BI, fit_bb_BB, "Beta-binomial data")
```

The formula for `mu` estimates the mean regression, while `sigma.formula` specifies the regression model for the beta-binomial dispersion parameter. In this simple example, `sigma.formula = ~ 1` fits a constant overdispersion parameter. More general models can include covariates in `sigma.formula`; however, this should be done when the sample size and distribution of the counts provide enough information to identify changes in dispersion separately from changes in the mean.

For this simulated data set, the beta-binomial model is strongly preferred. The fitted overdispersion is $\hat{\delta} \approx 0.318$, close to the true value $\delta = 0.30$. The AIC and BIC are much smaller for the beta-binomial fit than for the binomial fit. The residual analysis gives the same conclusion: under the binomial fit, the normalized quantile residuals have standard deviation about 2.63, which is much larger than one and indicates unmodeled overdispersion. Under the beta-binomial fit, the residual standard deviation is about 0.99, with residuals centered near zero and better agreement with the normal reference in the Q-Q plot.

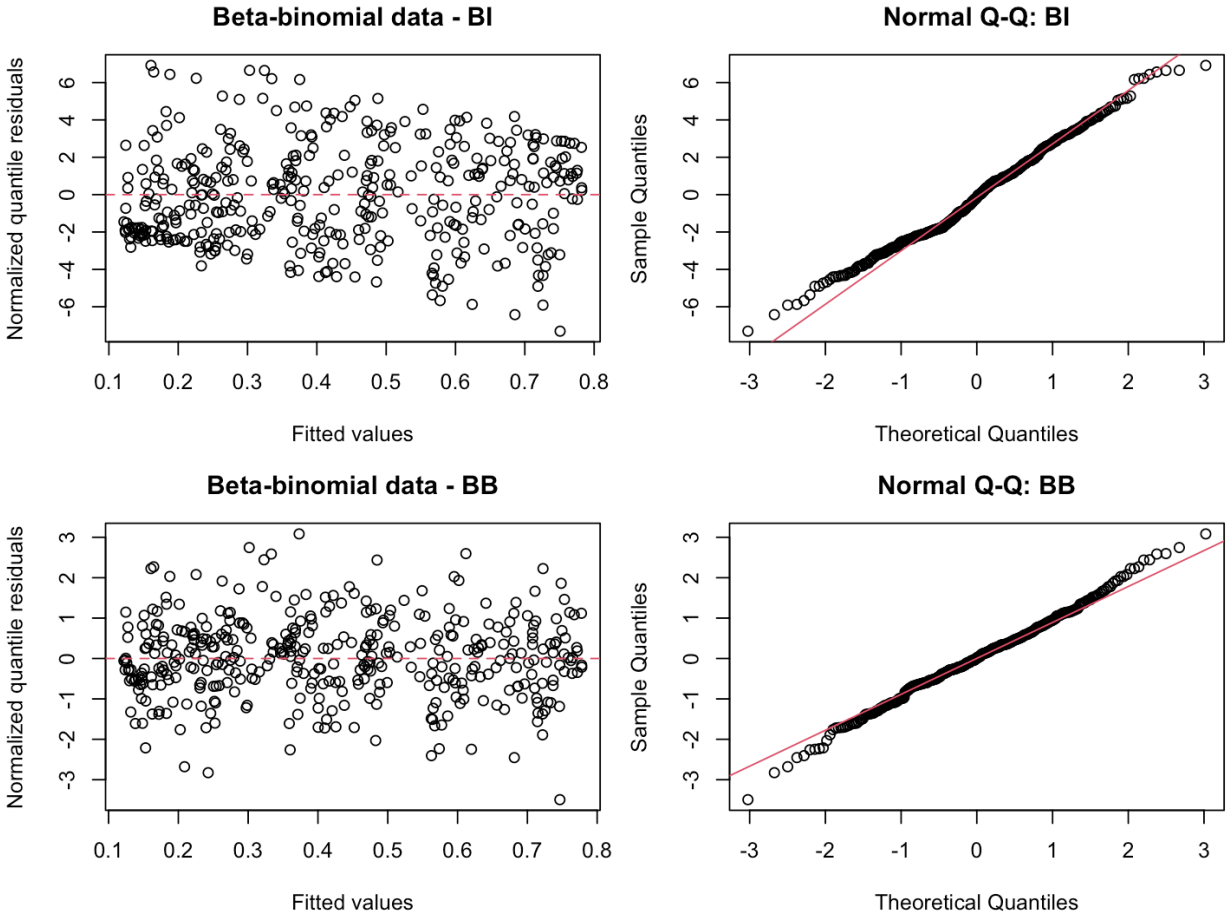


Figure 3: Residual diagnostics for the binomial and beta-binomial fits to data simulated from the beta-binomial regression model.

3.4.4 Example: Simulated B2B Regression Using `RobustDiscrete`

The `RobustDiscrete` package implements robust discrete regression families within the GAMLSS framework. For bounded counts, it provides the Beta-2-Binomial family `B2B()`, which extends the beta-binomial model by adding the shape parameter ν (Bayes et al. 2024, 2026). The package repository describes `RobustDiscrete` as a tool for robust regression with bounded and unbounded discrete responses, including the B2B model for bounded counts.

The package can be installed from the GitHub installer repository. The repository recommends the following installation commands.

For Windows:

```
install.packages(
  "https://github.com/cbayesr/RobustDiscrete_installer/raw/main/RobustDiscrete_0.2
    .0.zip"
)
```

For macOS or source installation:

```
install.packages(
  "https://github.com/cbayesr/RobustDiscrete_installer/raw/main/RobustDiscrete_0.2
    .0.tar.gz",
  repos = NULL,
  type = "source"
)
```

The B2B family uses logit links by default for `mu`, `sigma`, and `nu`. In `RobustDiscrete`, the B2B parameter `sigma` corresponds to the overdispersion parameter δ used in this paper. In contrast, `gamlss.dist::BB()` uses a beta-binomial dispersion parameter satisfying $\delta = \sigma/(1+\sigma)$, so the starting values from a BB fit must be transformed before fitting B2B.

The next example simulates a bounded count response from the B2B hierarchical model,

$$Y_i \mid W_i = w_i \sim \text{BetaBinomial}(m, \mu_i, w_i), \qquad W_i \sim \text{Beta}(\delta, \nu),$$

with $m = 20$, $\delta = 0.30$, $\nu = 0.60$, and

$$\text{logit}(\mu_i) = -0.4 + 1.1 x_i.$$

Then the same data are fitted using binomial, beta-binomial, and B2B regression models.

```
if (dir.exists("Rlib")) {
  .libPaths(c("Rlib", .libPaths()))
}

library(numDeriv)
library(RobustDiscrete)

rb2b <- function(n, bd, mu, delta, nu) {
  shape1 <- delta * (1 - nu) / nu
  shape2 <- (1 - delta) * (1 - nu) / nu
  w <- rbeta(n, shape1 = shape1, shape2 = shape2)
  sigma_w <- w / (1 - w)
  rBB(n, bd = bd, mu = mu, sigma = sigma_w)
}

set.seed(2028)
```

```r
n <- 200
m <- 20
x <- runif(n, -1.5, 1.5)

eta <- -0.4 + 1.1 * x
mu <- plogis(eta)
delta <- 0.30
nu <- 0.60
y <- rb2b(n, bd = m, mu = mu, delta = delta, nu = nu)

dat_b2b <- data.frame(
  y = y,
  failures = m - y,
  x = x
)

control_b2b <- gamlss.control(n.cyc = 100, trace = FALSE)

fit_b2b_BI <- gamlss(
  cbind(y, failures) ~ x,
  family = BI,
  data = dat_b2b,
  control = control_b2b
)

fit_b2b_BB <- gamlss(
  cbind(y, failures) ~ x,
  sigma.formula = ~ 1,
  family = BB,
  data = dat_b2b,
  control = control_b2b
)

fit_b2b_B2B <- gamlss(
  cbind(y, failures) ~ x,
  sigma.formula = ~ 1,
  nu.formula = ~ 1,
  family = B2B,
  data = dat_b2b,
  mu.start = fit_b2b_BB$mu.fv,
  sigma.start = fit_b2b_BB$sigma.fv / (1 + fit_b2b_BB$sigma.fv),
  nu.start = rep(0.5, n),
  control = control_b2b
)

fit_b2b_BI
```

```
##
## Family:  c("BI", "Binomial")
## Fitting method: RS()
##
## Call:  gamlss(formula = cbind(y, failures) ~ x, family = BI,
##     data = dat_b2b, control = control_b2b)
##
## Mu Coefficients:
## (Intercept)            x
##     -0.2743       1.1147
```

```
##
##  Degrees of Freedom for the fit: 2 Residual Deg. of Freedom   198
## Global Deviance:     1897.79
##             AIC:     1901.79
##             SBC:     1908.39
```

```
summary(fit_b2b_BI)
```

```
## ******************************************************************
## Family:  c("BI", "Binomial")
##
## Call:  gamlss(formula = cbind(y, failures) ~ x, family = BI, data = dat_b2b,
##     control = control_b2b)
##
## Fitting method: RS()
##
## ------------------------------------------------------------------
## Mu link function:  logit
## Mu Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.27427    0.03584  -7.652 8.49e-13 ***
## x            1.11470    0.04310  25.862  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## ------------------------------------------------------------------
## No. of observations in the fit:  200
## Degrees of Freedom for the fit:  2
##       Residual Deg. of Freedom:  198
##                       at cycle:  2
##
## Global Deviance:     1897.793
##             AIC:     1901.793
##             SBC:     1908.39
## ******************************************************************
```

```
fit_b2b_BB
```

```
##
## Family:  c("BB", "Beta Binomial")
## Fitting method: RS()
##
## Call:  gamlss(formula = cbind(y, failures) ~ x, sigma.formula = ~1,
##     family = BB, data = dat_b2b, control = control_b2b)
##
## Mu Coefficients:
## (Intercept)            x
##     -0.1766       0.9926
## Sigma Coefficients:
## (Intercept)
```

```
##     -0.7547
##
##  Degrees of Freedom for the fit: 3 Residual Deg. of Freedom   197
## Global Deviance:     1109
##              AIC:     1115
##              SBC:     1124.89
```

```
summary(fit_b2b_BB)
```

```
## ******************************************************************
## Family:  c("BB", "Beta Binomial")
##
## Call:  gamlss(formula = cbind(y, failures) ~ x, sigma.formula = ~1,
##     family = BB, data = dat_b2b, control = control_b2b)
##
## Fitting method: RS()
##
## ------------------------------------------------------------------
## Mu link function:  logit
## Mu Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.17660    0.08721  -2.025   0.0442 *
## x            0.99258    0.09806  10.122   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## ------------------------------------------------------------------
## Sigma link function:  log
## Sigma Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.75469    0.08358  -9.029   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## ------------------------------------------------------------------
## No. of observations in the fit:  200
## Degrees of Freedom for the fit:  3
##       Residual Deg. of Freedom:  197
##                       at cycle:  5
##
## Global Deviance:     1108.997
##              AIC:     1114.997
##              SBC:     1124.892
## ******************************************************************
```

```
fit_b2b_B2B
```

```
##
## Family:  c("BB", "Beta-2-Binomial")
## Fitting method: RS()
```

```
##
## Call:  gamlss(formula = cbind(y, failures) ~ x, sigma.formula = ~1,
##     nu.formula = ~1, family = B2B, data = dat_b2b,
##     mu.start = fit_b2b_BB$mu.fv, sigma.start = fit_b2b_BB$sigma.fv/(1 +
##         fit_b2b_BB$sigma.fv), nu.start = rep(0.5, n),
##     control = control_b2b)
##
## Mu Coefficients:
## (Intercept)            x
##     -0.4079       1.1246
## Sigma Coefficients:
## (Intercept)
##     -0.9712
## Nu Coefficients:
## (Intercept)
##      0.7168
##
##  Degrees of Freedom for the fit: 4 Residual Deg. of Freedom   196
## Global Deviance:     1013.7
##             AIC:     1021.7
##             SBC:     1034.89
```

```
summary(fit_b2b_B2B)
```

```
## ******************************************************************
## Family:  c("BB", "Beta-2-Binomial")
##
## Call:  gamlss(formula = cbind(y, failures) ~ x, sigma.formula = ~1,
##     nu.formula = ~1, family = B2B, data = dat_b2b, mu.start =
    fit_b2b_BB$mu.fv,
##     sigma.start = fit_b2b_BB$sigma.fv/(1 + fit_b2b_BB$sigma.fv),
##     nu.start = rep(0.5, n), control = control_b2b)
##
## Fitting method: RS()
##
## ------------------------------------------------------------------
## Mu link function:  logit
## Mu Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.40791    0.05726  -7.124  1.9e-11 ***
## x            1.12460    0.06901  16.295  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## ------------------------------------------------------------------
## Sigma link function:  logit
## Sigma Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -0.9712     0.1478   -6.57 4.31e-10 ***
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## ------------------------------------------------------------------
## Nu link function:  logit
## Nu Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.7168     0.2973   2.411   0.0168 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## ------------------------------------------------------------------
## No. of observations in the fit:  200
## Degrees of Freedom for the fit:  4
##       Residual Deg. of Freedom:  196
##                       at cycle:  4
##
## Global Deviance:     1013.697
##             AIC:     1021.697
##             SBC:     1034.89
## ******************************************************************
```

```
coef(fit_b2b_BI, what = "mu")
```

```
## (Intercept)           x
##  -0.2742707   1.1147030
```

```
coef(fit_b2b_BB, what = "mu")
```

```
## (Intercept)           x
##  -0.1766013   0.9925844
```

```
coef(fit_b2b_B2B, what = "mu")
```

```
## (Intercept)           x
##  -0.4079099   1.1245983
```

```
delta_hat_b2b <- plogis(coef(fit_b2b_B2B, what = "sigma"))
nu_hat_b2b <- plogis(coef(fit_b2b_B2B, what = "nu"))
c(delta_hat = delta_hat_b2b, nu_hat = nu_hat_b2b)
```

```
## delta_hat.(Intercept)    nu_hat.(Intercept)
##             0.2746351             0.6719038
```

```
compare_three_models(fit_b2b_BI, fit_b2b_BB, fit_b2b_B2B, n)
```

```
##           model     AIC     BIC
## 1      Binomial 1901.79 1908.39
## 2 Beta-binomial 1115.00 1124.89
## 3           B2B 1021.70 1034.89
```

```
deviance(fit_b2b_BI)
```

```
## [1] 1897.793
```

```
deviance(fit_b2b_BB)
```

```
## [1] 1108.997
```

```
deviance(fit_b2b_B2B)
```

```
## [1] 1013.697
```

```
round(rbind(
  Binomial = residual_summary(fit_b2b_BI),
  `Beta-binomial` = residual_summary(fit_b2b_BB),
  B2B = residual_summary(fit_b2b_B2B)
), 3)
```

```
##                   mean    sd    min median   max
## Binomial      -0.020 2.551 -7.677 -0.380 7.577
## Beta-binomial -0.019 1.021 -3.014 -0.119 2.918
## B2B            0.092 1.015 -2.956  0.069 2.319
```

```
plot_residual_diagnostics_three(fit_b2b_BI, fit_b2b_BB, fit_b2b_B2B, "B2B data")
```

For this simulated data set, the B2B model gives the smallest AIC and BIC. The binomial model is inadequate because it cannot represent the extra-binomial variation; its normalized quantile residuals have standard deviation much larger than one. The BB model substantially improves the fit by modeling overdispersion, but the B2B model improves the information criteria further because it also models the additional shape induced by the random dispersion component. In this run, $\hat{\delta} \approx 0.275$ and $\hat{\nu} \approx 0.672$, close to the data-generating values $\delta = 0.30$ and $\nu = 0.60$.

3.4.5 Repository Example: Beta-2-Binomial Model

The `RobustDiscrete_installer` repository also includes an applied Beta-2-Binomial example from Appendix C of Bayes et al. (2024). The example uses the `aep` data set and compares a beta-binomial model with a B2B model. The BB model is fitted first and then used to provide initial values for the B2B fit (Bayes et al. 2026).

The repository code is reproduced below. The B2B model is more expensive to fit than the BB model because its likelihood involves numerical integration. For that reason, this chunk is shown as code but is not evaluated during the rendering of this notebook; the repository reports that model `M5` required 212 seconds on an Intel Core i7 2.80 GHz computer with 16 GB RAM.

```
if (dir.exists("Rlib")) {
  .libPaths(c("Rlib", .libPaths()))
}

library(gamlss)
library(numDeriv)
library(RobustDiscrete)

data(aep, package = "gamlss.data")

M2 <- gamlss(
  y ~ ward + loglos + year,
  sigma.formula = ~ year + ward,
  family = BB,
  data = aep
)

M5 <- gamlss(
  y ~ ward + loglos + year,
```

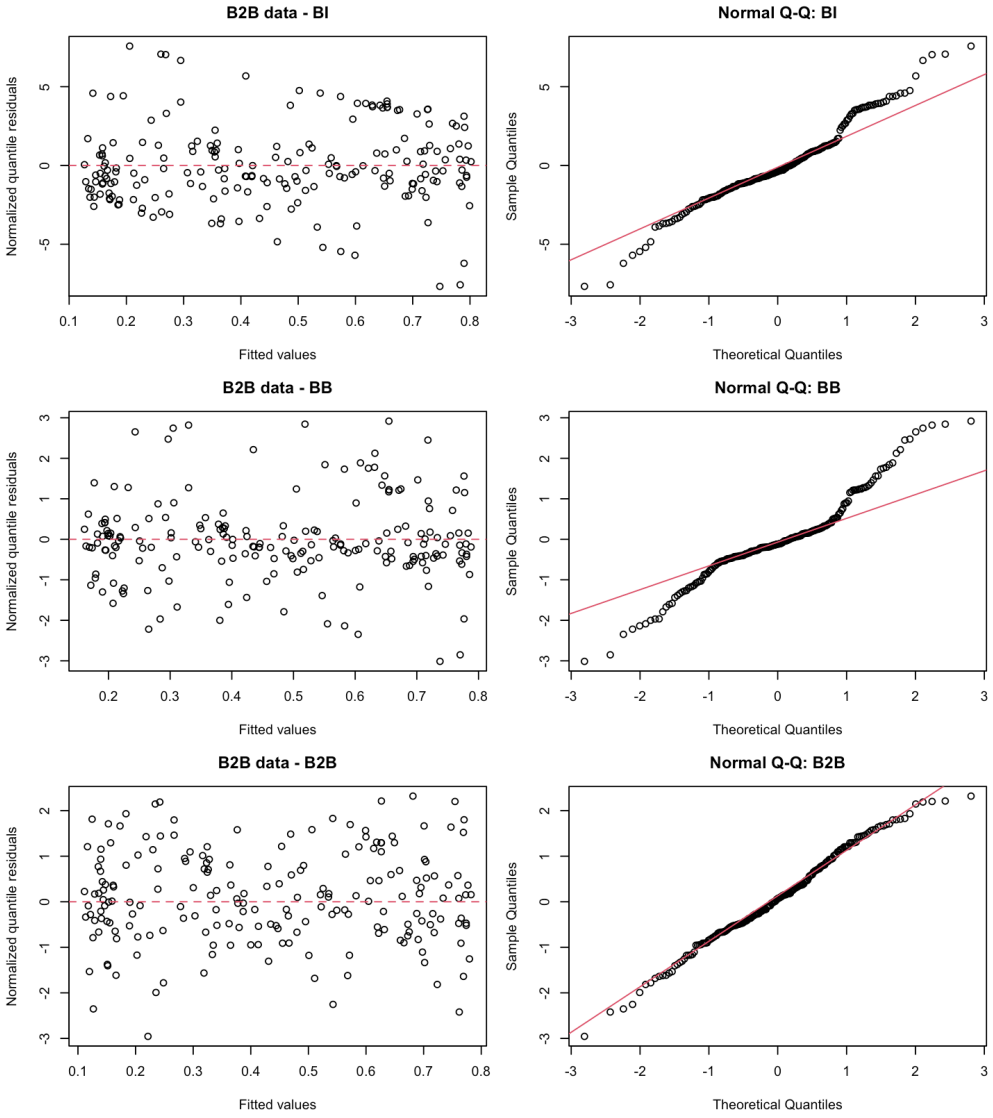


Figure 4: Residual diagnostics for the binomial, beta-binomial, and B2B fits to data simulated from the B2B regression model.

```
  sigma.formula = ~ year + ward,
  nu.formula = ~ year,
  family = B2B,
  data = aep,
  mu.start = M2$mu.fv,
  sigma.start = M2$sigma.fv / (1 + M2$sigma.fv)
)

summary(M5)
AIC(M2, M5)
GAIC(M2, M5, k = log(nrow(aep)))
```

The reported B2B fit has 1,383 observations, 11 fitted degrees of freedom, residual degrees of freedom equal to 1,372, and convergence at cycle 6. Its global deviance is 4449.109, AIC is 4471.109, and SBC is 4528.661. The mean model suggests lower expected counts for wards 2 and 3 relative to the baseline ward, a positive association with `loglos`, and a positive effect for `year90`. The dispersion model indicates lower dispersion for `year90` and ward 2. The shape model for ν includes a positive `year90` coefficient, suggesting that the additional B2B shape component changes between the two years.

Parameter	Term	Estimate	Std. error	p-value
μ	Intercept	-0.98203	0.07976	< 0.001
μ	ward2	-0.52860	0.09046	< 0.001
μ	ward3	-0.73761	0.27748	0.00795
μ	loglos	0.55101	0.05554	< 0.001
μ	year90	0.19811	0.08991	0.02774
δ	Intercept	0.29456	0.09492	0.00195
δ	year90	-0.45151	0.12546	< 0.001
δ	ward2	-0.69533	0.12588	< 0.001
δ	ward3	-0.82951	0.61858	0.18014
ν	Intercept	-1.95630	0.57420	< 0.001
ν	year90	1.25220	0.63650	0.04932

This example illustrates the practical estimation strategy used in the package: fit the simpler BB model first, use its fitted mean and dispersion values as initial values, and then fit the B2B model with an additional regression equation for ν. The comparison with the BB model should then be based on likelihood-based criteria, residual diagnostics, and the interpretability of the extra shape parameter.

3.5 Conclusions

The binomial, beta-binomial, and beta-2-binomial distributions form a natural sequence of models for bounded count responses. The binomial model is the most parsimonious specification: once m and μ are fixed, both the mean and variance are determined. This simplicity is useful when the bounded count can reasonably be interpreted as a sum of conditionally independent Bernoulli trials with a common success probability, but it is often too restrictive for real data.

The beta-binomial model keeps the same marginal mean, $m\mu$, but adds the dispersion parameter δ. This parameter inflates the binomial variance and provides a direct way to model extra-binomial variation. The BB model is therefore a natural first extension when residual diagnostics, information criteria, or empirical variance indicate that the binomial model underestimates variability.

The B2B model preserves the same mean and variance structure as the BB model but adds the shape parameter ν. This additional parameter does not change the first two moments; instead, it affects higher-order features such as skewness, kurtosis, tail behavior, and boundary mass. Consequently, the B2B model is most useful when the data show overdispersion together with distributional shapes that remain poorly represented by the BB model.

For regression modeling, monotonicity in μ is well defined when the remaining distributional parameters are held fixed. If δ or ν vary jointly with μ, larger values of μ do not by themselves guarantee stochastic ordering of the full response distribution. For this reason, regression models with covariates in multiple parameters should be interpreted through the full fitted probability mass function, fitted moments, residual diagnostics, and model comparison criteria.

The estimation examples show a practical workflow. First fit the simpler binomial model, then compare it with the BB model to assess overdispersion. When the BB model improves the fit but residual diagnostics or substantive knowledge suggest remaining tail or boundary structure, the B2B model can be fitted using the BB estimates as initial values. In applications, the final model should balance goodness of fit, parameter interpretability, numerical stability, and the amount of information available in the bounded count data.

3.6 Exercises

1. Fit the bounded count regression models studied here to the `aep` data set from the application in Section 5.3 of Bayes et al. (2024), which concerns inappropriate days spent in hospital and is available in `gamlss.data` (Stasinopoulos and Rigby 2025). Use the workflow illustrated in this notebook: define binomial, BB, and B2B candidate models; specify the same mean regression structure where comparisons require it; fit the BB model before the B2B model to obtain stable starting values; and compare the fitted models using AIC, BIC, normalized randomized quantile residuals, and convergence information. Choose the best model using the joint evidence from information criteria and diagnostics. Then interpret the estimated coefficients separately for the mean, dispersion, and shape submodels, making clear which parameters are on the real-line link scale and which response-scale quantities are obtained after inverse-link transformation.

2. Simulate a data set from the beta-binomial regression model and compare the fitted results obtained with the available R implementations. At a minimum, fit the model using `gamlss` and `gamlss.dist` (Stasinopoulos and Rigby 2007; Stasinopoulos and Rigby 2023), and compare the results with the implementations available in `aod` (Lesnoff and Lancelot 2012) and `VGAM` (Yee 2010, 2015), when these packages are installed. Take care with the different parameterizations, because the meaning of location, precision, scale, overdispersion, or shape parameters may differ across packages. The comparison should report not only the estimated coefficients and fit criteria, but also the parameterization used by each package and any transformations needed to put the estimates on a common interpretive scale.

3. Use one grouped binary or bounded count data set available in the `aod` package, such as `orob1` or `orob2`, to compare classical beta-binomial regression with the binomial, BB, and B2B regression models considered in this paper (Lesnoff and Lancelot 2012). Compare AIC, BIC, fitted response-scale mean curves, and normalized randomized quantile residuals. Discuss whether the best bounded count regression model gives a visibly different interpretation of covariate effects from the simpler alternatives, especially near the lower or upper boundary

of the response.

4. Use a proportional or grouped binary response data set available in `VGAM`, such as `sdac`, to compare beta-binomial regression with the bounded count regression models proposed here (Yee 2010, 2015). Fit the beta-binomial model with `VGAM` using an appropriate mean link, and fit the other models studied here with the same covariates in the mean parameter. Compare the selected models using fit criteria and residual diagnostics, and explain how the interpretation of dispersion differs between packages and parameterizations.
5. Implement a Bayesian approach for the models studied here.

3.7 Appendix: Proofs of Moment-Shape Relationships

Proof of Theorem 1. For part (a), the beta-binomial skewness coefficient is

$$\mathrm{Skew}_{\mathrm{BB}}(m,\mu,\delta) = \frac{(1-2\mu)\{1+(2m-1)\delta\}}{(1+\delta)\sqrt{mqA_\delta}}.$$

Since the binomial skewness coefficient is

$$\mathrm{Skew}_{\mathrm{Bin}}(m,\mu) = \frac{1-2\mu}{\sqrt{mq}},$$

we factor the first expression as

$$\mathrm{Skew}_{\mathrm{BB}}(m,\mu,\delta) = \mathrm{Skew}_{\mathrm{Bin}}(m,\mu)\frac{1+(2m-1)\delta}{(1+\delta)\sqrt{A_\delta}}.$$

For part (b), by the hierarchical definition of the B2B distribution,

$$Y \mid W = w \sim \mathrm{BetaBinomial}(m,\mu,w), \qquad W \sim \mathrm{Beta}(\delta,\nu).$$

The B2B distribution has variance mqA_δ, whereas the conditional BB distribution has variance mqA_W. Since both distributions have mean $m\mu$, the third centered moment of the B2B distribution is

$$\mathbb{E}\{(Y-m\mu)^3\} = \mathbb{E}\left[\mathbb{E}\{(Y-m\mu)^3 \mid W\}\right].$$

For a fixed $W = w$,

$$\mathbb{E}\{(Y-m\mu)^3 \mid W = w\} = \mathrm{Skew}_{\mathrm{BB}}(m,\mu,w)(mqA_w)^{3/2}.$$

Dividing the unconditional third centered moment by $(mqA_\delta)^{3/2}$ gives

$$\mathrm{Skew}_{\mathrm{B2B}}(m,\mu,\delta,\nu) = \frac{\mathbb{E}\left[\mathrm{Skew}_{\mathrm{BB}}(m,\mu,W)A_W^{3/2}\right]}{A_\delta^{3/2}}.$$

(□)

Proof of Theorem 2. For part (a), the beta-binomial kurtosis coefficient is

$$\mathrm{Kurt}_{\mathrm{BB}}(m,\mu,\delta) = \frac{k_0+k_1\delta+k_2\delta^2+k_3\delta^3}{(1+\delta)(1+2\delta)mqA_\delta^2}.$$

The binomial kurtosis coefficient satisfies

$$mq\ \mathrm{Kurt}_{\mathrm{Bin}}(m,\mu) = mq\left(3+\frac{1-6q}{mq}\right) = 3mq+1-6q = 3q(m-2)+1 = k_0.$$

Substituting $k_0 = mq\ \mathrm{Kurt}_{\mathrm{Bin}}(m,\mu)$ into the beta-binomial kurtosis formula gives

$$\mathrm{Kurt}_{\mathrm{BB}}(m,\mu,\delta) = \frac{mq\ \mathrm{Kurt}_{\mathrm{Bin}}(m,\mu) + k_1\delta + k_2\delta^2 + k_3\delta^3}{(1+\delta)(1+2\delta)mqA_\delta^2}.$$

For part (b), using the same hierarchical representation,

$$Y \mid W = w \sim \mathrm{BetaBinomial}(m,\mu,w), \qquad W \sim \mathrm{Beta}(\delta,\nu).$$

Again, both the conditional BB distribution and the unconditional B2B distribution have mean $m\mu$. Hence,

$$\mathbb{E}\{(Y-m\mu)^4\} = \mathbb{E}\left[\mathbb{E}\{(Y-m\mu)^4 \mid W\}\right].$$

For a fixed $W = w$,

$$\mathbb{E}\{(Y-m\mu)^4 \mid W = w\} = \mathrm{Kurt}_{\mathrm{BB}}(m,\mu,w)(mqA_w)^2.$$

Dividing the unconditional fourth centered moment by $(mqA_\delta)^2$ gives

$$\mathrm{Kurt}_{\mathrm{B2B}}(m,\mu,\delta,\nu) = \frac{\mathbb{E}\left[\mathrm{Kurt}_{\mathrm{BB}}(m,\mu,W)A_W^2\right]}{A_\delta^2}.$$

(□)

Chapter 4

Bounded Item Response Theory

4.1 Introduction

Item response theory (IRT) usually treats item scores as dichotomous or ordinal. Many measurements, however, produce continuous scores with known lower and upper bounds. Visual analogue scales, slider responses, proportions, bounded response times, and approximately continuous summed scores are common examples. After transformation to the unit interval, a response from person i to item j is written as

$$Y_{ij} = \frac{Y'_{ij} - L}{U - L}, \qquad 0 \leq Y_{ij} \leq 1,$$

where $L < U$ are the theoretical lower and upper bounds.

The main measurement objective is to relate the conditional distribution of Y_{ij} to a latent person position θ_i and item parameters τ_j. In the models considered here, an item discrimination parameter $a_j > 0$ and an easiness parameter $b_j \in \mathbb{R}$ enter through the linear predictor

$$\eta_{ij} = a_j \theta_i + b_j.$$

Molenaar et al. (2022) develop a common Bayesian framework for bounded continuous IRT models and extend it to responses on the closed interval $[0, 1]$. This matters because replacing exact zeros and ones by arbitrary interior values can distort person and item parameter estimates, even when boundary responses are relatively uncommon.

This document introduces the response distributions and their moment restrictions, develops the bounded IRT models and boundary-inflation mechanism, and presents Bayesian estimation and model comparison. An Abasement-scale example illustrates the workflow using `BoundIRT` (Molenaar 2026), followed by exercises and comparative density figures.

4.2 Bounded Continuous Distributions

Let Y be a continuous response variable with support $0 < Y < 1$. Three distributions are considered for the interior responses in this notebook: the Johnson S_B (or logit-normal), beta, and simplex

distributions. All three respect the bounded response scale, but they differ in their parameterization, variance structure, skewness, and behavior near zero and one. Consequently, distributions with similar means can imply substantially different response densities.

Throughout this section, $m = E(Y)$ and $v = \mathrm{Var}(Y)$ denote response-scale moments. The symbol μ denotes a logit-scale location for Johnson S_B, but the response mean for the beta and simplex families. Likewise, ϕ is a precision for the beta family and a dispersion for the simplex family. Their numerical values therefore have different meanings. The section first describes the families, then establishes the common moment bounds and compares feasible matched distributions, before discussing identification, monotonicity, and higher moments.

4.2.1 Johnson S_B Distribution

The Johnson S_B distribution (Johnson 1949) is obtained by transforming a real-valued normal random variable. Let W have distribution

$$W \sim N(\mu, \sigma^2), \qquad \mu \in \mathbb{R}, \quad \sigma > 0,$$

and define

$$Y = \frac{\exp(W)}{1 + \exp(W)} = \mathrm{logit}^{-1}(W).$$

Because the inverse-logit transformation maps the real line onto $(0, 1)$, Y has density

$$f(y \mid \mu, \sigma) = \frac{1}{\sigma y(1-y)\sqrt{2\pi}} \exp\left[-\frac{\{\mathrm{logit}(y) - \mu\}^2}{2\sigma^2}\right], \qquad 0 < y < 1.$$

An alternative parameterization uses the logit-scale variance $\delta = \sigma^2 > 0$. This is the dispersion parameterization used by Molenaar et al. (2022), whose item-specific parameter is δ_j. Writing $W \sim N(\mu, \delta)$, the corresponding probability density is

$$f(y \mid \mu, \delta) = \frac{1}{y(1-y)\sqrt{2\pi\delta}} \exp\left[-\frac{\{\mathrm{logit}(y) - \mu\}^2}{2\delta}\right], \qquad 0 < y < 1, \quad \mu \in \mathbb{R}, \quad \delta > 0.$$

Thus, $\sigma = \sqrt{\delta}$; δ is the variance on the logit scale, rather than the variance of Y on the response scale. In the median parameterization below, the same density follows by replacing μ with $\mathrm{logit}(\xi)$.

Its cumulative distribution function is

$$F(y \mid \mu, \sigma) = \Phi\left\{\frac{\mathrm{logit}(y) - \mu}{\sigma}\right\}, \qquad 0 < y < 1,$$

where $\Phi(\cdot)$ is the standard normal distribution function. The parameter μ controls location on the logit scale and σ controls dispersion. Although the normal baseline is symmetric, the transformed density can be symmetric or skewed on the response scale.

Because the inverse-logit transformation is strictly increasing, it preserves quantiles. The median of W is μ, so the median of Y is

$$\xi = \mathrm{Med}(Y) = \mathrm{logit}^{-1}(\mu) = \frac{\exp(\mu)}{1+\exp(\mu)}.$$

This gives an alternative parameterization in terms of the response-scale median $0 < \xi < 1$ and scale $\sigma > 0$. Substituting $\mu = \mathrm{logit}(\xi)$ into the density gives

$$f(y \mid \xi, \sigma) = \frac{1}{\sigma y(1-y)\sqrt{2\pi}} \exp\left[-\frac{\{\mathrm{logit}(y) - \mathrm{logit}(\xi)\}^2}{2\sigma^2}\right], \qquad 0 < y < 1.$$

Unlike the mean, the median is therefore available in closed form and has a direct interpretation on the original response scale.

Let $s(w) = \mathrm{logit}^{-1}(w)$ and let φ denote the standard normal density. Define the raw moments

$$m_r = E(Y^r) = \int_{-\infty}^{\infty} \{s(w)\}^r \frac{1}{\sigma}\varphi\left(\frac{w-\mu}{\sigma}\right) dw, \qquad r = 1, 2, \ldots .$$

Then $m = m_1$ and $v = m_2 - m_1^2$. Neither the mean nor the variance has an elementary closed-form expression; both can be evaluated numerically. When $\mu = 0$, reflection symmetry gives $m = \xi = 1/2$. In general, the median $\xi = s(\mu)$ and mean m differ. For finite σ, the density tends to zero at both endpoints, even when it has high peaks very close to them; this differs from a beta density that diverges at a boundary.

4.2.2 Beta Distribution

Let Y follow a beta distribution with positive shape parameters c and d. Its density is

$$f(y \mid c, d) = \frac{y^{c-1}(1-y)^{d-1}}{B(c,d)}, \qquad 0 < y < 1, \quad c > 0, \quad d > 0,$$

where $B(c,d)$ is the beta function. Its mean and variance have explicit expressions:

$$E(Y) = \frac{c}{c+d}, \qquad \mathrm{Var}(Y) = \frac{cd}{(c+d)^2(c+d+1)}.$$

Following Ferrari and Cribari-Neto (2004), an alternative parameterization uses the response mean μ and a precision parameter ϕ:

$$\mu = \frac{c}{c+d}, \qquad \phi = c + d, \qquad 0 < \mu < 1, \quad \phi > 0.$$

Consequently, $c = \mu\phi$ and $d = (1-\mu)\phi$, and the density becomes

$$f(y \mid \mu, \phi) = \frac{\Gamma(\phi)}{\Gamma(\mu\phi)\Gamma\{(1-\mu)\phi\}} y^{\mu\phi-1}(1-y)^{(1-\mu)\phi-1}, \qquad 0 < y < 1.$$

Its moments are

$$E(Y) = \mu, \qquad \mathrm{Var}(Y) = \frac{\mu(1-\mu)}{1+\phi}.$$

For fixed μ, increasing ϕ reduces the variance and concentrates responses around the mean. As $\phi \to 0$, the variance approaches the upper bound $\mu(1-\mu)$ and probability concentrates near the boundaries. The relative dispersion is $\rho = 1/(1+\phi) \in (0,1)$, so equivalently $\mathrm{Var}(Y) = \mu(1-\mu)\rho$. The notation $\phi = c + d$ is used for beta precision throughout this document.

This beta precision ϕ differs from the simplex dispersion in Section 2.3 and the beta item parameter λ_j in Section 3. There, $c_{ij}d_{ij} = \lambda_j$, whereas the response-specific beta precision is $\phi_{ij} = c_{ij} + d_{ij}$.

The beta family can represent symmetric, skewed, uniform, J-shaped, and U-shaped densities. In the IRT formulation developed below, c and d become response-specific shape parameters determined by the latent predictor and an item dispersion parameter.

4.2.3 Simplex Distribution

The simplex distribution was introduced by Barndorff-Nielsen and Jørgensen (1991). Let Y follow a simplex distribution with mean parameter μ and dispersion parameter ϕ. Its density is

$$f(y \mid \mu, \phi) = \{2\pi\phi[y(1-y)]^3\}^{-1/2} \exp\left[-\frac{(y-\mu)^2}{2\phi y(1-y)\mu^2(1-\mu)^2}\right],$$

with support and parameter space

$$0 < y < 1, \qquad 0 < \mu < 1, \qquad \phi > 0.$$

The mean is explicit and equals the location parameter,

$$E(Y) = \mu.$$

The variance is

$$\mathrm{Var}(Y) = \mu(1-\mu) - \frac{1}{\sqrt{2\phi}} \exp(A)\Gamma\left(\frac{1}{2}, A\right), \qquad A = \frac{1}{2\phi\mu^2(1-\mu)^2},$$

where $\Gamma(s,x) = \int_x^\infty t^{s-1}\exp(-t)\,dt$ is the upper incomplete gamma function. Thus, the simplex variance depends jointly on μ and ϕ.

The simplex parameter ϕ is a dispersion parameter, but it is not equal to $\mathrm{Var}(Y)$. For fixed μ, increasing ϕ increases the spread of the distribution and generally places more probability toward the boundaries; decreasing ϕ concentrates responses around μ. In particular,

$$\mathrm{Var}(Y) \approx \phi\mu^3(1-\mu)^3 \qquad \text{as } \phi \longrightarrow 0,$$

so $\mu^3(1-\mu)^3$ is the simplex variance function. At the opposite limit,

$$\mathrm{Var}(Y) \longrightarrow \mu(1-\mu) \qquad \text{as } \phi \longrightarrow \infty,$$

which is the largest variance permitted for a variable on $[0,1]$ with mean μ. Thus, the same dispersion ϕ can yield different response variances at different means. The small-dispersion approximation should not replace the exact variance when fitting the high-variance cases in Section 2.5.

The simplex distribution accommodates heteroscedastic and skewed responses, but its variance structure differs from that of the beta distribution. This difference can be especially relevant for observations near the boundaries. The Johnson S_B, beta, and simplex densities are defined only on $(0,1)$; all three assign zero probability to exact zeros and ones. Boundary observations therefore require the inflation mechanism introduced later.

4.2.4 Variance Restrictions on the Unit Interval

Proposition 1 (Variance bounds on the unit interval). Let Y be a random variable taking values in $[0,1]$, with mean $m = E(Y)$ and variance $v = \mathrm{Var}(Y)$. Then

$$0 \le v \le m(1-m) \le \frac{1}{4}.$$

The equality $v = m(1-m)$ holds if and only if Y is Bernoulli with $P(Y=1) = m$, including the degenerate cases $m = 0$ and $m = 1$. The overall bound $v = 1/4$ is attained only when $P(Y=0) = P(Y=1) = 1/2$. For a nondegenerate continuous distribution on $(0,1)$,

$$0 < m < 1, \qquad 0 < v < m(1-m).$$

Proof. Since $0 \le Y \le 1$, we have $Y^2 \le Y$. Taking expectations and subtracting m^2 gives

$$\mathrm{Var}(Y) = E(Y^2) - m^2 \le m - m^2 = m(1-m) = \frac{1}{4} - \left(m - \frac{1}{2}\right)^2 \le \frac{1}{4}.$$

Moreover, $m(1-m) - \mathrm{Var}(Y) = E\{Y(1-Y)\}$. Because $Y(1-Y) \ge 0$, this expectation is zero if and only if $Y \in \{0,1\}$ almost surely; the mean then determines the stated Bernoulli probabilities. The bound $1/4$ is attained precisely when, in addition, $m = 1/2$. For a distribution supported on $(0,1)$, $Y(1-Y) > 0$ almost surely, so $\mathrm{Var}(Y) < m(1-m)$; nondegeneracy gives $\mathrm{Var}(Y) > 0$. □

For $m = E(Y) > 0$, the coefficient of variation (CV) is the standard deviation divided by the mean:

$$\mathrm{CV}(Y) = \frac{\sqrt{\mathrm{Var}(Y)}}{m}.$$

The proposition therefore implies

$$0 \le \mathrm{CV}(Y) \le \sqrt{\frac{1-m}{m}}, \qquad 0 < m \le 1.$$

For a nondegenerate continuous distribution on $(0,1)$, both inequalities are strict. The upper bound is attained by a Bernoulli distribution with mean m. At $m = 0$, $Y = 0$ almost surely and the CV

is undefined, since its formula gives $0/0$. As $m \to 0^+$, the upper bound diverges; at $m = 1$, $Y = 1$ almost surely and the CV is zero.

The following figures show upper envelopes for all distributions on $[0, 1]$, rather than variances or CVs for a particular family. Here m (sometimes denoted by μ when it is a distribution's mean parameter) is the response mean. Both figures use linear axes; the CV curve is evaluated from $m = 0.001$ because its upper bound is infinite at the left limit.

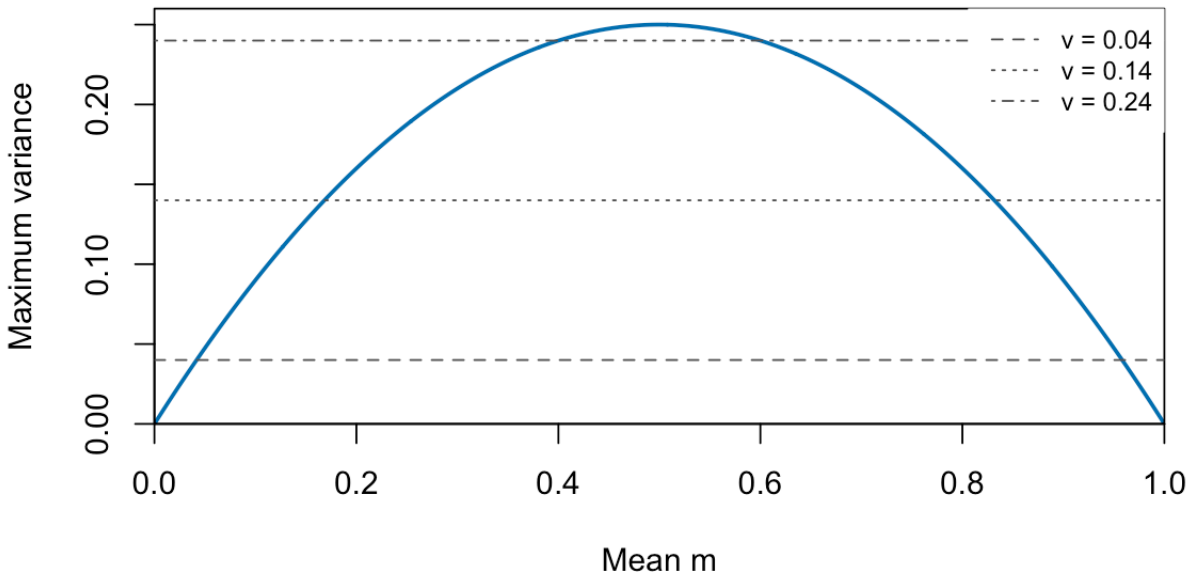


Figure 1: Maximum variance versus response mean on the unit interval. The dashed lines mark the target variances in Table 1; continuous interior distributions require values strictly below the curve.

The variance envelope $m(1-m)$ is symmetric around $m = 0.50$, where it reaches 0.25, and vanishes at the endpoints. For a fixed positive variance $v < 1/4$, feasible continuous response means satisfy

$$\frac{1 - \sqrt{1 - 4v}}{2} < m < \frac{1 + \sqrt{1 - 4v}}{2}.$$

Unlike the variance bound, the CV bound is not symmetric: it decreases from infinity near zero to zero at one. Dividing by a small mean can produce a large CV even when the absolute variance is small. For example, the maximum CV is 2 at $m = 0.20$, 1 at $m = 0.50$, and approximately 0.6547 at $m = 0.70$.

In the beta family, the variance approaches its bound as the precision $c + d \to 0$. In the simplex family, it approaches the bound as $\phi \to \infty$ at fixed mean. For Johnson S_B, boundary concentration arises as $\delta = \sigma^2 \to \infty$; maintaining a prescribed mean generally requires adjusting the logit-scale location jointly with the dispersion.

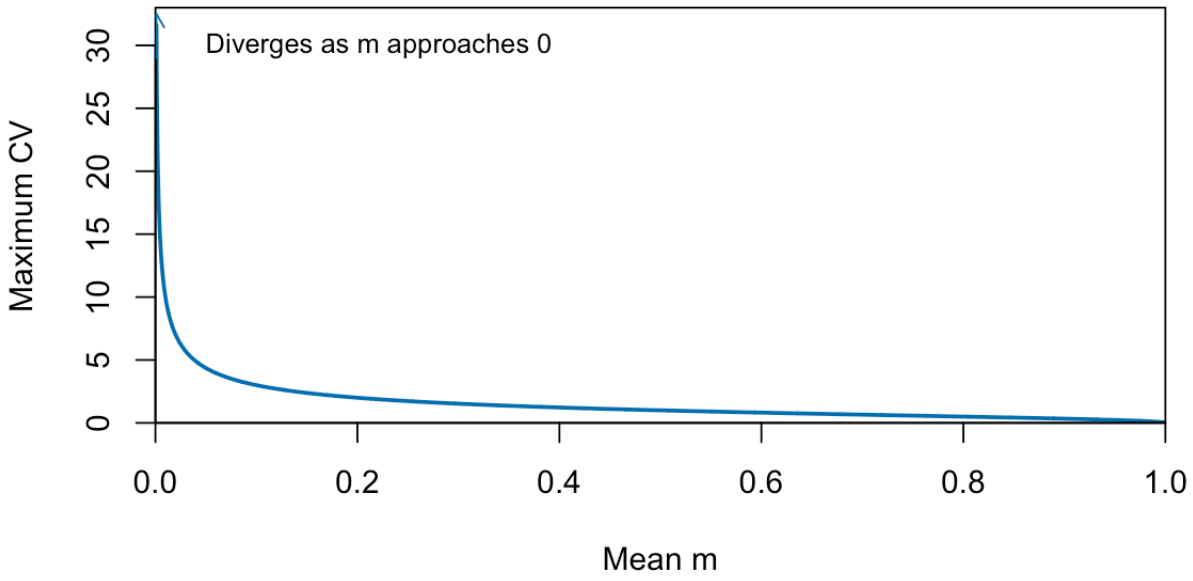


Figure 2: Maximum coefficient of variation versus response mean. The curve is evaluated for means from 0.001 to 1; the mean axis spans 0 to 1. The bound diverges as the mean approaches zero, where CV is undefined.

4.2.5 Moment Matching and Parameter Comparison

To compare distributional shape independently of the first two moments, consider target means $m \in \{0.20, 0.50, 0.70\}$ and target variances $v \in \{0.04, 0.14, 0.24\}$. Table 1 reports the distribution parameters for each feasible combination. By the proposition in Section 2.4, an interior continuous distribution requires $0 < v < m(1-m)$. The maximum variances for these means are 0.16, 0.25, and 0.21, respectively. Consequently, $v = 0.04$ and $v = 0.14$ are feasible for all three means, whereas $v = 0.24$ is feasible only for $m = 0.50$. The other two combinations are marked as infeasible rather than assigned distribution parameters.

For the beta distribution, write $\phi = c + d$. Solving the mean and variance equations gives

$$\phi = \frac{m(1-m)}{v} - 1, \qquad c = m\phi, \qquad d = (1-m)\phi.$$

The Johnson S_B parameters (μ, σ) are obtained by numerically solving

$$\int_{-\infty}^{\infty} s(w) g(w \mid \mu, \sigma)\, dw = m, \qquad \int_{-\infty}^{\infty} s(w)^2 g(w \mid \mu, \sigma)\, dw - m^2 = v,$$

where $s(w) = \text{logit}^{-1}(w)$ and g is the $N(\mu, \sigma^2)$ density. For the simplex distribution, $\mu = m$ and ϕ is found by equating its variance expression to v.

For the simplex distribution, numerical calculations use the exact simplex variance and verify that the achieved moments agree with their targets.

Table 1 uses (μ, δ) for Johnson S_B, with $\delta = \sigma^2$ as in Section 2.1; (μ, ϕ) for beta, where $\phi = c + d$ is the precision in Section 2.2; and (μ, ϕ) for simplex, where ϕ is the dispersion in Section 2.3. Family labels distinguish symbols with different meanings. The following chunk formats the computed parameters without repeating the moment-matching calculations.

Table 1: Moment-matched parameters in the parameterizations of Sections 2.1–2.3. For Johnson SB, $\delta = \sigma^2$ is the logit-scale variance; beta ϕ is precision, and simplex ϕ is dispersion. μ is the response mean for beta and simplex, and the logit of the median for Johnson SB. NA denotes an infeasible mean–variance combination.

m	v	SB μ	SB δ	Beta μ	Beta ϕ	Simplex μ	Simplex ϕ	Status
0.2	0.04	-1.9304	2.3226	0.2	3.0000	0.2	20.6791	Feasible
0.5	0.04	0.0000	0.8947	0.5	5.2500	0.5	4.1033	Feasible
0.7	0.04	1.0619	1.2977	0.7	4.2500	0.7	7.5930	Feasible
0.2	0.14	-11.7809	192.6619	0.2	0.1429	0.2	3316.3804	Feasible
0.5	0.14	0.0000	10.0692	0.5	0.7857	0.5	63.2240	Feasible
0.7	0.14	2.6037	21.4700	0.7	0.5000	0.7	193.0834	Feasible
0.2	0.24	NA	NA	NA	NA	NA	NA	Infeasible
0.5	0.24	0.0000	1588.2616	0.5	0.0417	0.5	14913.8022	Feasible
0.7	0.24	NA	NA	NA	NA	NA	NA	Infeasible

NA means *not applicable*: $v = 0.24$ exceeds the variance bounds 0.16 and 0.21 for means 0.20 and 0.70, respectively. These combinations are mathematically infeasible, not estimation failures. For $m = 0.50$, $v = 0.24 < 0.25$ is feasible. Continuous distributions require $0 < v < m(1-m)$; equality at the upper bound requires a Bernoulli distribution.

Table 1 separates changes in the mean from changes in the variance. At a fixed mean, increasing v increases the Johnson S_B scale σ and the simplex dispersion ϕ, while decreasing the beta precision $\phi = m(1-m)/v-1$. Both beta shapes decrease in the same proportion, so their ratio $c/d = m/(1-m)$ remains constant. In these scenarios, larger variance produces greater boundary concentration without changing the response mean.

At a fixed feasible variance, $m = 0.50$ gives symmetric distributions: the Johnson S_B location is $\mu = 0$, the beta shapes satisfy $c = d$, and the simplex mean is $\mu = 0.50$. The means 0.20 and 0.70 give asymmetric distributions directed toward the lower and upper boundaries, respectively. The Johnson S_B location is negative for $m = 0.20$ and positive for $m = 0.70$; it must be adjusted jointly with σ and is not generally equal to $\text{logit}(m)$. The simplex location, by contrast, equals the response mean exactly.

The same absolute variance represents different relative dispersions $v/\{m(1-m)\}$. For $v = 0.04$, these ratios are 0.25, 0.16, and approximately 0.1905 for the three means in ascending order. For $v = 0.14$, they increase to 0.875, 0.56, and approximately 0.6667. The mean 0.20 is therefore closest to its variance bound at either fixed variance and requires the largest Johnson S_B scale and simplex dispersion among these cases. At $v = 0.24$ and $m = 0.50$, the relative dispersion is 0.96, indicating an especially strong concentration near both boundaries. Dispersion values from different families are not directly comparable, because their parameterizations differ.

For the beta family, at $v = 0.04$ the shape pairs (c, d) are $(0.60, 2.40)$, $(2.625, 2.625)$, and $(2.975, 1.275)$ for means 0.20, 0.50, and 0.70, respectively. Thus, the first density decreases from a singularity at zero, while the other two have an interior mode. At $v = 0.14$, the pairs become approximately $(0.0286, 0.1143)$, $(0.3929, 0.3929)$, and $(0.35, 0.15)$: both shapes are below one in every case, giving U-shaped densities. At $v = 0.24$, the only feasible pair is $(0.0208, 0.0208)$, a symmetric U-shaped density with particularly strong boundary concentration.

Comparative density figures for all feasible combinations are provided in the appendix, **Moment-Matched Density Figures**. They show how distributions with equal means and variances can still differ in their interior and boundary behavior.

4.2.6 Identification

Identification means that two distinct parameter vectors cannot generate the same probability density on $(0, 1)$. For the Johnson S_B distribution, the transformation

$$Y = s(W) = \frac{\exp(W)}{1 + \exp(W)}$$

is one-to-one and strictly increasing, with inverse $s^{-1}(y) = \text{logit}(y)$. If two parameter vectors $\vartheta_1 = (\mu_1, \sigma_1)$ and $\vartheta_2 = (\mu_2, \sigma_2)$ produce the same bounded density,

$$f(y \mid \vartheta_1) = f(y \mid \vartheta_2), \qquad 0 < y < 1,$$

then multiplying both sides by $y(1-y)$ and setting $w = \text{logit}(y)$ gives equality of the corresponding normal densities on the real line. Because the normal location-scale family is identifiable, $\mu_1 = \mu_2$ and $\sigma_1 = \sigma_2$. Thus, the Johnson S_B family inherits identification from its normal baseline distribution.

For the beta family, the mean and variance determine the parameters: $\phi = m(1-m)/v-1$, $c = m\phi$, and $d = (1-m)\phi$. Thus, equality of the distributions implies equality of their shape parameters. For the simplex family, equality of the distributions first implies equality of their means μ. At a fixed mean, the density ratio for two dispersions contains a nonconstant function of y unless the dispersions are equal. Hence its dispersion is identifiable as well.

These are statements about identification within each distributional family. They do not imply that the mean and variance identify the family itself: Table 1 gives different families with the same first two moments. Nor do they establish identification of an IRT model, which additionally requires constraints on the latent location and scale. Even an identifiable model can yield imprecise estimates in a finite sample.

4.2.7 Monotonicity

For the Johnson S_B distribution, monotonicity in μ is stochastic rather than pointwise monotonicity of the density. For fixed σ, increasing μ shifts the normal distribution of W to the right. Because the inverse-logit transformation is increasing, for every fixed cutoff $0 < y < 1$,

$$\Pr(Y > y) = \Pr\{W > \text{logit}(y)\}$$

is nondecreasing as μ increases. Larger values of μ therefore assign more probability to larger response values, even though density curves for different values of μ may cross.

For the beta distribution, fix the precision $\phi = c + d$. If $m_b > m_a$, the ratio of the corresponding densities has the form

$$\frac{f(y \mid m_b, \phi)}{f(y \mid m_a, \phi)} = C \left\{ \frac{y}{1-y} \right\}^{\phi(m_b - m_a)}, \qquad C > 0.$$

This ratio increases with y, establishing monotone likelihood ratios and therefore stochastic ordering in the mean. For the simplex family, $E(Y) = \mu$ establishes monotonicity of the mean. That statement alone does not establish stochastic ordering of the entire distribution; changing μ also changes its variance and shape.

In the beta and simplex IRT models introduced later, the response mean is $m = \text{logit}^{-1}(\eta)$, which increases strictly in η. For Johnson S_B, the logit-scale location is $\mu = \eta$, so the response median is $\text{logit}^{-1}(\eta)$; its mean also increases but is generally not that logistic function (Molenaar et al. 2022). In the beta IRT parameterization, precision can vary with η, so the fixed-precision ordering argument above should not be applied to that model without further analysis. Monotonicity of a mean, stochastic ordering, and pointwise changes in density are distinct properties.

4.2.8 Skewness and Kurtosis

Skewness and kurtosis describe aspects of a bounded distribution that are not captured by its location and dispersion alone. Let $m_r = E(Y^r)$, $r = 1, 2, 3, 4$, and define $v = \text{Var}(Y) = m_2 - m_1^2$. The standardized skewness and kurtosis are, for $v > 0$,

$$\text{Skew}(Y) = \frac{E\left[\{Y - E(Y)\}^3\right]}{v^{3/2}}, \qquad \text{Kurt}(Y) = \frac{E\left[\{Y - E(Y)\}^4\right]}{v^2}.$$

Here kurtosis denotes the standardized fourth central moment rather than excess kurtosis. For Johnson S_B, the raw moments defined in Section 2.1 give skewness and kurtosis by numerical integration. For the beta distribution, writing $\phi = c + d$, the closed-form expressions are

$$\text{Skew}(Y) = \frac{2(d-c)\sqrt{\phi+1}}{(\phi+2)\sqrt{cd}},$$

$$\text{Kurt}(Y) = 3 + \frac{6\{(c-d)^2(\phi+1) - cd(\phi+2)\}}{cd(\phi+2)(\phi+3)}.$$

Equal beta shapes give zero skewness. For the simplex family, these shape measures depend jointly on μ and ϕ and can be computed by integrating its density. Symmetry also forces zero skewness for Johnson S_B with $\mu = 0$ and simplex with $\mu = 1/2$. Zero skewness does not, in general, establish symmetry.

4.2.8.1 Moment-Matched Comparison

For the parameter values obtained in Section 2.5, the three distributions in each scenario have the same mean and variance, but their third and fourth standardized moments need not agree. Table 2 reports skewness and kurtosis computed from the first four raw moments, together with the achieved means and variances as a numerical check.

Table 2: Skewness and kurtosis of the moment-matched distributions for the feasible mean-variance combinations in Table 1.

Mean	Variance	Distribution	Achieved mean	Achieved variance	Skewness	Kurtosis
0.2	0.04	Johnson SB	0.2	0.04	1.4049	4.4097
0.2	0.04	Beta	0.2	0.04	1.2000	3.8000
0.2	0.04	Simplex	0.2	0.04	1.3335	3.9179
0.5	0.04	Johnson SB	0.5	0.04	0.0000	2.1840
0.5	0.04	Beta	0.5	0.04	0.0000	2.2727
0.5	0.04	Simplex	0.5	0.04	0.0000	2.0292
0.7	0.04	Johnson SB	0.7	0.04	-0.7720	2.8711
0.7	0.04	Beta	0.7	0.04	-0.6400	2.7021
0.7	0.04	Simplex	0.7	0.04	-0.7364	2.6102
0.2	0.14	Johnson SB	0.2	0.14	1.5005	3.3908
0.2	0.14	Beta	0.2	0.14	1.4967	3.3818
0.2	0.14	Simplex	0.2	0.14	1.4985	3.3515
0.5	0.14	Johnson SB	0.5	0.14	0.0000	1.3901
0.5	0.14	Beta	0.5	0.14	0.0000	1.4151
0.5	0.14	Simplex	0.5	0.14	0.0000	1.3101
0.7	0.14	Johnson SB	0.7	0.14	-0.8726	2.0857
0.7	0.14	Beta	0.7	0.14	-0.8552	2.0694
0.7	0.14	Simplex	0.7	0.14	-0.8646	2.0005
0.5	0.24	Johnson SB	0.5	0.24	0.0000	1.0272
0.5	0.24	Beta	0.5	0.24	0.0000	1.0274
0.5	0.24	Simplex	0.5	0.24	0.0000	1.0205

The achieved means and variances verify the numerical moment matching for the feasible combinations in Table 1. At $m = 0.50$, symmetry forces skewness to zero at all three variances. For the asymmetric means, the higher moments vary with both the family and the target variance. As variance approaches $m(1-m)$, boundary concentration becomes dominant. At $m = 0.20$ and

$v = 0.04$, for example, skewness ranges from approximately 1.20 to 1.40, and kurtosis from 3.80 to 4.41, despite equal means and variances. At $m = 0.50$ and $v = 0.24$, all three kurtoses are close to 1, consistent with an approximately symmetric two-boundary distribution. Thus, the differences in higher moments can be substantial at moderate relative dispersion and smaller near the variance bound.

These shape measures matter because distributions with the same or similar means can have very different tail behavior and concentration near zero or one. Distributional choice should therefore consider the full fitted density, not only the conditional mean.

4.3 Bounded Continuous IRT Models

Consider a test administered to N individuals and composed of J bounded continuous items. Let $Y_{ij} \in (0, 1)$ denote the response of individual $i = 1, \dots, N$ to item $j = 1, \dots, J$. Conditional on the person parameter θ_i and the item parameter vector τ_j, responses are assumed to be locally independent. A bounded continuous IRT model can be written hierarchically as

$$\begin{aligned} Y_{ij} \mid \theta_i, \tau_j &\sim \mathcal{D}_j(\eta_{ij}, \psi_j), \\ \eta_{ij} &= a_j\theta_i + b_j, \qquad i = 1, \dots, N, \quad j = 1, \dots, J, \\ \theta_i &\sim N(0, 1), \end{aligned}$$

where $\mathcal{D}_j$ is the bounded response distribution for item j and ψ_j contains its additional distribution-specific item parameters. The discrimination parameter $a_j > 0$ determines how strongly the response changes with θ_i, the easiness parameter $b_j \in \mathbb{R}$ locates the item on the latent scale, and $\theta_i \in \mathbb{R}$ is the latent position of individual i. Fixing $\theta_i \sim N(0, 1)$ identifies the origin and scale of the latent variable.

The linear predictor above uses the intercept or easiness parameterization $a_j\theta_i + b_j$. In conventional IRT, the same predictor is often written with an item difficulty parameter b_j^* as

$$\eta_{ij} = a_j(\theta_i - b_j^*).$$

The two forms are algebraically equivalent because

$$a_j\theta_i + b_j = a_j(\theta_i - b_j^*) \quad \iff \quad b_j = -a_j b_j^* \quad \iff \quad b_j^* = -\frac{b_j}{a_j}.$$

Thus, estimates reported in the (a_j, b_j) parameterization can be converted to the usual discrimination-difficulty parameterization by computing $b_j^* = -b_j/a_j$. Conversely, given (a_j, b_j^*), the easiness/intercept is $b_j = -a_j b_j^*$. Larger values of b_j shift the response curve to the left and make an item easier, whereas larger values of b_j^* shift it to the right and make an item more difficult. The two location parameters therefore have opposite directions, and b_j is not simply the negative of b_j^* unless $a_j = 1$.

At $\theta_i = b_j^*$, the linear predictor is zero. Consequently, the logistic location is one half for the beta and simplex conditional means and for the Johnson S_B conditional median. This interpretation is one reason the $a_j(\theta_i - b_j^*)$ form is common in IRT.

For each interior model, write $\tau_j = (a_j, b_j, \zeta_j)$, where $\zeta_j > 0$ is a generic family-specific scale parameter and $\psi_j = (\zeta_j)$. The model-specific notation is

$$\zeta_j = \begin{cases} \delta_j, & \text{Johnson } S_B \text{ (logit-scale variance)}, \\ \lambda_j, & \text{beta (product of the two shape parameters)}, \\ \phi_j, & \text{simplex (dispersion)}, \\ \sigma_j^2, & \text{normal (response-scale variance)}. \end{cases}$$

The beta precision remains $\phi_{ij} = c_{ij} + d_{ij}$, as in Section 2.2; it can depend on the person as well as the item. This precision is distinct from the item parameter $\lambda_j = c_{ij} d_{ij}$. The beta precision and simplex dispersion share the conventional symbol ϕ, so the family must always be specified when interpreting it.

Let

$$\text{logit}^{-1}(z) = \frac{1}{1 + \exp(-z)}$$

denote the logistic function. The Samejima, beta, and simplex models share the latent predictor $\eta_{ij} = a_j \theta_i + b_j$ but differ in their conditional response distribution and in the role played by ζ_j.

4.3.1 Samejima or Johnson S_B IRT Model

The Samejima model uses a Johnson S_B response distribution, or equivalently a normal distribution after applying the logit transformation (Samejima 1973; Johnson 1949). In the notation introduced above, its hierarchical specification uses the variance parameterization (μ, δ) from Section 2.1:

$$\begin{aligned} Y_{ij} \mid \theta_i, \tau_j &\sim \text{Johnson } S_B(\eta_{ij}, \delta_j), \\ \eta_{ij} &= a_j \theta_i + b_j, \\ \theta_i &\sim N(0, 1), \\ \tau_j &= (a_j, b_j, \delta_j), \end{aligned} \qquad a_j > 0, \quad b_j \in \mathbb{R}, \quad \delta_j > 0.$$

Equivalently, $\text{logit}(Y_{ij}) \mid \theta_i, \tau_j \sim N(\eta_{ij}, \delta_j)$. The conditional median is explicit:

$$\xi_{ij} = \text{Med}(Y_{ij} \mid \theta_i) = \text{logit}^{-1}(\eta_{ij}) = \text{logit}^{-1}(a_j \theta_i + b_j).$$

The model can therefore be parameterized equivalently as

$$Y_{ij} \mid \theta_i, \tau_j \sim \text{Johnson } S_B^{(M)}(\xi_{ij}, \delta_j), \qquad \text{logit}(\xi_{ij}) = a_j \theta_i + b_j,$$

where the superscript (M) indicates the median parameterization and its second argument is the logit-scale variance δ_j. The conditional mean does not have a simple closed form. The median equals one half at $\theta_i = -b_j / a_j$. Without boundary inflation, item information is constant over the latent scale and equals a_j^2 / δ_j.

4.3.2 Beta IRT Model

The beta IRT model has the hierarchical specification

$$
\begin{aligned}
Y_{ij} \mid \theta_i, \tau_j &\sim \text{Beta}(c_{ij}, d_{ij}), \\
c_{ij} &= \exp\left\{\frac{\eta_{ij} + \log(\lambda_j)}{2}\right\}, \\
d_{ij} &= \exp\left\{\frac{-\eta_{ij} + \log(\lambda_j)}{2}\right\}, \qquad a_j > 0, \quad b_j \in \mathbb{R}, \quad \lambda_j > 0, \\
\eta_{ij} &= a_j\theta_i + b_j, \\
\theta_i &\sim N(0,1), \\
\tau_j &= (a_j, b_j, \lambda_j),
\end{aligned}
$$

following Noel and Dauvier (2007) and Molenaar et al. (2022).

Here $c_{ij} > 0$ and $d_{ij} > 0$ are response-distribution shape parameters, not item discrimination or easiness parameters. Because $c_{ij}/(c_{ij} + d_{ij}) = \text{logit}^{-1}(\eta_{ij})$,

$$
E(Y_{ij} \mid \theta_i) = \text{logit}^{-1}(a_j\theta_i + b_j).
$$

In the mean–precision parameterization of Section 2.2, this is equivalent to

$$
\mu_{ij} = \text{logit}^{-1}(\eta_{ij}), \qquad \phi_{ij} = c_{ij} + d_{ij} = 2\sqrt{\lambda_j}\cosh(\eta_{ij}/2),
$$

$$
\text{Var}(Y_{ij} \mid \theta_i) = \frac{\mu_{ij}(1-\mu_{ij})}{1+\phi_{ij}}.
$$

Thus, a fixed λ_j does not imply fixed beta precision across persons.

The item information is typically unimodal and decreases toward the extremes of the latent scale. The exact shape also depends on λ_j.

4.3.3 Simplex IRT Model

The simplex IRT model has the hierarchical specification

$$
\begin{aligned}
Y_{ij} \mid \theta_i, \tau_j &\sim \text{Simplex}(\mu_{ij}, \phi_j), \\
\mu_{ij} &= \text{logit}^{-1}(\eta_{ij}), \\
\eta_{ij} &= a_j\theta_i + b_j, \qquad\qquad a_j > 0, \quad b_j \in \mathbb{R}, \quad \phi_j > 0. \\
\theta_i &\sim N(0,1), \\
\tau_j &= (a_j, b_j, \phi_j),
\end{aligned}
$$

Here μ_{ij} is the conditional mean and ϕ_j is the dispersion parameter. This IRT specification follows Molenaar et al. (2022); Flores et al. (2020) use the simplex distribution in a joint model for bounded response time and response accuracy.

Thus,

$$E(Y_{ij} \mid \theta_i) = \mu_{ij}.$$

The simplex and beta models have the same logistic conditional mean, but their conditional variances and item information functions differ. The simplex model can place considerable information near the extremes of the latent scale.

4.3.4 Normal IRT Model

The normal continuous-response model of Mellenbergh (1994) serves as an unbounded reference:

$$Y_{ij} \mid \theta_i \sim N(a_j\theta_i + b_j, \sigma_j^2).$$

Here $\sigma_j > 0$ is the response-scale standard deviation. Its simplicity is useful for comparison, but its support is the real line, so it can assign probability to impossible scores below zero or above one. Its item information is constant at a_j^2/σ_j^2.

4.3.5 Comparing the Parameterizations

The person parameters are comparable when every model identifies the latent scale in the same way, commonly $\theta_i \sim N(0, 1)$. For beta and simplex IRT, a_j and b_j enter the same logistic mean function. In the Samejima model, the conditional mean is not exactly logistic, so comparisons should use the conditional median as the corresponding logistic location curve. A useful location index is

$$b_j^* = -\frac{b_j}{a_j},$$

For the beta and simplex models, b_j^* is the latent position at which the conditional mean is one half. For the Johnson S_B model, it is the position at which the conditional median is one half. Discrimination can be compared through the maximum slope of the appropriate logistic location curve.

The following figures separate the effects of difficulty and discrimination using the difficulty parameterization

$$L(\theta; a, b) = \text{logit}^{-1}\{a(\theta - b)\}, \qquad a > 0, \quad b \in \mathbb{R}.$$

Here a is discrimination and b is difficulty, corresponding to b_j^* above. This difficulty b is distinct from the easiness/intercept b_j in $a_j\theta + b_j$; their relation is $b_j = -a_j b$. For beta and simplex, L is the conditional response mean. For Johnson S_B, it is the conditional median. In inflated models these curves describe the interior response distribution.

With $a = 1$ fixed, increasing difficulty from $b = -1$ to 0 and 1 shifts the curve to the right without changing its shape. Every curve reaches $L = 0.5$ at $\theta = b$. Thus, a larger b requires a higher latent trait to achieve the same response location and gives a smaller L at any fixed θ. The midpoint slope remains $a/4 = 0.25$.

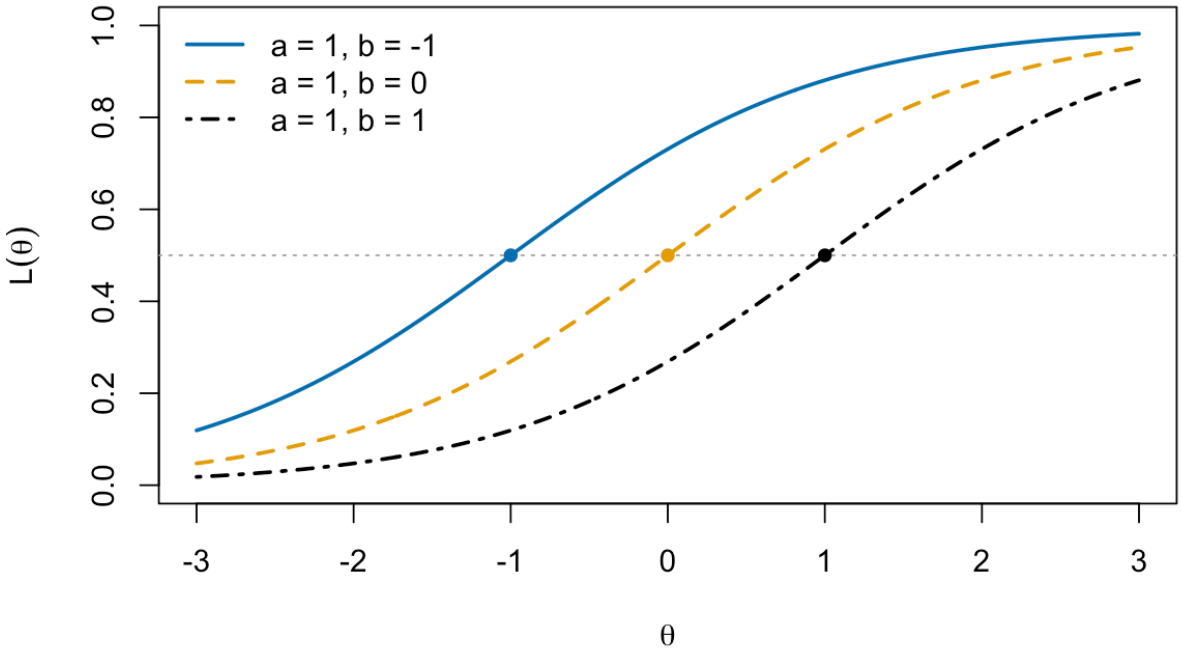


Figure 3: Effect of difficulty: fixed discrimination a = 1 and difficulties b = -1, 0, 1 in the parameterization logistic{a(theta-b)}. Marked points identify the half-response locations.

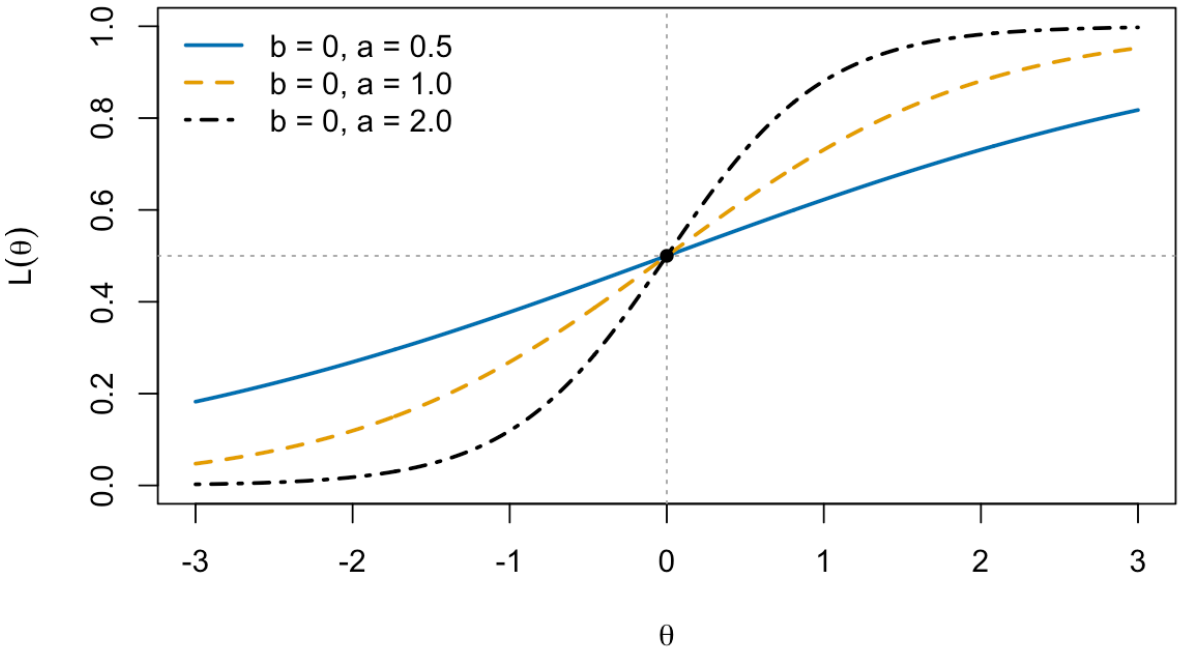


Figure 4: Effect of discrimination: fixed difficulty b = 0 and discriminations a = 0.5, 1, 2. All curves share the midpoint theta = 0, L = 0.5.

With $b = 0$ fixed, increasing discrimination from $a = 0.5$ to 1 and 2 steepens the transition around the common midpoint. Nearby latent positions therefore produce larger differences in response location. In particular,

$$\frac{\partial L}{\partial \theta} = aL(1 - L), \qquad \left.\frac{\partial L}{\partial \theta}\right|_{\theta=b} = \frac{a}{4}.$$

The midpoint slopes are 0.125, 0.25, and 0.50, respectively. Above b, larger a gives larger L; below b, it gives smaller L. Discrimination changes steepness while difficulty determines the midpoint. These comparisons concern logistic location curves, not the normal IRT mean or the Johnson S_B mean, and do not imply equal information functions across the three bounded families.

4.3.6 Zero and One Inflation Bounded IRT models

Conventional bounded densities are defined on $(0, 1)$. They cannot assign positive probability to exact endpoint responses. Molenaar et al. introduce a three-part response mechanism for the closed interval (Molenaar et al. 2022). Let Z_{ij} indicate whether a response is at zero, in the interior, or at one:

$$Z_{ij} = \begin{cases} 0, & Y_{ij} = 0, \\ 1, & 0 < Y_{ij} < 1, \\ 2, & Y_{ij} = 1. \end{cases}$$

For item-specific boundary parameters γ_{0j} and γ_{1j}, with $\gamma_{1j} > \gamma_{0j}$, define

$$\begin{aligned} P(Z_{ij} = 0 \mid \theta_i) &= \text{logit}^{-1}(\gamma_{0j} - a_j\theta_i), \\ P(Z_{ij} = 1 \mid \theta_i) &= \text{logit}^{-1}(\gamma_{1j} - a_j\theta_i) - \text{logit}^{-1}(\gamma_{0j} - a_j\theta_i), \\ P(Z_{ij} = 2 \mid \theta_i) &= 1 - \text{logit}^{-1}(\gamma_{1j} - a_j\theta_i). \end{aligned}$$

For an inflated model, the item vector is expanded to $\tau_j = (a_j, b_j, \zeta_j, \gamma_{0j}, \gamma_{1j})$, including only the boundary parameters required by the inflation structure. If $f_j(y \mid \theta_i)$ denotes one of the bounded interior densities, the complete mixed distribution is

$$k_j(y \mid \theta_i) = \begin{cases} \text{logit}^{-1}(\gamma_{0j} - a_j\theta_i), & y = 0, \\ \left[\text{logit}^{-1}(\gamma_{1j} - a_j\theta_i) - \text{logit}^{-1}(\gamma_{0j} - a_j\theta_i)\right] f_j(y \mid \theta_i), & 0 < y < 1, \\ 1 - \text{logit}^{-1}(\gamma_{1j} - a_j\theta_i), & y = 1. \end{cases}$$

Ospina and Ferrari (2012) model zero-or-one inflation in beta regression through a separate mixture probability. In the IRT formulation of Molenaar et al. (2022), the boundary probabilities depend on the latent trait. Here the endpoint responses carry information about θ_i, a_j, and the item-specific boundary parameters. It therefore avoids replacing zeros and ones by arbitrary values such as 10^{-5} and $1 - 10^{-5}$.

Figure 5 illustrates how the latent trait determines the probabilities of zero, interior, and one responses for an item with $a_j = 1.2$, $\gamma_{0j} = -1.8$, and $\gamma_{1j} = 1.4$. The ordered boundary parameters

ensure that the interior probability is positive and that the three probabilities sum to one at every latent position.

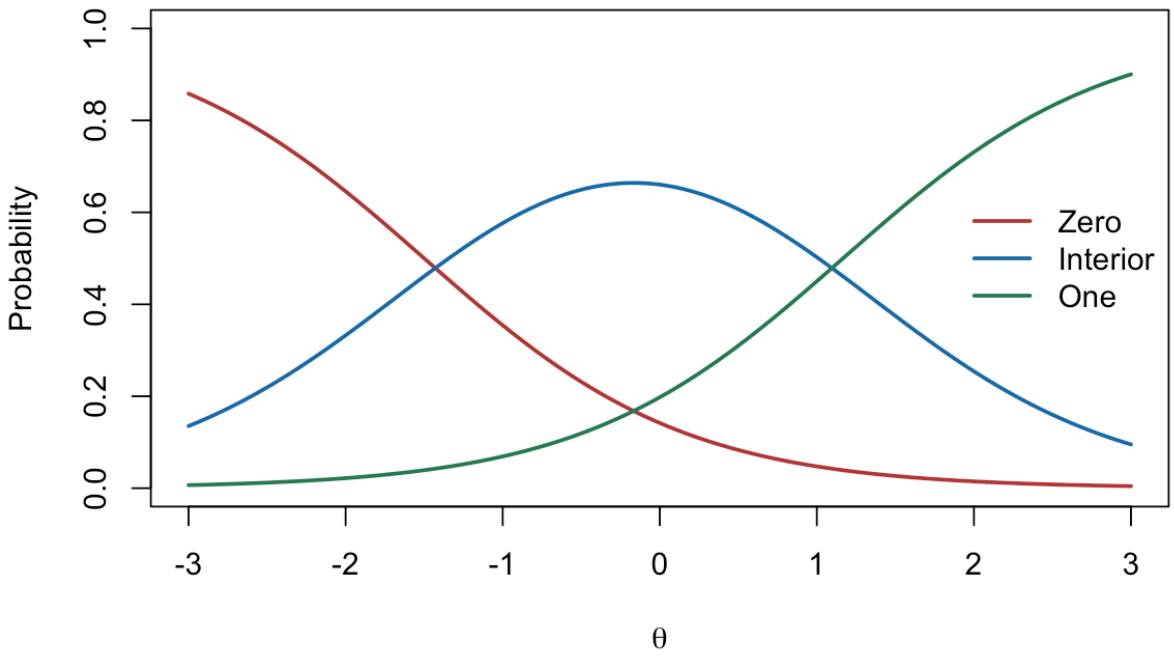


Figure 5: Zero, interior, and one response probabilities as functions of the latent trait, with a = 1.2, gamma0 = -1.8, and gamma1 = 1.4.

In Figure 5, low values of θ favor zero responses, whereas high values favor one responses. The zero-response probability decreases with θ and equals 0.5 at $\theta = \gamma_{0j}/a_j = -1.50$. The one-response probability increases and equals 0.5 at $\theta = \gamma_{1j}/a_j \approx 1.17$. In the central region, interior responses are most likely; their probability reaches its maximum near $\theta = (\gamma_{0j} + \gamma_{1j})/(2a_j) \approx -0.17$. At $\theta = 0$, the probabilities of zero, interior, and one responses are approximately 0.142, 0.660, and 0.198, respectively.

The boundary parameters locate the transitions between response categories, while a_j controls how sharply their probabilities change with the latent trait. These boundary transitions are distinct from the interior difficulty b_j^*. The curves describe the probability of each response category, not the density within $(0, 1)$; the same boundary mechanism can accompany any of the interior families. Thus, an exact zero or one provides information about the latent trait rather than requiring an arbitrary replacement value.

4.4 Bayesian Estimation and Model Comparison Using BoundIRT

The R package `BoundIRT` implements the bounded continuous IRT framework of Molenaar et al. (2022) through Bayesian estimation in Stan (Molenaar 2026). Its main function, `fitBIRT()`, sup-

ports Samejima (Johnson S_B), beta, simplex, and normal models, with optional zero and one inflation. It also provides simulation, posterior summaries, predictive checks, and model comparison. This section explains the estimation workflow underlying the Abasement example in Section 5.

4.4.1 Posterior Distribution and Identification

Molenaar et al. (2022) formulate estimation in a Bayesian Markov chain Monte Carlo framework. Let $\theta = (\theta_1, \ldots, \theta_N)$ denote the person parameters, let $\mathbf{T} = (\tau_1, \ldots, \tau_J)$ collect the item parameters, and let $\mathcal{O}$ be the set of observed person-item pairs. The posterior distribution can be written as

$$p(\theta, \mathbf{T} \mid \mathbf{Y}) \propto \left\{ \prod_{(i,j)\in\mathcal{O}} k_j(Y_{ij} \mid \theta_i, \tau_j) \right\} \left\{ \prod_{i=1}^{N} g(\theta_i) \right\} \left\{ \prod_{j=1}^{J} h(\tau_j) \right\},$$

where $k_j(\cdot)$ is either the interior density or the zero-and-one-inflated mixed distribution, $g(\cdot)$ is the person-parameter prior, and $h(\cdot)$ is the item-parameter prior. Missing responses contribute no likelihood term.

The latent scale is identified by

$$\theta_i \sim N(0, 1).$$

This fixes the origin and variance of the person distribution. The discrimination parameters are constrained to be positive, which prevents the orientation of the latent scale from switching during sampling. In the Stan implementation, this is achieved by sampling $\log(a_j)$ and transforming back with $a_j = \exp\{\log(a_j)\}$.

When both endpoints are modeled, $\gamma_{1j} > \gamma_{0j}$ ensures a positive interior probability. `BoundIRT` enforces this through a lower-truncated prior for γ_{1j}. With `inflated = "auto"`, boundary parameters are included only for endpoints observed in each item (Molenaar 2026). The paper's original Stan implementations are also available in the authors' code archive.

4.4.2 Prior Distributions

The analysis uses standard-normal priors for person parameters and diffuse normal priors with standard deviation 10 for item parameters on their sampling scales, following Molenaar et al. (2022). The defaults in `fitBIRT()` are summarized below; the two boundary priors are subject to the ordering constraint when both endpoints are modeled:

Parameter	Sampling-scale prior	Default SD argument
θ_i	$N(0, s_T^2)$	`sT = 1`
$\log(a_j)$	$N(0, s_A^2)$	`sA = 10`
b_j	$N(0, s_B^2)$	`sB = 10`
$\log(\zeta_j)$	$N(0, s_P^2)$	`sP = 10`
γ_{0j}, γ_{1j}	$N(0, s_G^2)$, with ordering when both occur	`sG = 10`

In the prior table, ζ_j denotes the family-specific positive parameter defined in Section 3. The package calls this parameter `phi`: it corresponds to δ_j for Samejima, λ_j for beta, ϕ_j for simplex,

and σ_j^2 for normal. The software name is preserved in code. The package documentation uses `alpha` and `beta` for discrimination and easiness; these correspond to a_j and b_j in this notebook. Prior scales can be changed selectively by passing a named list. For example,

```
fit <- BoundIRT::fitBIRT(
  z = response_matrix,
  model = "beta",
  inflated = "auto",
  prior = list(sA = 2, sB = 5, sP = 3, sG = 5)
)
```

Because the default item priors are deliberately broad, sensitivity analysis is advisable when the sample is small, endpoint responses are rare, or an item has little variation. This is especially important for Bayes factors, which depend on the prior distribution as well as on the likelihood.

4.4.3 Posterior Sampling with `fitBIRT()`

`fitBIRT()` uses `rstan::sampling()` and supports the beta, Samejima, simplex, and normal models, each with `"none"`, `"zero"`, `"one"`, `"both"`, or item-adaptive `"auto"` inflation. Its main sampling defaults are `iter = 2000`, `warmup = floor(iter/2)`, and `nchains = 1`. Additional Stan controls are passed through `...` (Molenaar 2026). For substantive analyses, multiple chains are needed:

```
fit <- BoundIRT::fitBIRT(
  z = response_matrix,
  model = "beta",
  inflated = "auto",
  iter = 4000,
  warmup = 2000,
  nchains = 4,
  prior = list(sA = 2, sB = 5, sP = 3, sG = 5),
  seed = 2022,
  control = list(adapt_delta = 0.95, max_treedepth = 12)
)
```

The returned `BoundIRT` fitted object contains posterior summaries of item parameters in `output`, person summaries in `theta`, the complete `stanfit` object in `mcmc`, and the prior scales in `prior`. Useful extraction methods include `print()`, `coef()`, `summary()`, and `plot()`.

Before interpreting parameters or comparing models, inspect at least:

- rank-normalized $\widehat{R}$, which should be close to one;
- bulk and tail effective sample sizes;
- trace plots and mixing across chains;
- divergent transitions and maximum-treedepth warnings;
- energy diagnostics where available;
- posterior concentration near parameter constraints.

Increasing `adapt_delta` can reduce divergences, but persistent divergences or poor mixing should prompt investigation of the data, priors, inflation structure, and parameterization rather than only longer sampling.

4.4.4 Posterior Predictive Checking

Convergence diagnostics assess whether sampling adequately explores the posterior; posterior predictive checks assess whether the model reproduces important features of the data. The package

function `repBIRT()` simulates replicated datasets from posterior draws. Its default is `nrep = 1000`. It returns a `repBIRT` object whose `z_rep` component is an array indexed by replication, person, and item.

```
replicated <- BoundIRT::repBIRT(fit, nrep = 500)
y_rep <- replicated$z_rep

# Example checks for the first item
observed_zero_rate <- mean(response_matrix[, 1] == 0, na.rm = TRUE)
replicated_zero_rate <- apply(y_rep[, , 1] == 0, 1, mean, na.rm = TRUE)

observed_one_rate <- mean(response_matrix[, 1] == 1, na.rm = TRUE)
replicated_one_rate <- apply(y_rep[, , 1] == 1, 1, mean, na.rm = TRUE)
```

Useful discrepancy measures include itemwise zero and one proportions, interior means and variances, quantiles, skewness, boundary-near frequencies, and person-level total scores. Density overlays can reveal lack of fit in the interior distribution even when endpoint frequencies are reproduced.

4.4.5 Marginal and Conditional Log Likelihoods

For latent-variable models, the observational unit used in predictive criteria must be chosen carefully. `loglikBIRT()` provides two forms:

- `marginal = TRUE` integrates over each person parameter and returns one log likelihood contribution per person and posterior draw. The integral is approximated with Gaussian quadrature; `nquad = 15` is the default.
- `marginal = FALSE` conditions on sampled person parameters and returns contributions for person-item combinations.

The marginal form is generally preferable for comparing measurement models at the person level because it evaluates prediction for new persons rather than prediction of additional responses for persons whose latent positions have already been estimated. The resulting matrix can be passed to the `loo` package:

```
log_lik <- BoundIRT::loglikBIRT(
  fit,
  marginal = TRUE,
  nquad = 15
)

loo_result <- loo::loo(log_lik)
waic_result <- loo::waic(log_lik)
```

Smaller LOOIC and WAIC indicate better expected predictive fit. Differences should be interpreted together with their standard errors, Pareto-k diagnostics for LOO, and posterior predictive checks (Vehtari et al. 2017).

4.4.6 Fully Marginalized Likelihood and Bayes Factors

The paper emphasizes the fully marginalized likelihood, which integrates over both person and item parameters:

$$L(\mathbf{Y}) = \iint p(\mathbf{Y} \mid \theta, \mathbf{T}) p(\theta) p(\mathbf{T}) \, d\theta \, d\mathbf{T}.$$

`bridgeBIRT()` approximates $\log L(\mathbf{Y})$ by bridge sampling with a multivariate normal proposal distribution (Gronau et al. 2017). Its defaults are `niter = 10000` and `tol = 1e-7`; the iterative algorithm stops when the change in its ratio estimate is below the tolerance. The implementation centers calculations at the median posterior log-likelihood for numerical stability (Molenaar 2026).

```
bridge_beta <- BoundIRT::bridgeBIRT(
  beta_fit,
  niter = 10000,
  tol = 1e-7
)

bridge_simplex <- BoundIRT::bridgeBIRT(simplex_fit)

log_bf_beta_simplex <-
  bridge_beta$log_ml - bridge_simplex$log_ml
```

The model with the larger log marginal likelihood is preferred. For models M_1 and M_2, the log Bayes factor is

$$\log BF_{12} = \log p(\mathbf{Y} \mid M_1) - \log p(\mathbf{Y} \mid M_2).$$

Bridge estimates should be rerun or checked for numerical stability when models are high dimensional or posterior geometry is difficult. Since the marginal likelihood averages over the prior, all candidate models should use priors that are substantively defensible and comparable in informativeness.

4.4.7 Evidence from the Paper

The simulation study in Molenaar et al. (2022) compared the fully marginalized log likelihood with DIC and WAIC. Under the study conditions with $N = 50$, the log marginal likelihood selected the generating SB, beta, and simplex models in 0.88, 0.98, and 0.91 of replications, respectively. Its advantage was largest for the smaller sample; differences diminished as sample size increased, and all three criteria selected the generating model in all reported conditions at $N = 200$. These results support the fully marginalized likelihood for discrimination among the candidate bounded IRT families, but they do not eliminate the need for predictive and graphical checks.

The same study found that misspecifying the interior distribution affected discrimination estimates more strongly than easiness estimates. The beta and SB models were comparatively robust to one another, whereas fitting simplex data with those models, or fitting beta/SB data with the simplex model, could produce more noticeable discrimination bias. The unbounded normal model was biased across the bounded-data scenarios, particularly for discrimination. Thus, model comparison is consequential even when fitted central response curves appear similar.

4.4.8 Recommended Comparison Strategy

No single statistic answers every model-comparison question. A defensible workflow is:

1. Fit the same inflation structure and substantively comparable priors across candidate interior distributions.
2. Establish satisfactory MCMC diagnostics for every candidate model.
3. Use `repBIRT()` to assess endpoint frequencies and the shape of interior responses.
4. Use marginal `loglikBIRT()` output for person-level LOOIC or WAIC.

5. Use `bridgeBIRT()` when evidence based on the fully marginalized likelihood or Bayes factors is desired.
6. Examine whether substantive conclusions about a_j, b_j^*, and θ_i are robust across well-fitting models.

Preference should be given to a model that combines stable computation, adequate posterior predictive behavior, and favorable out-of-sample or marginal-likelihood evidence, rather than to a model selected by one numerical criterion alone.

4.4.9 Data Structure and Initial Checks

Bounded IRT data are stored as an $N \times J$ matrix: rows represent persons, columns represent items, and entries contain scores in $[0, 1]$. Missing values may occur, but every observed value must respect the bounds.

Before fitting a model, inspect the use of the boundaries item by item. The following helper returns the number of observations, missing values, zeros, ones, and interior responses.

```
boundary_summary <- function(x) {
  stopifnot(is.matrix(x) || is.data.frame(x))
  x <- as.matrix(x)

  data.frame(
    item = colnames(x) %||% paste0("Item", seq_len(ncol(x))),
    observed = colSums(!is.na(x)),
    missing = colSums(is.na(x)),
    zeros = colSums(x == 0, na.rm = TRUE),
    ones = colSums(x == 1, na.rm = TRUE),
    interior = colSums(x > 0 & x < 1, na.rm = TRUE),
    prop_zero = colMeans(x == 0, na.rm = TRUE),
    prop_one = colMeans(x == 1, na.rm = TRUE),
    row.names = NULL
  )
}

`%||%` <- function(x, y) if (is.null(x)) y else x
```

The helper also supports the modeling decision used by `fitBIRT()`:

- no observed boundary values: `inflated = "none"` is admissible;
- only zeros: use `inflated = "zero"` or `"auto"`;
- only ones: use `inflated = "one"` or `"auto"`;
- both boundaries: use `inflated = "both"` or `"auto"`.

The default `inflated = "auto"` adds boundary parameters only for items that actually contain zeros or ones. This is a convenient starting point when endpoint use differs across items.

4.4.10 Recommended Workflow for practical analysis

1. Confirm the theoretical score bounds and transform responses to $[0, 1]$.
2. Summarize exact zeros, exact ones, missingness, and interior distributions for every item.
3. Use `inflated = "auto"` unless theory requires a common inflation structure across all items.
4. Fit more than one plausible interior distribution. Beta, Samejima, and simplex models may yield similar central item curves but differ in tails, information, and person estimates.
5. Treat the normal model as a reference, not as the default for bounded data.

6. Check MCMC diagnostics before interpreting coefficients or fit criteria.
7. Compare models using predictive fit and, where feasible, bridge sampling.
8. Examine item characteristic curves, information curves, fitted densities, and boundary probabilities.
9. Report the chosen interior distribution, inflation rule, priors, chain settings, convergence diagnostics, and model-comparison criterion.

4.5 Practical Estimation Using BoundIRT: Abasement Scale Example

The Abasement scale is part of the Adjective Check List (ACL), a personality assessment developed by Gough and Heilbrun (1980). It concerns self-critical attitudes and expressions of inferiority, including guilt and a sense of inadequacy in social situations. Higher responses, after reverse scoring where required, indicate greater endorsement of these characteristics.

This example uses the continuous-response version analyzed by Molenaar et al. (2022) and distributed as `Abasement` in `BoundIRT` (Molenaar 2026). It contains responses from 244 people to 10 adjectives. Participants indicated how well each adjective described them on a 60-millimeter line segment; these ratings were rescaled to $[0, 1]$. This response format differs from the original ACL checklist. The item marked with an asterisk, `self-confident*`, is contra-indicative and has already been reverse scored in the package data.

The dataset contains 2,440 item responses, approximately 10.74% of which are exact zeros, with no exact ones. This makes it useful for illustrating zero-inflated bounded IRT: the model accounts separately for the probability of a zero and the distribution of interior responses. The analysis compares candidate interior families and then interprets item and person estimates. The package also supplies `out_beta`, a previously fitted zero-inflated beta model, which provides a starting point for the comparison.

4.5.1 Installation of the `BoundIRT` Package

The CRAN package is named `BoundIRT` (rather than `BoundedIRT`). Install it once, then load it before using the Abasement data and fitting models. The installation command is shown but not run when this document is rendered.

```
install.packages("BoundIRT", repos = "https://cloud.r-project.org")
library(BoundIRT)
packageVersion("BoundIRT")
```

`BoundIRT` uses `rstan` and compiled Stan support, so installation may require a working C++ toolchain. This notebook also supports the project-local `Rlib` library configured in its setup chunk. The analyses below use the installed package and saved fits; rendering them does not reinstall or refit the models.

The principal public functions in `BoundIRT` version 0.6.0 are summarized below (Molenaar 2026).

Function	Purpose
`fitBIRT()`	Fit Samejima, beta, simplex, or normal bounded IRT models
`rBIRT()`	Simulate bounded IRT data, with optional zero/one inflation
`coef()`	Extract posterior means and optional posterior standard deviations
`plot()`	Plot ICCs, item information, test information, or fitted densities

Function	Purpose
`repBIRT()`	Generate posterior predictive response datasets
`loglikBIRT()`	Compute person-marginal or conditional log likelihoods
`bridgeBIRT()`	Estimate the log marginal likelihood by bridge sampling
`latregBIRT()`	Regress the latent person variable on observed predictors

4.5.2 Item Analysis

Before fitting the IRT models, the ten original item scores are examined for internal consistency and boundary responses. Cronbach's coefficient alpha is

$$\hat{\alpha} = \frac{J}{J-1}\left(1 - \frac{\sum_{j=1}^{J} s_j^2}{s^2_{\sum_j Y_j}}\right), \qquad J = 10,$$

where s_j^2 is the sample variance of item j and the denominator is the sample variance of the respondent-level total score.

For the 244 complete respondents, Cronbach's alpha is 0.775. This describes internal consistency of the observed item scores; it is not a model-fit criterion. The boundary summary shows exact endpoint use before any transformation:

Table 3: Boundary summary of the original Abasement item scores.

item	observed	missing	zeros	ones	interior	prop_zero	prop_one
anxious	244	0	26	0	218	0.107	0
cowardly	244	0	39	0	205	0.160	0
despondent	244	0	27	0	217	0.111	0
submissive	244	0	40	0	204	0.164	0
pessimistic	244	0	30	0	214	0.123	0
gloomy	244	0	29	0	215	0.119	0
self-punishing	244	0	10	0	234	0.041	0
self-denying	244	0	30	0	214	0.123	0
self-confident*	244	0	2	0	242	0.008	0
weak	244	0	29	0	215	0.119	0

The corrected item–total correlation is the Pearson correlation between an item and the sum of the other nine items. Descriptive statistics and corrected item–total correlations, computed from complete respondents, are:

Table 4: Item descriptive statistics and corrected item-total correlations.

Item	Min	Max	Mean	SD	Corrected item-total r
anxious	0	0.800	0.227	0.199	0.372
cowardly	0	0.683	0.150	0.152	0.405
despondent	0	0.950	0.220	0.215	0.611
submissive	0	0.817	0.175	0.187	0.328
pessimistic	0	0.950	0.231	0.217	0.546
gloomy	0	0.950	0.257	0.226	0.649
self-punishing	0	0.950	0.392	0.254	0.269
self-denying	0	0.967	0.220	0.223	0.505

Item	Min	Max	Mean	SD	Corrected item-total r
self-confident*	0	0.950	0.435	0.230	0.332
weak	0	0.850	0.199	0.184	0.452

The histograms below show the original response distributions, including exact zeros. All panels use the same horizontal and vertical scales.

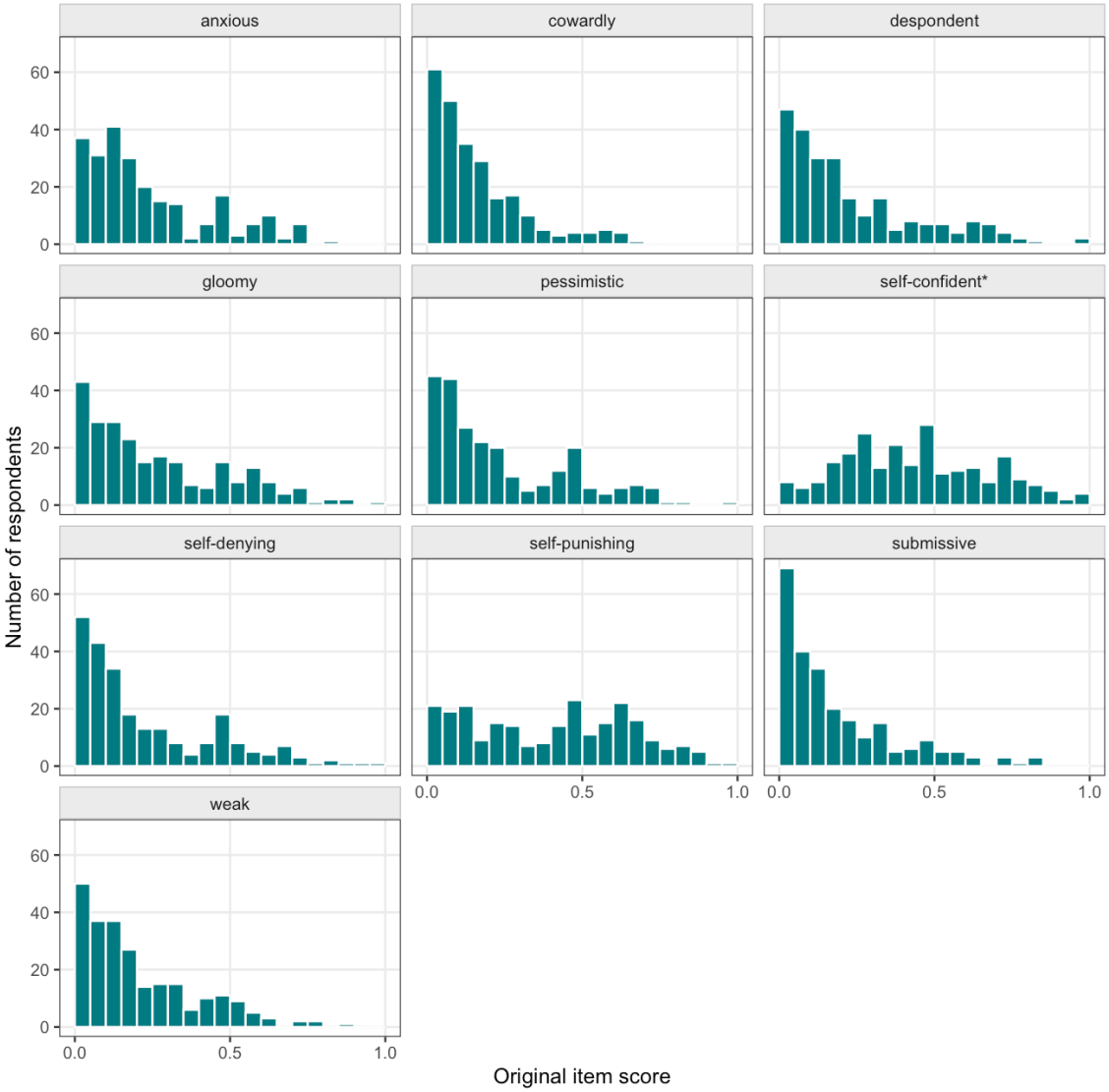


Figure 6: Observed Abasement item-score distributions before boundary transformation. Each panel contains 244 responses; zero responses are included.

4.5.3 Fitted Models

Three zero-inflated models are fitted to the ten Abasement items. For person $i = 1, \ldots, 244$ and item $j = 1, \ldots, 10$, let

$$\eta_{ij} = a_j\theta_i + b_j,$$

where $a_j > 0$ is discrimination, b_j is easiness, and $\theta_i \sim N(0, 1)$. Because these data contain zeros but no ones, the boundary component is

$$\Pr(Y_{ij} = 0 \mid \theta_i) = \text{logit}^{-1}(\gamma_{0j} - a_j\theta_i).$$

For an interior response, the **beta IRT model** used by `BoundIRT` is

$$Y_{ij} \mid 0 < Y_{ij} < 1, \theta_i \sim \text{Beta}(c_{ij}, d_{ij}),$$

$$c_{ij} = \exp\left\{\frac{\eta_{ij} + \log(\lambda_j)}{2}\right\}, \qquad d_{ij} = \exp\left\{\frac{-\eta_{ij} + \log(\lambda_j)}{2}\right\}.$$

The corresponding **Johnson S_B IRT model**, called the Samejima model in `BoundIRT`, assumes

$$\text{logit}(Y_{ij}) \mid 0 < Y_{ij} < 1, \theta_i \sim N(\eta_{ij}, \delta_j),$$

where the second argument denotes the variance; hence the logit-scale standard deviation is $\sqrt{\delta_j}$. The **simplex IRT model** uses $Y_{ij} \mid 0 < Y_{ij} < 1, \theta_i \sim \text{Simplex}\{\text{logit}^{-1}(\eta_{ij}), \phi_j\}$. These are the same family-specific parameters defined in Section 3. Each interior density is multiplied by $1 - \Pr(Y_{ij} = 0 \mid \theta_i)$. Exact zeros preclude fitting the corresponding non-inflated models directly. For a sensitivity analysis, all responses are transformed to the open interval using

$$Y_{ij}^{\text{new}} = \frac{Y_{ij}(n-1) + 0.5}{n}, \qquad n = 244,$$

where n is the number of respondents. Former zeros become 0.5/244, and all 2,440 transformed scores are used to fit beta, Johnson S_B, and simplex models with `inflated = "none"`. The two groups of models have different response data and likelihoods, so information criteria are comparable *within* each group, not across the original-data and transformed-data groups.

The package supplies the zero-inflated beta fit. The Samejima and simplex fits use 2,000 iterations, including 1,000 warmup iterations, and one chain. Saved fits and person-marginal log-likelihood matrices are read from the `rds` folder beside this R Markdown file. The optional refitting chunks also write their RDS files to this folder. These saved results keep rendering inexpensive. These fits illustrate the workflow; a single chain cannot establish convergence across independent chains. Substantive conclusions should be checked with the multiple-chain workflow in Section 4.3.

In `BoundIRT` 0.6.0, `loglikBIRT()` omits initialization of the empty zero/one-inflation masks when `inflated = "none"`. The calculation used for the cached transformed-data log-likelihood matrices initializes only those masks before evaluating the package function body:

The package documentation reports that `out_beta` was obtained with the code below. It is left unevaluated because this section uses the distributed fitted object and does not need to rerun MCMC.

The Johnson S_B fit used in this section was obtained analogously. The fitted object is stored in `rds/section8_out_jsb.rds`, so this computationally expensive chunk is not rerun when the document is rendered.

Table 5: Person-marginal predictive criteria. Compare values only within each data group.

data	model	LOOIC	LOOIC_SE	WAIC	WAIC_SE
Original; zero-inflated	Beta	-517.98	168.23	-517.93	168.29
Original; zero-inflated	Johnson S_B	-675.85	165.33	-676.20	165.31
Original; zero-inflated	Simplex	-521.52	185.17	-521.94	185.09
Transformed; non-inflated	Beta	-3031.09	185.83	-3031.26	185.82
Transformed; non-inflated	Johnson S_B	-2770.90	209.61	-2771.19	209.61
Transformed; non-inflated	Simplex	-793.71	245.63	-781.70	245.80

Both criteria are on the deviance scale, so smaller values indicate better expected out-of-sample fit *on the same response data*. The Johnson S_B model has the smaller LOOIC and WAIC on the original data. Among the transformed-data non-inflated fits, beta has the smaller criteria. These are separate within-group comparisons; the transformed beta value cannot be used to declare it superior to the zero-inflated Johnson S_B fit. WAIC produces influential-observation warnings. The original-data simplex fit has one observation with Pareto-$k > 1$; the transformed-data simplex fit has two such observations and weak MCMC diagnostics (maximum $\widehat{R} = 1.027$, minimum effective sample size 55). Its criteria are therefore only provisional.

4.5.4 Item Parameters for the Selected Model

The Johnson S_B model is retained for the illustrative item and person interpretations below, subject to the sampling limitations noted above. Only its item-parameter estimates are shown below.

Table 6: Item-parameter posterior means and standard deviations for the selected Johnson S_B model.

Item	a_j	SD(a_j)	b_j	SD(b_j)	δ_j	SD(δ_j)	γ_{0j}	SD(γ_{0j})
item1	0.495	0.093	-1.414	0.082	1.204	0.128	-2.238	0.223
item2	0.581	0.086	-1.953	0.075	0.912	0.100	-1.675	0.174
item3	1.125	0.097	-1.632	0.091	0.841	0.106	-2.387	0.232
item4	0.541	0.101	-1.832	0.090	1.460	0.163	-1.632	0.179
item5	0.909	0.094	-1.499	0.087	1.043	0.114	-2.170	0.210
item6	1.168	0.096	-1.377	0.089	0.680	0.105	-2.321	0.219
item7	0.467	0.105	-0.602	0.090	1.845	0.178	-3.015	0.293
item8	0.919	0.115	-1.622	0.102	1.374	0.152	-2.191	0.206
item9	0.467	0.085	-0.336	0.076	1.296	0.122	-3.682	0.405
item10	0.724	0.093	-1.662	0.082	1.017	0.110	-2.169	0.216

The item characteristic curves (ICCs) below show selected items under that same fitted model. Their different slopes and locations reflect the estimated discrimination and easiness parameters.

The item-parameter diagram below displays the posterior means of the item discriminations and easiness parameters. Items farther to the right discriminate more strongly between nearby values of θ_i, whereas items higher on the vertical axis have larger easiness parameters. Holding discrimination and logit-scale variance fixed, increasing easiness raises the conditional response location at a given θ_i. Differences in the other item parameters must also be considered when comparing items.

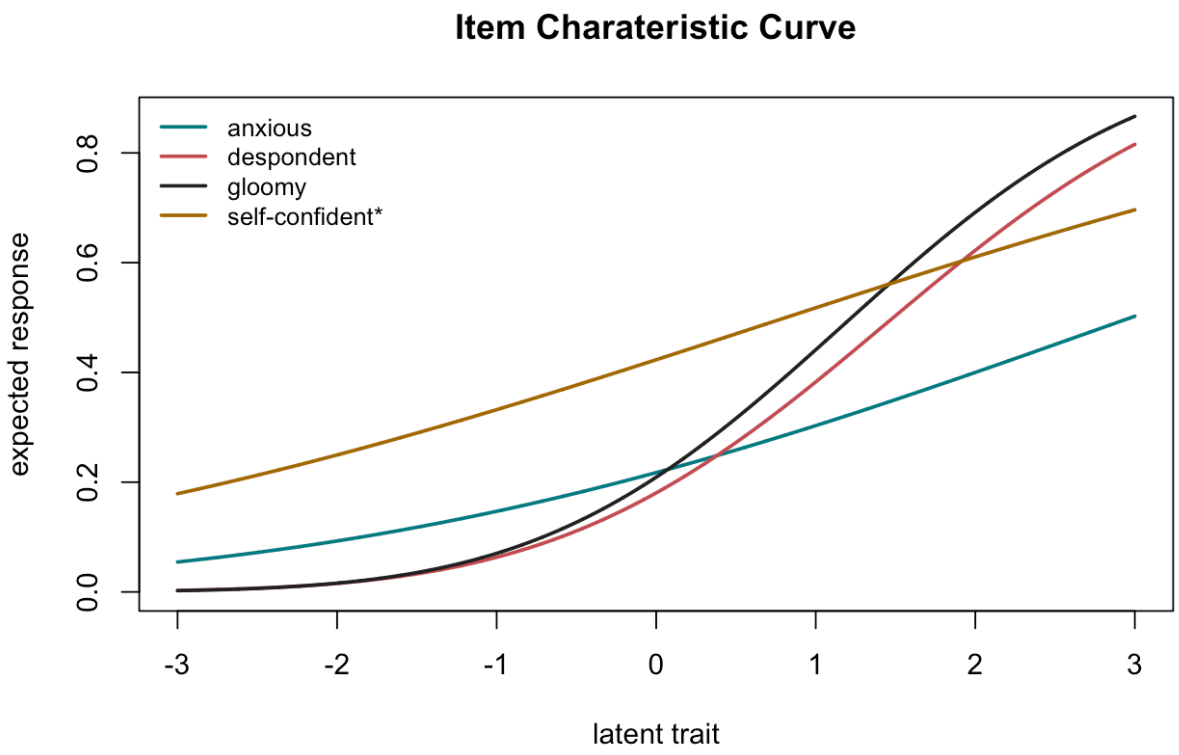


Figure 7: Item characteristic curves for selected Abasement items under the zero-inflated Johnson S_B IRT model.

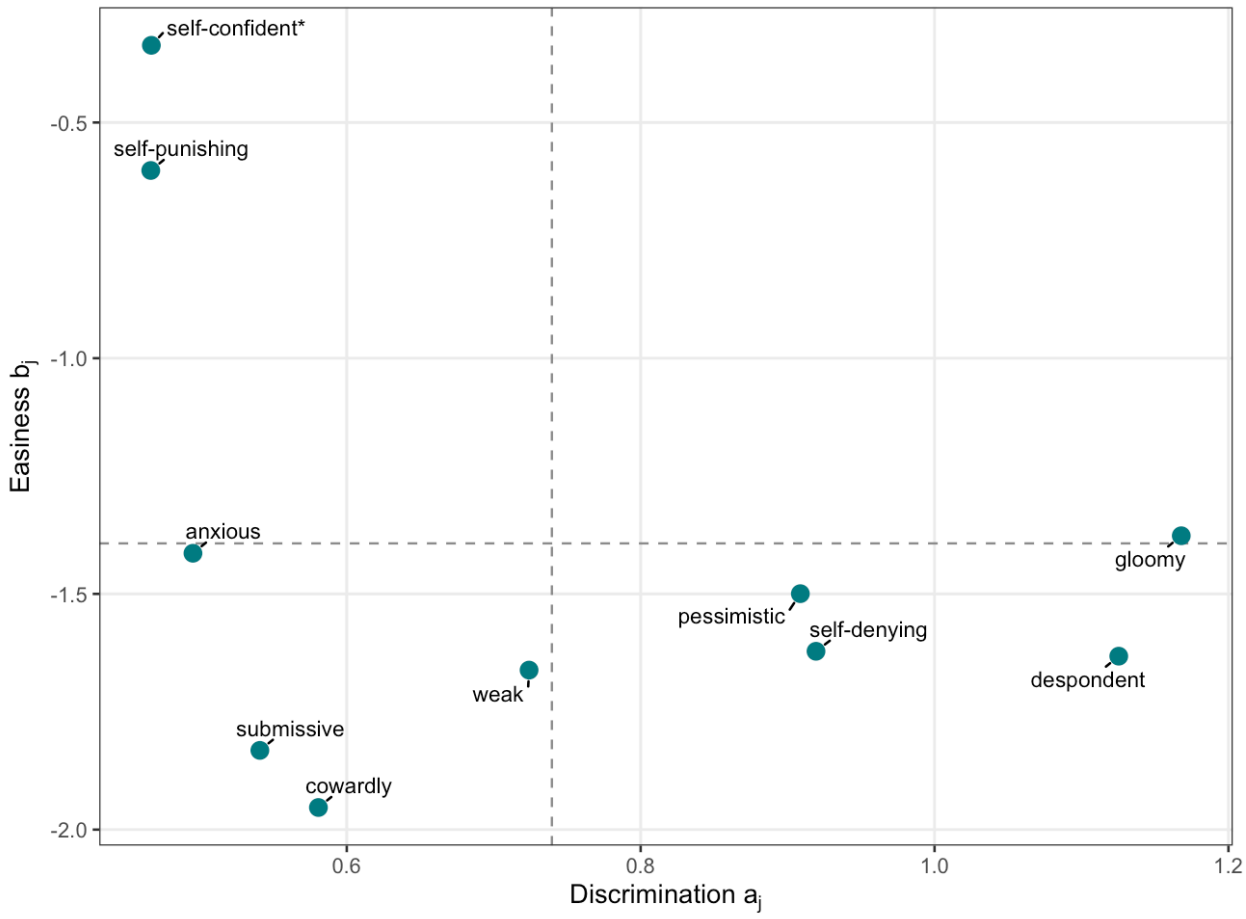


Figure 8: Posterior mean discrimination and easiness parameters under the selected Johnson S_B IRT model. Dashed lines mark the respective item-parameter means.

4.5.5 Person Parameters for the Selected Model

The selected model also produces a posterior distribution for each person parameter. The 244 observed total scores, posterior means, and posterior standard deviations are written to `section8_theta_estimates.csv`.

Descriptive statistics of the 244 posterior means are:

Table 7: Descriptive statistics of the person-parameter posterior means.

statistic	value
N	244.000
Mean	0.002
Standard deviation	0.922
Minimum	-2.995
2.5%	-2.074
First quartile	-0.522
Median	0.054
Third quartile	0.617
97.5%	1.664
Maximum	2.394

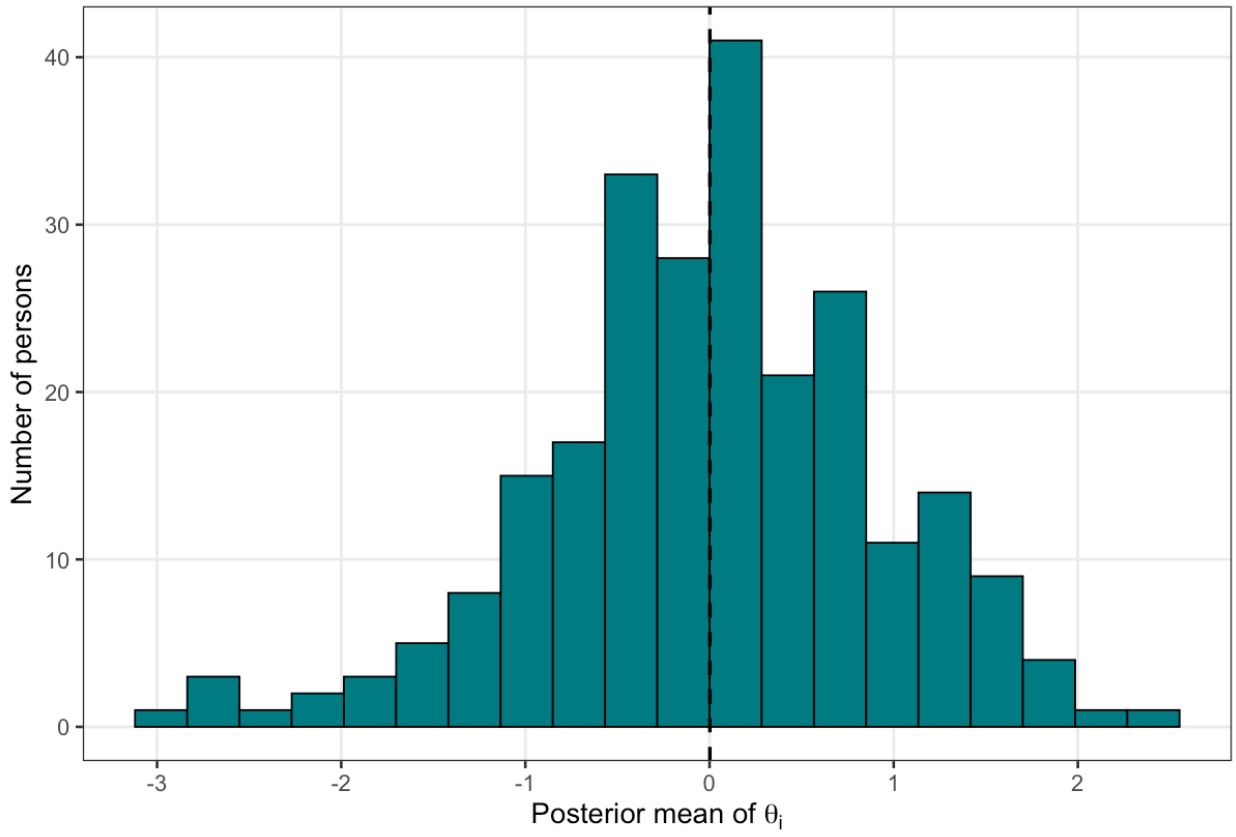


Figure 9: Distribution of posterior mean person parameters under the selected Johnson S_B IRT model.

4.5.5.1 Total Score and Estimated Trait

The observed total score is the unweighted sum of the ten original item responses, $S_i = \sum_{j=1}^{10} Y_{ij}$. Its Pearson correlation with the posterior mean of θ_i under the selected model is reported below.

The correlation is 0.921. The score and theta estimates are strongly associated, but the relationship is not one-to-one.

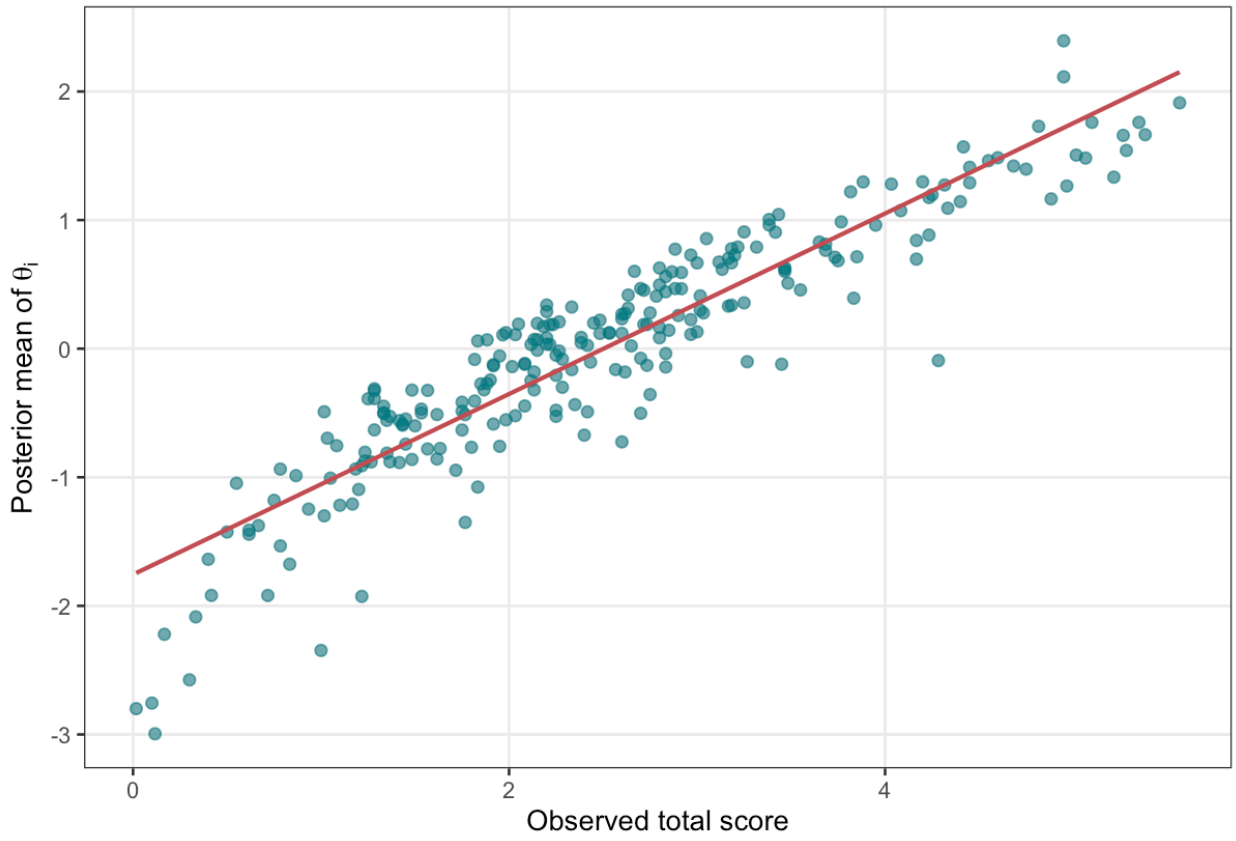


Figure 10: Observed ten-item total score versus posterior mean theta under the selected Johnson S_B IRT model.

To illustrate why equal totals need not imply equal IRT estimates, the next two displays use the two tied-score groups with the largest within-group spread in posterior mean theta. The lowest- and highest-theta respondent in each group are shown; the table includes posterior SDs for context.

Table 8: Respondents with matching total scores but different theta estimates.

Score group	Person	Total score	Theta mean	Theta SD
Total score = 1.833	238	1.833	-1.076	0.489
Total score = 1.833	59	1.833	0.060	0.363
Total score = 1.217	94	1.217	-1.926	0.592
Total score = 1.217	46	1.217	-0.910	0.367

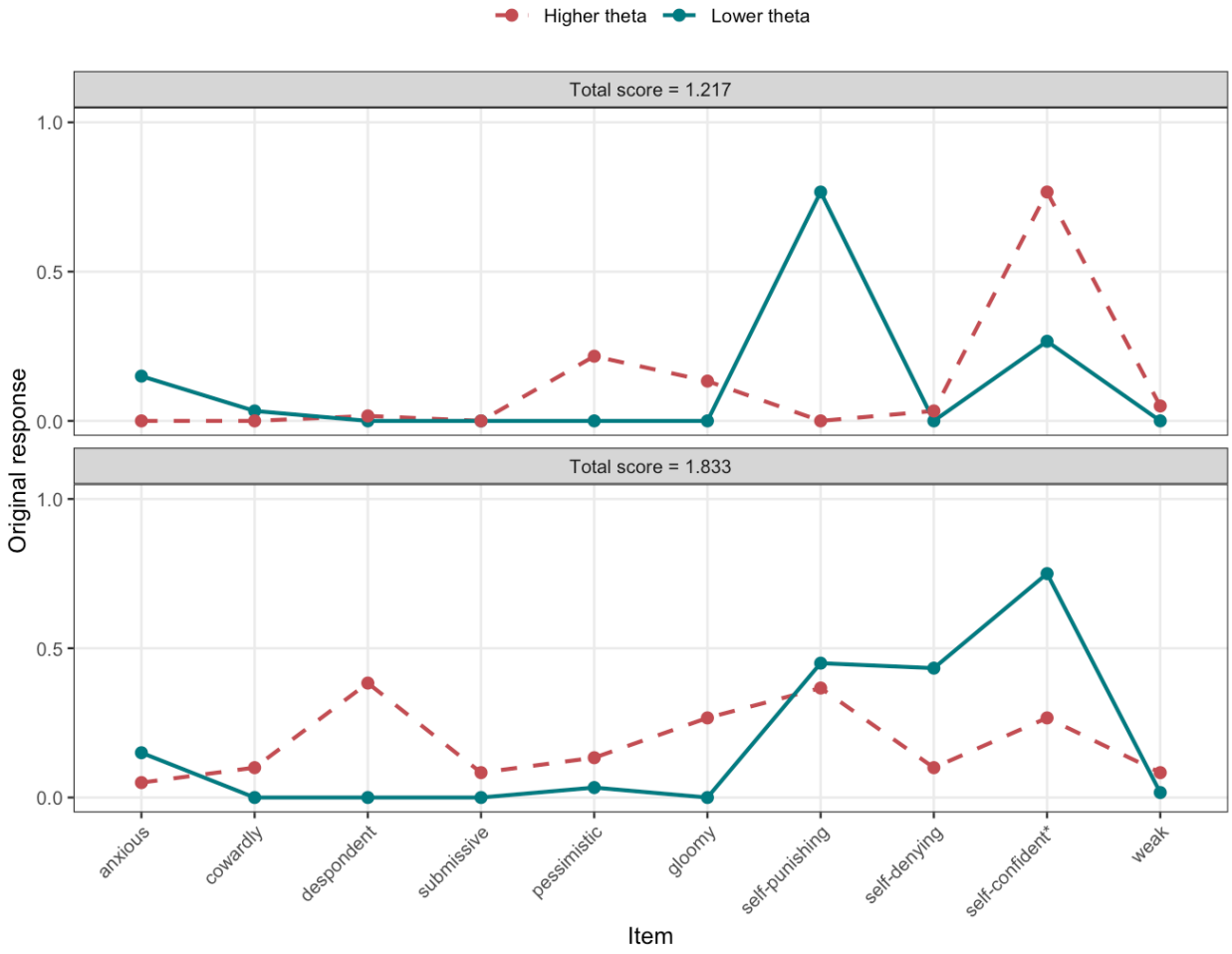


Figure 11: Item-response profiles for two pairs of respondents with equal total scores but different posterior mean theta values.

4.6 Exercises

1. Fit Samejima's continuous response model to the Abasement data using `EstCRM` (Samejima 1973; Zopluoglu 2022). Compare its item and person estimates with the Bayesian results, aligning the latent scale and parameterizations. Explain how you handle exact endpoint responses.
2. The `SelfEff` dataset in `EstCRM` contains responses from 307 pre-service teachers to 10 teaching self-efficacy items (Zopluoglu 2022). Rescale the scores using their theoretical bounds, inspect endpoint use, fit the three bounded IRT models, and interpret the best-supported model.
3. Simulate a dataset from each of the Johnson S_B, beta, and simplex IRT models. Fit all three candidate families to each dataset and compare recovery of the generating parameters and response distributions.
4. Extend the simulation to investigate whether the model favored by predictive criteria also best recovers the person parameters. Report uncertainty and convergence diagnostics as well as average recovery.
5. Analyze the municipal indices for economy, education, and health described by Oliveira et al. (2026a). Fit the three bounded IRT families and interpret the best-supported model, explaining whether a single latent dimension is substantively appropriate across these domains.
6. Assess dimensionality and local dependence for the Abasement and exercise datasets. Discuss how violations affect interpretation of a unidimensional bounded IRT model.
7. Implement a Marginal Likelihood approach for estimate Item parameters of the models studied here.

4.7 Conclusions

Bounded continuous IRT requires attention to both the interior distribution and the endpoints of the response scale. Samejima, beta, and simplex IRT models respect the open interval but imply different conditional variances and information functions. A normal IRT model can serve as an unbounded benchmark, although it assigns positive probability outside the response interval.

The zero-and-one-inflated framework of Molenaar et al. (2022) extends these models to $[0, 1]$ without arbitrary recoding. Because the boundary probabilities depend on the latent trait and item discrimination, endpoint responses contribute to measurement rather than being treated as a separate nuisance component.

The `BoundIRT` package supplies a unified Stan implementation, simulation facilities, posterior summaries, item and test plots, bridge sampling, and latent regression tools (Molenaar 2026). A sound analysis combines these tools with careful boundary inspection, multiple-chain diagnostics, and comparison of plausible response distributions.

4.8 Appendix: Moment-Matched Density Figures

The figures compare the seven feasible mean-variance combinations from Table 1. No density is drawn for $v = 0.24$ with $m = 0.20$ or $m = 0.70$, because these combinations exceed the variance bound. Empty panels in the first figure identify these infeasible combinations.

Both figures use a base-10 logarithmic density axis. High boundary peaks would compress the interior curves on a linear axis; the logarithmic scale makes both large and small density values visible. Equal vertical distances represent equal ratios, such as 0.1 to 1 and 1 to 10. Common

limits permit comparisons across panels. The response axis remains linear, and areas under the displayed curves do not directly represent probabilities. Densities below 10^{-3} are displayed at 10^{-3}, including numerical zeros; this plotting floor does not affect the moment calculations. Colors and distinct line types provide redundant encoding.

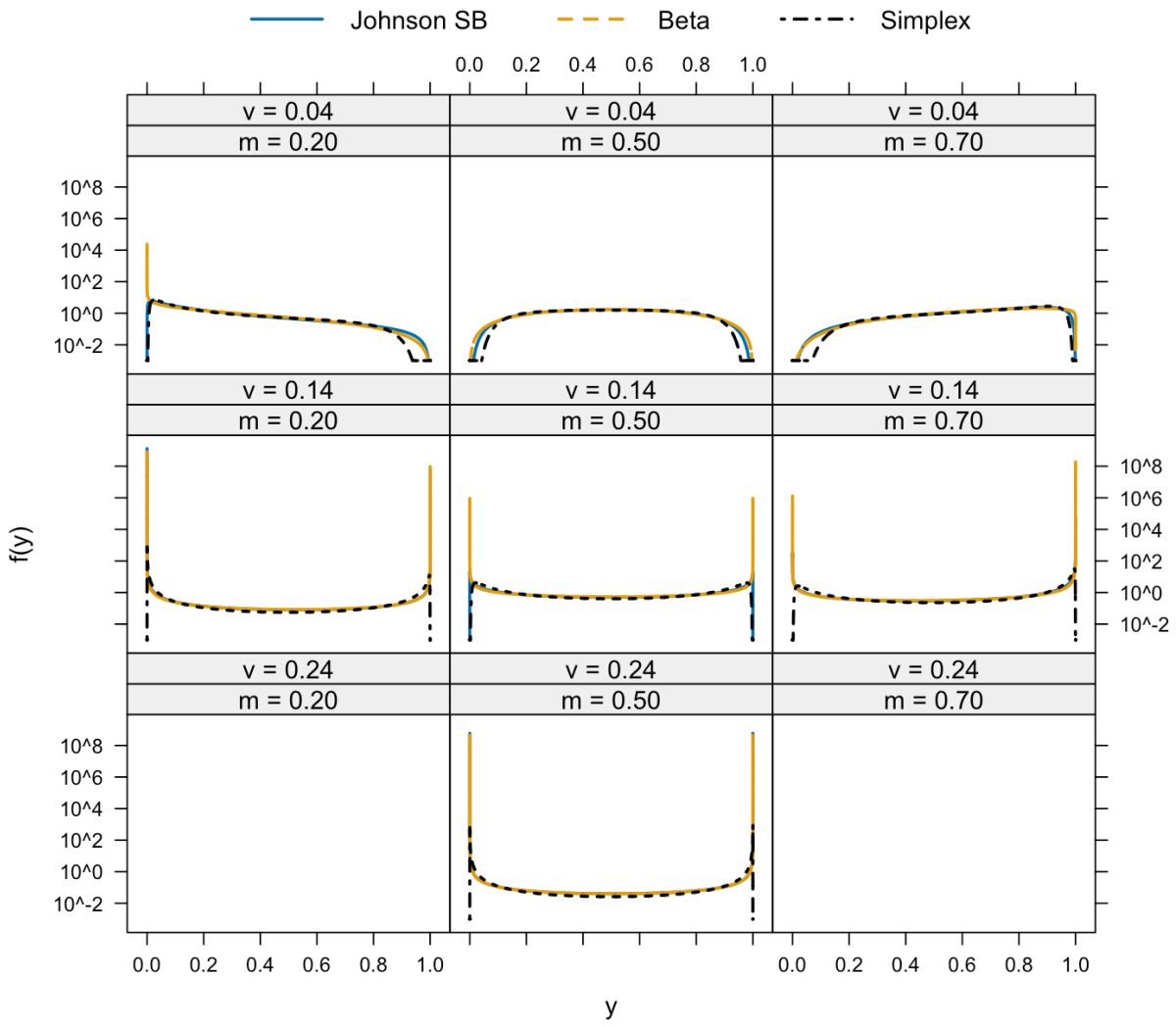


Figure 12: Johnson SB, beta, and simplex densities for the feasible combinations in Table 1. Columns vary the mean; rows vary the target variance. Empty panels are infeasible. The density axis is logarithmic with a display floor of 0.001.

At $v = 0.04$, the curves show more interior concentration, although the beta density with $m = 0.20$ diverges at zero. At $v = 0.14$, boundary concentration is more pronounced for all three means. At $v = 0.24$, the symmetric densities concentrate strongly near both endpoints. The families can differ sharply near the endpoints even when their first two moments agree.

This arrangement shows how changing the mean alters each family's density at a fixed variance. At variance 0.24, each panel contains only the $m = 0.50$ curve, reflecting the feasibility restriction rather than a missing fit.

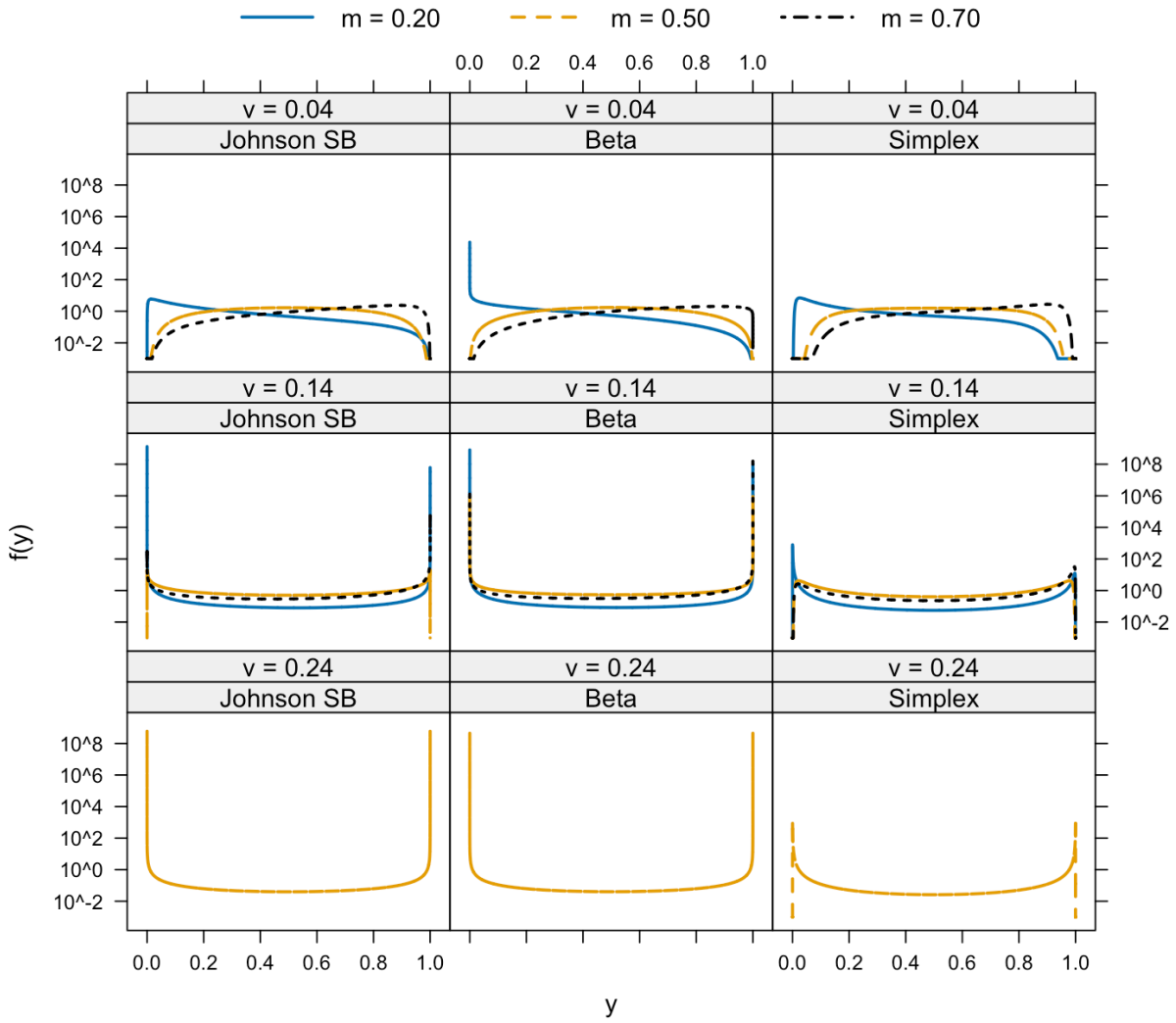


Figure 13: Moment-matched densities grouped by distribution (columns) and target variance (rows). Curves identify the mean; only mean 0.50 is feasible at variance 0.24. The density axis is logarithmic with a display floor of 0.001.

Chapter 5

Bounded Cognitive Diagnosis Models

5.1 Introduction

Cognitive diagnosis models (CDMs), also called diagnostic classification models, describe individuals through combinations of discrete latent attributes. These attributes may represent skills, competencies, or substantively defined conditions. A Q-matrix specifies which attributes each item measures, allowing the response model to connect observed performance to an interpretable attribute profile. The intended output is a set of attribute-specific diagnoses and their uncertainty, rather than only an overall score.

An influential foundation is the work of Junker and Sijtsma (2001) on conjunctive cognitive assessment models and their connections with nonparametric IRT. The deterministic inputs, noisy "and" gate (DINA) model is a conjunctive model for binary item responses: an individual belongs to an item's mastery group only if they possess every attribute required by that item's Q-matrix row. For example, an item requiring two skills places individuals with both skills in the mastery group and all others in the nonmastery group. Responses remain probabilistic: the item-specific slipping parameter is the probability of an incorrect response in the mastery group, whereas the guessing parameter is the probability of a correct response in the nonmastery group. Thus, the probability of a correct response is $1 - s_j$ for the mastery group and g_j for the nonmastery group. The "and" gate expresses the requirement that all relevant attributes be present, while "noisy" allows observed responses to depart from this ideal rule. This structure is useful when success requires every component skill, but can be restrictive when partial mastery also improves performance. The generalized DINA (G-DINA) framework of Torre (2011) relaxes this restriction through attribute main effects and interactions, and includes several reduced CDMs under appropriate constraints. Thus, CDM selection requires a substantive account of how attributes combine, as well as an assessment of model fit.

Although these foundational models were formulated for binary responses, many applications involve measurements that require a richer response model. Visual analogue scales, proportions, and social-development indices may retain meaningful variation within a bounded continuous interval. Converting such measurements into binary categories discards within-category information and makes the analysis depend on selected cutoffs. Continuous-response CDMs provide a route to diagnostic classification that retains the observed response scale; the continuous DINA model of Minchen et al. (2017) and the classification model of Oliveira et al. (2023) illustrate this broader development. Bounded responses, as discussed in the preceding chapter, additionally require a

distribution compatible with their finite support.

For the observed response Y'_{ij} of individual i to item j, with theoretical bounds $L < U$, define

$$Y_{ij} = (Y'_{ij} - L)/(U - L).$$

Here the beta response model assumes $0 < Y_{ij} < 1$. Its mean–precision parameterization separates the expected response from conditional variability and accommodates asymmetric densities (Ferrari and Cribari-Neto 2004). The boundedness concerns the observed indicators; the diagnostic attributes remain discrete.

Recent work makes this connection explicit. Jiang et al. (2025) propose a family of CDMs for continuous bounded responses motivated by visual analogue scales and examine parameter recovery, model misspecification, and model selection through simulations and a real-data illustration. Oliveira et al. (2026a) introduce the Bounded DINA (B-DINA) model, which uses two beta response distributions, Bayesian estimation, and posterior probabilities for latent attribute profiles. Their municipal application interprets the attributes as dimensions of economy, education, and health, extending diagnostic classification beyond individual educational assessment. In the B-DINA formulation developed below, the conjunctive gate selects between two item-specific beta distributions, allowing both the mean and precision to differ between regimes. A theoretical and empirical comparison of the models proposed by Jiang et al. (2025) and Oliveira et al. (2026a) is provided in the latter study's Supporting Information (Oliveira et al. 2026b).

The development of bounded CDMs parallels advances in bounded item response theory (IRT), which also provides models for continuous indicators with finite support. Noel and Dauvier (2007) develop a beta item-response model, while Molenaar et al. (2022) present a framework that also accommodates exact zeros and ones. In the bounded IRT models, responses depend on a continuous latent position θ_i and item parameters. Their principal purpose is measurement along a latent scale. A bounded CDM instead targets a discrete vector $\alpha_i \in \{0, 1\}^K$ and its posterior profile probabilities. The choice therefore concerns the desired latent representation, not merely the choice of beta versus another bounded response distribution.

The distinction between continuous measurement and diagnostic classification is particularly relevant to the present application, which uses municipal indicators from the Atlas of Human Development for Southeastern Brazil (Oliveira et al. 2026a). The local data snapshot contains 1,668 municipalities and 19 bounded continuous indicators (Bazán et al. 2026). The indicators describe demographic conditions, educational attainment, income inequality, poverty, income, unemployment, access to water, garbage collection, electricity and sanitation, social vulnerability, and the education, longevity and income components of the Municipal Human Development Index. The observational unit is a municipality, and the indicators play the role of items in the measurement model. The paper's introductory count of 1,688 differs from the 1,668 rows in the supplied data; this document uses the observed dataset size, with the discrepancy discussed in the application section.

Responses are coded on the unit interval so that larger values are intended to indicate more favorable conditions. Adverse indicators therefore require appropriate reversal; for example, the application represents income equality using $1 - \text{GINI}$. These are continuous observed measurements, not observed mastery labels. A substantive Q-matrix links each indicator to one or more of three latent attributes: Economy, Health, and Education. Some indicators are assigned to a single domain, whereas others, such as access to water and garbage collection, are assigned jointly to Economy

and Health. The Q-matrix expresses the proposed measurement relationships; it does not establish causal effects of one domain on another.

The purpose of a bounded CDM is to estimate which combination of favorable latent conditions best describes each municipality. With three binary attributes, there are eight possible profiles, from $(0, 0, 0)$ to $(1, 1, 1)$, in the order Economy, Health, Education. For example, $(1, 0, 1)$ represents favorable economic and educational conditions with an unfavorable health condition. B-DINA estimates posterior probabilities for all eight profiles, as well as their population proportions, from the full vector of continuous indicators. Under its conjunctive assumption, an indicator assigned to several domains has the higher-mean response regime only when all its required attributes are present. The labels favorable and unfavorable are model-based interpretations of the latent classes; they are not externally validated policy thresholds or direct observations of municipal conditions.

A bounded IRT analysis of the same indicators would instead estimate a municipality's position on a continuous latent scale, such as overall socioeconomic development, if a common-trait interpretation were defensible. That output is useful for ranking municipalities and describing differences in degree. A multidimensional bounded IRT model could estimate separate continuous positions for Economy, Health, and Education. The bounded CDM addresses a different substantive question: which combination of domain conditions characterizes a municipality, and how certain is that diagnosis? Municipalities with similar aggregate scores can have different domain profiles, suggesting different priorities for further investigation or policy planning. Reporting profile probabilities can also distinguish a clear diagnosis from an ambiguous one. Such results support descriptive targeting and monitoring; they do not by themselves demonstrate which intervention will improve an indicator.

Both approaches preserve the bounded response scale, so the motivation for bounded CDM in this application rests on the need to identify interpretable combinations of domain conditions. This diagnostic interpretation requires additional substantive and statistical assumptions. Binary attributes must be substantively justified, the Q-matrix must be credible, and the available indicators must distinguish the profiles. DINA additionally assumes that all required attributes are necessary for the higher-mean regime. If the construct varies gradually and the objective is ranking or continuous trait estimation, bounded IRT may be more appropriate. Classification uncertainty should be reported rather than treating an estimated profile as an observed fact. Furthermore, the B-DINA likelihood used here has support strictly inside $(0, 1)$; the endpoint mechanisms of Molenaar et al. (2022) address a separate response-distribution issue and are not automatically supplied by replacing IRT with a CDM.

Against this background, this document develops the beta B-DINA model of Oliveira et al. (2026a) and its supporting information. It covers the response distribution, Q-matrix, Bayesian estimation using JAGS, item discrimination, classification uncertainty. Municipal observations and original application code are supplied in the local snapshot of 1.

5.2 Bounded Continuous Distributions

5.2.1 Beta mean–precision parameterization

For $0 < \mu < 1$ and $\phi > 0$, let

$$Y \sim \text{Beta}(\mu\phi, (1 - \mu)\phi).$$

Its density and moments are

$$f(y \mid \mu, \phi) = \frac{y^{\mu\phi-1}(1-y)^{(1-\mu)\phi-1}}{B(\mu\phi, (1-\mu)\phi)}, \quad 0 < y < 1,$$

$$E(Y) = \mu, \qquad \mathrm{Var}(Y) = \frac{\mu(1-\mu)}{1+\phi}.$$

This parameterization separates the response mean from precision (Ferrari and Cribari-Neto 2004). The parameter μ is the expected response on the unit interval: for example, $\mu = 0.80$ means an expected coded indicator value of 0.80. On the original scale, the expected value is $L + (U - L)\mu$. In the municipal application, where indicators are oriented toward favorable conditions, a larger μ represents a more favorable expected indicator value within the relevant latent response regime. It is not the probability that a municipality possesses an attribute; attribute membership probabilities are estimated separately by the CDM.

The parameter ϕ controls concentration around the mean. At fixed μ, a larger ϕ gives a smaller conditional variance and more tightly clustered responses, whereas a smaller ϕ permits greater heterogeneity. Precision is therefore not an additional mean or a direct measure of attribute mastery. It is also not the reciprocal of the variance: rather,

$\mathrm{Var}(Y) = \mu(1-\mu)/(1+\phi)$ depends on both parameters. Consequently, equal precisions do not imply equal variances when the means differ. In B-DINA, regime-specific precisions allow municipalities in different latent regimes to have different within-regime variability.

In the conventional shape parameterization, $a = \mu\phi$ and $b = (1-\mu)\phi$, so that $\mu = a/(a+b)$ and $\phi = a + b$. The balance between a and b determines the direction of asymmetry: $a < b$ produces positive (right) skewness, $a > b$ produces negative (left) skewness, and $a = b$ produces symmetry. When $a, b > 1$, the density has an interior mode at $(a-1)/(a+b-2)$, which need not coincide with its mean. When both shapes are below one, the density is U-shaped; when one shape is below one, the density can rise sharply near the corresponding boundary. These features explain why the beta family can represent different patterns of bounded indicators without imposing a symmetric normal density.

Figure 1 illustrates the effect of changing the mean while holding precision fixed at $\phi = 12$. It compares $\mu = 0.25$, 0.50, and 0.80, corresponding to shape pairs $(a, b) = (3, 9)$, $(6, 6)$, and $(9.6, 2.4)$, respectively. All three densities have an interior mode. The horizontal axis gives the coded response, and the vertical axis gives its probability density; probabilities correspond to areas under a curve over response intervals.

In Figure 1, the blue solid curve ($\mu = 0.25$) places most mass toward the lower end of the scale and has a longer right tail. The orange dashed curve ($\mu = 0.50$) is symmetric around 0.50. The green dotted curve ($\mu = 0.80$) places most mass toward the upper end and has a longer left tail. Their modes are 0.20, 0.50, and 0.86, illustrating that a density's peak is generally not its expected value. Although the three curves share the same precision, their variances are approximately 0.0144, 0.0192, and 0.0123, respectively. The central curve therefore has the greatest variance, because $\mu(1-\mu)$ is largest at $\mu = 0.50$. The figure illustrates distributions for chosen parameter values; it does not display fitted municipal profiles.

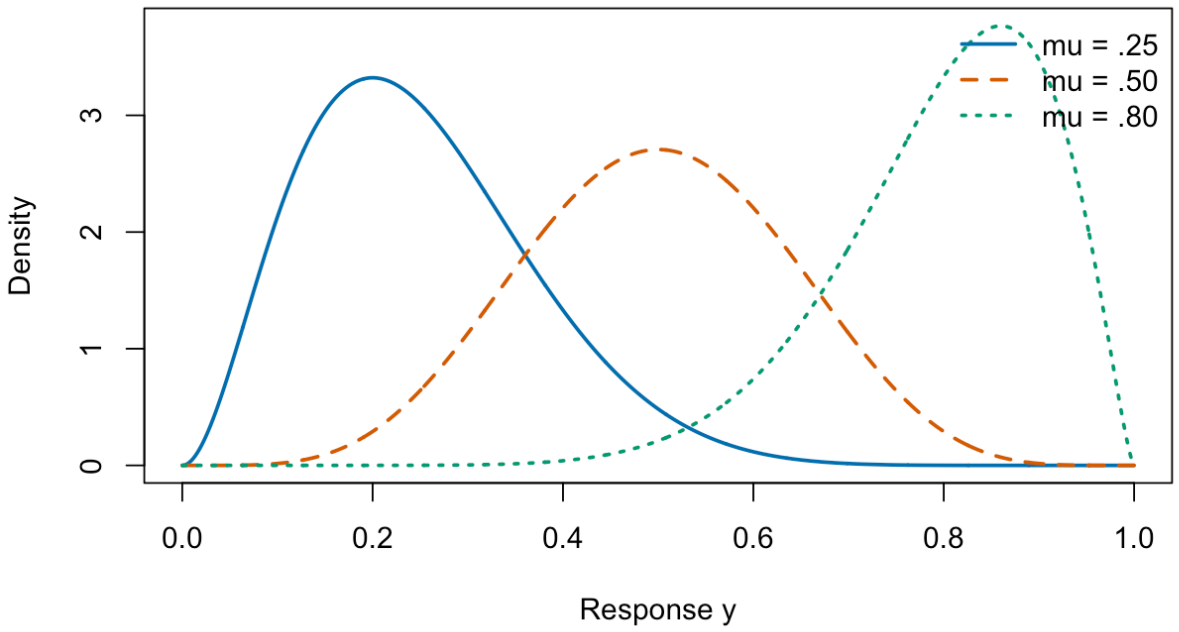


Figure 1: Beta densities at three means and common precision 12.

5.2.2 Moment restrictions and endpoints

The finite response scale imposes restrictions on the moments, regardless of the chosen distribution. As discussed in the chapter on bounded IRT, if $Y \in [0,1]$, with mean $m = E(Y)$ and variance $v = \mathrm{Var}(Y)$, then

$$0 \leq v \leq m(1-m) \leq \frac{1}{4}.$$

To obtain the first upper bound, note that $Y^2 \leq Y$, and hence $v = E(Y^2) - m^2 \leq m - m^2$. The second follows from $m(1-m) = 1/4 - (m-1/2)^2$. Thus, variability is most restricted when the mean is close to either endpoint. Equality $v = 0$ corresponds to a constant response, whereas $v = m(1-m)$ requires all probability to lie at zero and one, as in a Bernoulli distribution. The maximum variance $1/4$ occurs when these endpoints each have probability $1/2$.

For a beta distribution with finite positive shape parameters, the moment conditions are strict:

$$0 < m < 1, \qquad 0 < v < m(1-m).$$

Conversely, any pair satisfying these conditions determines a unique beta mean–precision parameterization through

$$\mu = m, \qquad \phi = \frac{m(1-m)}{v} - 1 > 0,$$

with shapes $a = m\phi$ and $b = (1-m)\phi$. At fixed mean, the beta variance approaches zero as $\phi \to \infty$ and approaches $m(1-m)$ as $\phi \to 0^+$; neither limiting variance is attained by a beta distribution with finite positive parameters. These relationships concern population moments. Empirical moments used for initialization must satisfy the same strict feasibility conditions to produce valid positive beta shapes; they do not establish that the beta likelihood fits the data.

Endpoint observations require separate attention. A beta density can become unbounded near zero or one when the corresponding shape parameter is below one, but this does not create probability mass at either endpoint:

$$P(Y = 0) = P(Y = 1) = 0.$$

The B-DINA formulation developed here therefore assumes responses strictly inside $(0,1)$. This restriction belongs to its beta observation model, rather than to cognitive diagnosis itself. Likewise, bounded IRT does not automatically accommodate endpoints: the zero-and-one-inflated models of Molenaar et al. (2022) add explicit probability masses to an interior response distribution. An analogous extension for a bounded CDM would require its own specification of endpoint probabilities and their relationship to latent attribute profiles.

The treatment of observed zeros and ones should reflect how the indicators were generated. Genuine boundary values may call for an endpoint-mass model, whereas rounding or censoring may call for a likelihood based on the recorded measurement intervals. Moving endpoints into the interior is a preprocessing choice that changes the data and must be documented and assessed for sensitivity; it is not part of the B-DINA likelihood. Rescaling should use theoretical bounds $L < U$, rather

than the sample minimum and maximum, which can manufacture endpoint observations and make the scale depend on the particular sample. These checks are especially relevant for municipal proportions and indices, where a boundary value may have a substantive meaning distinct from an interior value close to that boundary.

5.3 Bounded Continuous Cognitive Diagnosis Models

5.3.1 Attributes, profiles, and the Q-matrix

A cognitive diagnosis model starts with a substantive description of the latent skills or attributes that the indicators are intended to measure. A skill is latent because it is not directly observed: its presence is inferred from a unit's pattern of responses across several items. In an educational assessment, these attributes might represent specific abilities, such as recognizing fractions or performing arithmetic operations. In the municipal application, the broader term *attribute* is more appropriate: Economy, Health, and Education represent unobserved domain conditions rather than skills performed by a municipality. The observed bounded indicators provide evidence about these conditions, but are not themselves the latent attributes.

For unit i, let $\alpha_{ik} = 1$ denote possession or mastery of attribute k, and let $\alpha_{ik} = 0$ denote its absence or nonmastery. These binary states are a modeling assumption; they do not imply that the observed responses are binary. The vector

$$\alpha_i = (\alpha_{i1}, \dots, \alpha_{iK})^\top \in \{0, 1\}^K$$

is the unit's latent attribute profile. With K binary attributes there are $C = 2^K$ possible profiles, indexed by $c = 1, \dots, C$, when all combinations are allowed. Different units may share a profile, and the model estimates posterior probabilities of profile membership rather than observing mastery directly. In the municipal setting, the states are interpreted as favorable or unfavorable latent conditions within each domain, conditional on the model and its coding conventions.

The Q-matrix supplies the link between these latent attributes and the observed indicators (Junker and Sijtsma 2001; Torre 2011). Let $Q = (q_{jk})$ be a $J \times K$ binary matrix, with one row per item and one column per attribute. The entry $q_{jk} = 1$ indicates that item j requires attribute k, whereas $q_{jk} = 0$ indicates that this attribute is not required by that item. For example, the row $(1, 0, 1)$ assigns an indicator to Economy and Education but not Health. Q is treated as known in this document: it encodes substantive hypotheses about the measurement relationships and should be justified by domain knowledge and assessed against the data. Unlike the unit-specific profile α_i, it is shared across all units. Q specifies which attributes enter an item's response model; the CDM specifies how those attributes combine.

A proposed Q-matrix should begin with clearly defined attributes and a review of what each indicator measures. Domain experts can assign candidate rows, document the rationale for single- and multiple-attribute assignments, and resolve disagreements using the indicator definitions. For municipal data, this means examining whether an indicator reflects economic, health, or educational conditions, and whether a joint assignment supports the conjunctive interpretation. These assignments are measurement hypotheses, not conclusions drawn solely from correlations.

Empirical validation can complement this substantive construction. Torre (2008) proposes a DINA-based Q-matrix validation method that can be combined with expert knowledge. Torre and Chiu

(2016) extend empirical validation to a broader class of models within the G-DINA framework using a discrimination index to identify and replace potentially misspecified entries. Candidate revisions should be assessed for substantive plausibility, model fit, and classification stability. These methods were developed for their specified response models; applying them to continuous beta responses requires an appropriate adaptation rather than direct reuse of binary-response criteria. The distinction between proposing Q, validating its entries, and establishing statistical identifiability is developed in Section 3.3.

In DINA, the combination rule is conjunctive: a unit must possess every attribute required by the item to enter its mastery response regime (Junker and Sijtsma 2001). Writing $\mathbf{q}_j$ for the column vector containing row j of Q, the ideal-response indicator is

$$\eta_{ij} = \prod_{k:q_{jk}=1} \alpha_{ik} = \mathbf{1}\{\alpha_i^\top \mathbf{q}_j = \mathbf{q}_j^\top \mathbf{q}_j\}.$$

A unit with only some of the required attributes remains in the $\eta_{ij} = 0$ group. This is a strong substantive assumption; it should be assessed against compensatory alternatives. Here all-zero Q rows are rejected because their gate is always one and their nonmastery regime cannot be learned.

Table 1 presents an illustrative Q-matrix with $J = 12$ items and $K = 3$ attributes, labeled Economy, Health, and Education to parallel the application. Each row describes an item's requirements, rather than a municipality's characteristics. A value of one links the item to the corresponding attribute; a zero indicates that the attribute is not required by that item in the model. Items 1–9 consist of three repeated sets of single-attribute items: items 1, 4, and 7 measure Economy, items 2, 5, and 8 measure Health, and items 3, 6, and 9 measure Education. Items 10–12 require the pairs Economy–Health, Economy–Education, and Health–Education, respectively. Repeated Q rows mean that items share attribute requirements, not that their beta response parameters or observed values are identical. This matrix is a teaching example; the application uses the separate 19-indicator municipal Q-matrix.

Table 2 translates these requirements into ideal-response indicators for all $C = 2^3 = 8$ attribute profiles. Its first column gives the profile index $c = 1, \ldots, C$; these indices identify the profiles and do not rank their development levels. The next three columns give the latent attribute values in the order Economy, Health, Education, while its last three columns give the profile-specific gates $\eta_{c,10}$, $\eta_{c,11}$, and $\eta_{c,12}$ for the paired items in Table 1. Suppressing the profile index in this illustration, the gates reduce to

$$\eta_{10} = \alpha_1\alpha_2, \qquad \eta_{11} = \alpha_1\alpha_3, \qquad \eta_{12} = \alpha_2\alpha_3.$$

Thus, Table 1 specifies which attributes an item requires, and Table 2 shows whether each possible profile satisfies those requirements.

Table 1: Illustrative Q-matrix; not the municipal Q-matrix.

	Economy	Health	Education
Item 1	1	0	0
Item 2	0	1	0
Item 3	0	0	1
Item 4	1	0	0

	Economy	Health	Education
Item 5	0	1	0
Item 6	0	0	1
Item 7	1	0	0
Item 8	0	1	0
Item 9	0	0	1
Item 10	1	1	0
Item 11	1	0	1
Item 12	0	1	1

Table 2: Attribute profiles and ideal-response indicators for items 10–12 of the illustrative Q-matrix.

Profile (c)	Economy	Health	Education	Item 10	Item 11	Item 12
1	0	0	0	0	0	0
2	1	0	0	0	0	0
3	0	1	0	0	0	0
4	1	1	0	1	0	0
5	0	0	1	0	0	0
6	1	0	1	0	1	0
7	0	1	1	0	0	1
8	1	1	1	1	1	1

For example, the profile $(1, 0, 1)$ in Table 2 has the Economy and Education attributes but lacks Health. It therefore satisfies item 11's requirements, giving $(\eta_{10}, \eta_{11}, \eta_{12}) = (0, 1, 0)$. The profile $(1, 1, 1)$ satisfies all three paired items, whereas $(0, 0, 0)$ and each profile with only one attribute satisfy none. Consequently, the paired items alone cannot distinguish all eight profiles. The single-attribute items in Table 1 provide additional ideal-response distinctions: their gates equal the corresponding individual attributes.

An ideal-response indicator of one does not mean an observed score of one, and an indicator of zero does not mean an observed score of zero. In B-DINA, η_{ij} selects the item's mastery or nonmastery beta response regime. Both regimes generate continuous values inside $(0, 1)$, and their densities may overlap. The tables describe the model's structural mapping from profiles to regimes; actual classification uses the observed responses and estimates posterior profile probabilities, retaining uncertainty.

5.3.2 Conditional and marginal response models

Let $Y = (Y_{ij})$ be the $N \times J$ matrix of bounded responses, with $0 < Y_{ij} < 1$. For a fixed Q-matrix, the beta B-DINA model of Oliveira et al. (2026a) combines two item-specific beta regimes with a population distribution over latent attribute profiles. Write $\alpha^{(c)}$ for profile c, $c = 1, \ldots, C$, where $C = 2^K$, and let Z_i be unit i's profile index. The item parameters are

$$\Omega_j = (\mu_{j0}, \mu_{j1}, \phi_{j0}, \phi_{j1})^\top, \qquad 0 < \mu_{j0} \leq \mu_{j1} < 1, \quad \phi_{j0}, \phi_{j1} > 0.$$

The indices 0 and 1 label the nonmastery and mastery regimes. Let $\pi = (\pi_1, \ldots, \pi_C)^\top$ denote population profile proportions, with $\pi_c \geq 0$ and $\sum_c \pi_c = 1$; a full-support specification assumes $\pi_c > 0$ for all c. These proportions are distinct from beta response means and unit-specific posterior membership probabilities.

Starting with the observed response, the model can be written hierarchically as follows, for $i = 1, \dots, N$ and $j = 1, \dots, J$, conditional on Q:

$$\begin{aligned}
Y_{ij} \mid Z_i, \Omega_j, Q &\overset{\text{ind}}{\sim} \text{Beta}\big(\mu_{ij}\phi_{ij}, (1-\mu_{ij})\phi_{ij}\big), && 0 < Y_{ij} < 1, \\
\mu_{ij} &= (1-\eta_{ij})\mu_{j0} + \eta_{ij}\mu_{j1}, && \text{conditional mean}, \\
\phi_{ij} &= (1-\eta_{ij})\phi_{j0} + \eta_{ij}\phi_{j1}, && \text{conditional precision}, \\
\eta_{ij} &= \prod_{k:q_{jk}=1} \alpha_{ik}, && \text{conjunctive gate}, \\
\alpha_i &= \alpha^{(Z_i)}, && \text{attribute profile}, \\
Z_i \mid \pi &\overset{\text{ind}}{\sim} \text{Categorical}(\pi_1, \dots, \pi_C), && \text{latent profile allocation}.
\end{aligned}$$

The first line specifies the distribution of the observed bounded response. The next two lines define its conditional mean and precision, which depend on the conjunctive gate. The gate is determined by the unit's latent attributes, and the final two lines specify the profile and its population distribution. The mean, precision, gate, and profile relationships are deterministic; the stochastic layers are the response distribution and the profile allocation. Responses are independent across items conditional on the unit's profile and the model parameters. For municipalities, the response layer describes the 19 coded indicators, while the latent profile encodes Economy, Health, and Education.

For the Bayesian implementation used below, this hierarchy is completed by the priors from Oliveira et al. (2026a):

$$\begin{aligned}
\pi &\sim \text{Dirichlet}(1, \dots, 1), \\
U_{j1}, U_{j2} &\overset{\text{iid}}{\sim} \text{Uniform}(0,1), \\
\mu_{j0} &= \min(U_{j1}, U_{j2}), \qquad \mu_{j1} = \max(U_{j1}, U_{j2}), \\
\phi_{j0}, \phi_{j1} &\overset{\text{iid}}{\sim} \text{Gamma}(0.01, 0.01).
\end{aligned}$$

The Gamma parameters are shape and rate. Auxiliary means and precisions are independent across items and independent of π before the ordering transformation. Sorting induces dependence between the two ordered means and gives $0 < \mu_{j0} < \mu_{j1} < 1$ almost surely. Q is fixed and has no prior in this specification. The estimation section discusses the implications of these priors and their sensitivity. The conditional and marginal expressions below make explicit the likelihood implied by the hierarchy.

To express the likelihood using profile indices, define

$$\eta_{cj} = \prod_{k:q_{jk}=1} \alpha_k^{(c)}, \qquad \eta_{ij} = \eta_{Z_i, j}.$$

Conditional on $Z_i = c$, the gate selects regime $r = \eta_{cj}$, giving

$$E(Y_{ij} \mid Z_i = c, \Omega_j, Q) = \mu_{j\eta_{cj}}, \qquad \text{Var}(Y_{ij} \mid Z_i = c, \Omega_j, Q) = \frac{\mu_{j\eta_{cj}}(1-\mu_{j\eta_{cj}})}{1+\phi_{j\eta_{cj}}}.$$

The mean ordering makes the mastery regime's expected response at least as large as that of the nonmastery regime, consistent with favorable indicator coding. The general constraint permits equal means, whereas the sorted-mean prior above gives strict ordering almost surely. The precisions control within-regime concentration and are allowed to differ. Thus, a higher mean does not necessarily imply a smaller variance.

All profiles with the same gate value for an item share its conditional response distribution. In particular, profiles with partial mastery use the same nonmastery regime as profiles lacking all required attributes. This is the substantive DINA restriction; the model does not assign a separate beta distribution to every partial-mastery pattern. For an item requiring Economy and Education, for example, both $(1, 0, 1)$ and $(1, 1, 1)$ select regime 1, while $(1, 1, 0)$ selects regime 0. Conditional on a profile, the regime is selected deterministically, and the observed response remains continuous.

For $r \in \{0, 1\}$, write its density as

$$f_{jr}(y \mid \Omega_j) = \frac{y^{\mu_{jr}\phi_{jr}-1}(1-y)^{(1-\mu_{jr})\phi_{jr}-1}}{B\big(\mu_{jr}\phi_{jr}, (1-\mu_{jr})\phi_{jr}\big)}, \qquad 0 < y < 1.$$

Let $\Omega = (\Omega_1, \dots, \Omega_J)$. Assuming local independence, the joint conditional density of the response vector $\mathbf{y}_i = (y_{i1}, \dots, y_{iJ})^\top$ is

$$p(\mathbf{y}_i \mid Z_i = c, \Omega, Q) = \prod_{j=1}^{J} f_{j\eta_{cj}}(y_{ij} \mid \Omega_j) = \prod_{j=1}^{J} f_{j0}(y_{ij} \mid \Omega_j)^{1-\eta_{cj}} f_{j1}(y_{ij} \mid \Omega_j)^{\eta_{cj}}.$$

Integrating out the unobserved profile yields the finite-mixture density

$$p(\mathbf{y}_i \mid \Omega, \pi, Q) = \sum_{c=1}^{C} \pi_c \prod_{j=1}^{J} f_{j\eta_{cj}}(y_{ij} \mid \Omega_j).$$

For independent units, the observed-data likelihood is therefore

$$L(\Omega, \pi; Y, Q) = \prod_{i=1}^{N} \left\{ \sum_{c=1}^{C} \pi_c \prod_{j=1}^{J} f_{j\eta_{cj}}(y_{ij} \mid \Omega_j) \right\}.$$

The item product remains inside the profile sum because each unit has one profile shared across all its responses. The mixture is over complete response vectors, rather than over independently selected item regimes. Although the responses are independent conditional on the profile, they can be dependent marginally through their shared latent attributes.

For fixed model parameters, Bayes' rule gives the membership probability

$$P(Z_i = c \mid \mathbf{y}_i, \Omega, \pi, Q) = \frac{\pi_c \prod_{j=1}^{J} f_{j\eta_{cj}}(y_{ij} \mid \Omega_j)}{\sum_{d=1}^{C} \pi_d \prod_{j=1}^{J} f_{j\eta_{dj}}(y_{ij} \mid \Omega_j)}.$$

In Bayesian estimation, these probabilities are averaged over the posterior of the item parameters and profile proportions to account for parameter uncertainty. For the municipal application, the

resulting probabilities quantify support for each Economy–Health–Education profile. Posterior computation and model checking are developed in the estimation section below.

5.3.3 Identification and interpretation

Identifiability means that distinct model specifications cannot generate the same distribution of observed responses. Two questions must be separated: whether item parameters and class proportions are identifiable when Q is fixed, and whether Q itself can be learned jointly with those parameters. An estimated Q may fit well without being uniquely recoverable from the data.

For binary DINA with a fixed Q, Gu and Xu (2019) establish necessary and sufficient conditions under their parameter assumptions, including positive probabilities for all profiles and strictly separated item response probabilities. After permuting rows, Q must contain an identity block I_K, so it can be written as

$$Q = \begin{pmatrix} I_K \\ Q^* \end{pmatrix}.$$

This is the completeness condition: each attribute has a single-attribute item. In addition, every attribute must be required by at least three items, and the columns of the remaining matrix Q^* must be distinct. These requirements provide repeated measurement and distinguish attributes beyond the identity block. Completeness alone distinguishes ideal-response patterns but is insufficient to identify all unknown parameters in the noisy model. The illustrative Q in Table 1 satisfies these structural checks: it contains three identity blocks, each attribute occurs in five items, and removing one identity block leaves distinct columns.

When Q is unknown, Gu and Xu (2021) study joint identifiability of Q and model parameters in restricted latent class models, including DINA. Attribute column permutations must be accounted for, because relabeling unnamed attributes leaves the observational model unchanged. Domain definitions provide the substantive labels Economy, Health, and Education; the data alone do not supply those meanings. Joint identifiability is therefore a stronger question than parameter identifiability conditional on a supplied Q, and its conditions depend on the specified model and parameter space.

These binary-response results are useful design references, but do not by themselves prove identifiability of the beta B-DINA likelihood used here. The discussion in Oliveira et al. (2026a) considers analogous structural requirements and the limits of B-DINA-specific theory. Ordering the beta means anchors the two response regimes, but the weak constraint $\mu_{j0} \leq \mu_{j1}$ permits equality and does not establish identification of every item parameter or class proportion. A larger sample, informative priors, or successful MCMC convergence cannot substitute for a model-specific identifiability argument. Practical assessment should also consider weak separation between regime densities and uncertainty in rarely represented profiles.

Structural identifiability concerns uniqueness at the level of the population response distribution. It does not guarantee precise estimation in a finite sample or certain classification of each municipality. Even an identifiable model can have substantial uncertainty when classes are small or the regime densities overlap. If two profiles have identical gate vectors across all items, their conditional response distributions coincide, and the likelihood can identify only their combined population proportion. Distinct gate vectors avoid this particular ambiguity, but do not alone

establish identifiability of all unknown parameters. Likewise, if an item's two beta regimes are identical, that item carries no information about its gate.

Finally, mean ordering is an expectation restriction, not a statement that every response from the mastery regime exceeds a response from the nonmastery regime. With unequal precisions, beta CDFs can cross even when $\mu_{j1} > \mu_{j0}$, so first-order stochastic dominance need not hold. For municipalities, the attributes should therefore be interpreted as model-based domain conditions supported by a pattern of indicators, with posterior uncertainty reported alongside any assigned profile.

5.3.4 Item parameters and discrimination

The beta B-DINA model retains the distinction between mastery and nonmastery from traditional DINA, but replaces binary success probabilities with parameters of continuous response distributions. In binary DINA, the traditional item parameters are guessing g_j and slipping s_j (Junker and Sijtsma 2001):

$$g_j = P(X_{ij} = 1 \mid \eta_{ij} = 0), \qquad s_j = P(X_{ij} = 0 \mid \eta_{ij} = 1),$$

where $X_{ij} \in \{0, 1\}$ denotes a binary item response. Thus, $E(X_{ij} \mid \eta_{ij} = 0) = g_j$ and $E(X_{ij} \mid \eta_{ij} = 1) = 1 - s_j$. Guessing represents success without all required attributes, and slipping represents failure despite possessing them. These are probabilistic labels, not necessarily literal behavioral mechanisms. Under $g_j < 1 - s_j$, the conditional success-probability gap $1 - s_j - g_j$ measures how strongly the item separates its two regimes.

In beta B-DINA, the corresponding conditional expectations are μ_{j0} and μ_{j1}. They play analogous roles to g_j and $1-s_j$, respectively, but are expected continuous indicator values, not probabilities of a correct response. Likewise, $1-\mu_{j1}$ is a mean shortfall from the upper bound, not a slipping probability. The four item parameters have complementary interpretations: μ_{j0} describes the expected response when at least one required attribute is absent, μ_{j1} describes it when all are present, and ϕ_{j0} and ϕ_{j1} describe concentration within the respective regimes. For favorably coded municipal indicators, these means summarize expected conditions and the precisions describe residual heterogeneity among municipalities assigned to each regime. Higher precision means less variability at a fixed mean; comparing regime variances requires both their means and precisions.

Traditional IRT uses a different item interpretation. In the bounded IRT parameterization $a_j\theta_i + b_j$, a_j controls sensitivity to a continuous latent trait and b_j is a location or easiness intercept (Molenaar et al. 2022). Under a logistic mean link, $\theta_i = -b_j/a_j$ gives a conditional mean of 0.5. B-DINA has no continuous θ_i or corresponding item slope and threshold. Its parameters compare discrete attribute regimes, so IRT difficulty and discrimination interpretations cannot be transferred directly to μ_{jr} or ϕ_{jr}.

Following Oliveira et al. (2026a), the B-DINA item difficulty parameter is defined as the difference between the mastery and nonmastery conditional means:

$$\lambda_j = \mu_{j1} - \mu_{j0}.$$

This definition expresses item difficulty through the expected response contrast associated with satisfying all of the item's attribute requirements. The parameter lies in $[0, 1)$ under the stated

constraints and is positive under strict mean ordering. A larger λ_j indicates that the two attribute regimes differ more strongly in expected response: units lacking at least one required attribute have a larger expected shortfall relative to units possessing them all. A smaller λ_j indicates a smaller expected shortfall, so the item's expected response depends less strongly on complete mastery. In this sense, difficulty describes the magnitude of the mastery-related response contrast, rather than the overall level of the indicator.

For a favorably coded municipal indicator, suppose $\mu_{j0} = 0.40$ and $\mu_{j1} = 0.70$. Then $\lambda_j = 0.30$: municipalities lacking at least one required favorable domain condition have an expected indicator value 0.30 below that of municipalities possessing all required conditions. For a proportion, this is a difference of 30 percentage points; on the original bounded scale, it is $(U - L)\lambda_j$. The contrast describes the fitted regimes and does not estimate the causal effect of changing a municipality's attributes or the effort required for a policy intervention.

The term *item difficulty* is used here in this specific B-DINA sense. Unlike an IRT location parameter, λ_j is not a threshold on a continuous latent scale. It also does not indicate whether expected responses are high or low in both regimes: mean pairs $(0.20, 0.50)$ and $(0.60, 0.90)$ have the same difficulty $\lambda_j = 0.30$ but different response levels. Interpretation should therefore report λ_j together with μ_{j0} and μ_{j1}. Because difficulty is defined as a mean contrast, it also conveys separation of regime expectations; the distributional measures below assess separation of the full response distributions.

The mean gap alone does not describe how much the two response distributions overlap. Two items with the same gap may provide different classification information because their precisions differ. Moreover, equal means with unequal precisions can yield distinct densities despite $\lambda_j = 0$. Distributional discrimination measures therefore complement the mean gap. Let f_{jr} and F_{jr} denote the beta density and cumulative distribution function for regime $r \in \{0, 1\}$. The Kolmogorov–Smirnov distance is

$$KS_j = \sup_{0<y<1} |F_{j0}(y) - F_{j1}(y)|.$$

It measures the largest difference between regime probabilities of a response at or below a threshold. A value of zero means identical distributions; larger values, up to one, indicate greater separation by this criterion. The absolute difference remains appropriate when the CDFs cross.

A second comparison uses the Bhattacharyya affinity. With $a_{jr} = \mu_{jr}\phi_{jr}$ and $b_{jr} = (1 - \mu_{jr})\phi_{jr}$,

$$BC_j = \int_0^1 \sqrt{f_{j0}(y) f_{j1}(y)}\, dy = \frac{B\big((a_{j0} + a_{j1})/2, (b_{j0} + b_{j1})/2\big)}{\sqrt{B(a_{j0}, b_{j0}) B(a_{j1}, b_{j1})}}.$$

Affinity lies between zero and one and equals one for identical densities. To express separation on an increasing scale, this document uses the squared Hellinger distance and Hellinger distance

$$H_j^2 = 1 - BC_j, \qquad H_j = \sqrt{1 - BC_j}.$$

Both are zero for identical distributions and approach one as affinity approaches zero. The integral printed as Hellinger in Section 3.3 of Oliveira et al. (2026a) is the affinity, which decreases with

separation; the reported simulation values are consistent with H^2, rather than H. The convention used here is therefore stated explicitly.

Table 3 presents three illustrative types of beta B-DINA items with low, medium, and high separation between their nonmastery and mastery response regimes, using the simulation settings of Oliveira et al. (2026a). Each row represents an item parameter configuration, rather than a latent profile or an observed municipal indicator. The columns μ_{j0} and μ_{j1} give the two conditional means, ϕ_{j0} and ϕ_{j1} give their precisions, and λ_j, KS_j, H_j^2, and H_j summarize regime separation.

All three item types have $\mu_{j0} = 0.25$ and $\phi_{j0} = 5$. For the low-separation item, $\mu_{j1} = 0.40$ gives a mean gap of 0.15, so the regimes provide relatively similar expected responses. The medium-separation item has $\mu_{j1} = 0.60$ and a gap of 0.35, while the high-separation item has $\mu_{j1} = 0.80$ and a gap of 0.55. Mastery precision also increases from 7 to 10 to 12. The distributional distances in Table 3 capture the combined changes in means and precisions, and the following figure shows the resulting paired densities. These configurations illustrate different levels of item discrimination; they do not define universal cutoffs for classifying items as low, medium, or high.

Greater regime separation makes an item's response more informative about whether its required attributes are jointly present. It does not give a probability of correct profile classification, which also depends on the Q-matrix, the other indicators, and population profile proportions. Because means and precisions change together in these examples, the comparison does not isolate the effect of either parameter alone.

Table 3: Illustrative beta B-DINA item types: regime-specific means and precisions, difficulty, and distributional discrimination measures.

Item type	μ_{j0}	μ_{j1}	ϕ_{j0}	ϕ_{j1}	λ_j	KS_j	H_j^2	H_j
Low	0.25	0.4	5	7	0.15	0.355	0.108	0.329
Medium	0.25	0.6	5	10	0.35	0.701	0.423	0.650
High	0.25	0.8	5	12	0.55	0.902	0.745	0.863

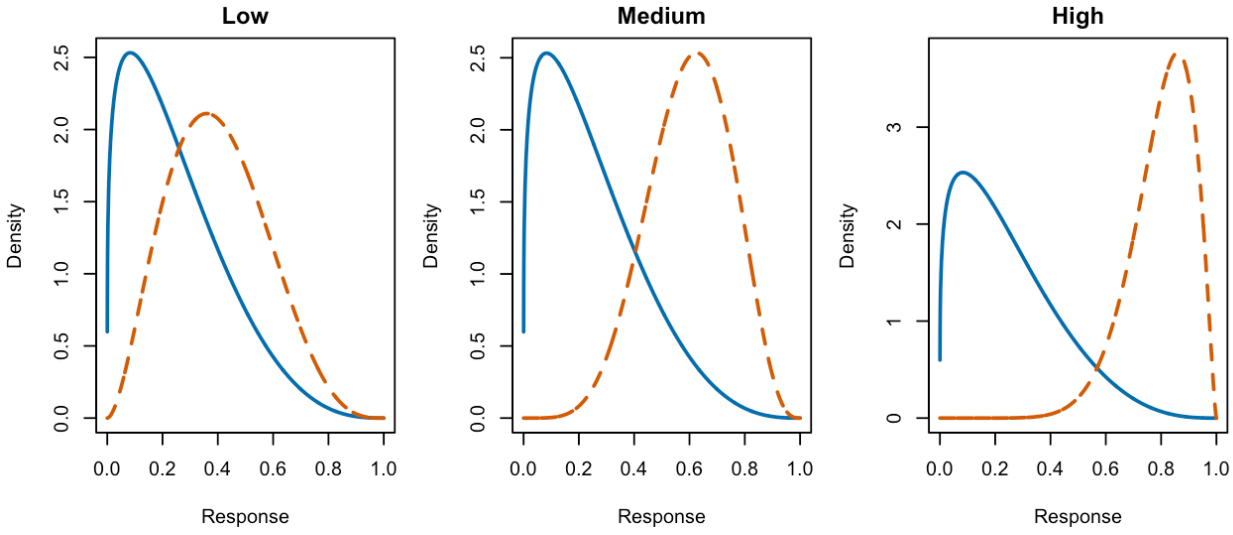


Figure 2: Nonmastery (solid) and mastery (dashed) beta densities for the paper's three scenarios.

5.4 Bayesian Estimation and Model Checking

The estimation strategy in Oliveira et al. (2026a) is Bayesian: item parameters, population profile proportions, and latent allocations are inferred jointly from their posterior distribution. The paper supplies JAGS model code and reproducible R scripts, rather than a dedicated fitting package for beta B-DINA. Accordingly, this document uses a self-contained implementation, not an assumed `BoundedCDM` package. `R2jags` is the general R interface to the JAGS engine; it does not provide a model-specific B-DINA estimator (Su and Yajima 2024).

A classical estimator for this beta B-DINA specification is not developed in the paper or implemented here. This is a statement about the scope of the available implementation, not a claim that classical estimation is impossible. The marginal likelihood in Section 3.2 could in principle be maximized, for example using an expectation–maximization strategy with numerical item-parameter updates. The present Bayesian approach instead propagates uncertainty in item parameters and population proportions into profile membership probabilities, and produces posterior intervals and predictive checks within the same framework.

JAGS is chosen to follow the paper's augmented hierarchy and code. It can retain the categorical allocations Z_i explicitly alongside continuous item parameters, and its BUGS syntax closely matches the model in Section 3.2 (Plummer 2017). Stan is also a possible implementation, but does not sample discrete parameters: Z_i would have to be summed out using the profile-mixture likelihood, with profile probabilities recovered from the continuous posterior draws (Stan Development Team 2026). That reformulation can be computationally advantageous, so the choice of JAGS here is justified by fidelity to the paper and direct representation of the latent allocations, rather than a claim of superior speed or accuracy over Stan. A Stan version would also need a suitable ordered-mean parameterization with the intended prior, instead of carrying over the auxiliary sorting operation unchanged.

5.4.1 Priors and posterior

Bayesian specification requires more than a response likelihood: it also requires a prior distribution for every unknown stochastic quantity. In beta B-DINA, the unknowns are the item means and precisions, the population profile proportions, and the unit-specific profile allocations. Q and the list of possible attribute profiles are fixed. Following Oliveira et al. (2026a), this document uses independent prior blocks for the item parameters and profile proportions. The allocations are conditionally independent given the proportions, rather than independent of them. These assumptions define how information is shared before observing the responses and must be considered part of the model.

The population profile probabilities have the prior

$$\pi \sim \text{Dirichlet}(\delta_1, \ldots, \delta_C), \qquad \delta_c = 1,$$

with $Z_i \mid \pi \sim \text{Categorical}(\pi)$. For general positive concentrations, writing $\delta_+ = \sum_c \delta_c$,

$$p(\pi) = \frac{\Gamma(\delta_+)}{\prod_c \Gamma(\delta_c)} \prod_{c=1}^{C} \pi_c^{\delta_c - 1}, \qquad E(\pi_c) = \frac{\delta_c}{\delta_+}, \qquad \text{Var}(\pi_c) = \frac{\delta_c(\delta_+ - \delta_c)}{\delta_+^2(\delta_+ + 1)}.$$

Setting all concentrations to one gives a uniform density over the probability simplex, with $E(\pi_c) = 1/C$. It expresses symmetry among profiles, not a requirement that their population proportions be equal. For the municipal model, each of the eight profiles has prior mean probability 1/8. Because the proportions sum to one, they are dependent; for $C > 1$, a marginal component has distribution $\text{Beta}(1, C-1)$ rather than $\text{Uniform}(0,1)$. The prior assigns positive mass to neighborhoods of all interior probability vectors, while exact zero proportions have probability zero. Thus, all profiles are retained in the full-support Bayesian model. A symmetric Dirichlet with concentration below one would favor sparser proportions, whereas larger concentrations would concentrate them toward equal proportions. Such changes alter assumptions about the latent population and should be examined through sensitivity analysis.

For each item, the paper generates ordered means from two auxiliary draws:

$$U_{j1}, U_{j2} \overset{\text{iid}}{\sim} \text{Uniform}(0,1), \qquad \mu_{j0} = \min(U_{j1}, U_{j2}), \quad \mu_{j1} = \max(U_{j1}, U_{j2}).$$

The resulting joint prior is

$$p(\mu_{j0}, \mu_{j1}) = 2\,\mathbf{1}\{0 < \mu_{j0} < \mu_{j1} < 1\}.$$

It gives strict mean ordering almost surely, assigning the higher expected response to the mastery regime. The prior is uniform over the ordered triangle, but its component distributions are not uniform:

$$\mu_{j0} \sim \text{Beta}(1,2), \qquad \mu_{j1} \sim \text{Beta}(2,1), \qquad E(\mu_{j0}) = \frac{1}{3}, \quad E(\mu_{j1}) = \frac{2}{3}.$$

The ordered means are dependent. In particular, this construction also induces a prior on the item difficulty parameter $\lambda_j = \mu_{j1} - \mu_{j0}$:

$$p(\lambda_j) = 2(1-\lambda_j), \qquad 0 < \lambda_j < 1, \qquad E(\lambda_j) = \frac{1}{3}.$$

Thus, small mean gaps have higher prior density than large gaps, even though the auxiliary means were drawn uniformly. Sorting is a substantive orientation constraint with consequences for prior expectations; it is not an assertion that the regimes are well separated in the observed data. There is no hierarchical pooling across items in this prior: each ordered pair is drawn independently using the same distribution.

The regime-specific precisions are independently assigned

$$\phi_{jr} \sim \text{Gamma}(a_\phi, b_\phi), \qquad a_\phi = b_\phi = 0.01, \quad r \in \{0,1\},$$

using the shape–rate density

$$p(\phi_{jr}) = \frac{b_\phi^{a_\phi}}{\Gamma(a_\phi)} \phi_{jr}^{a_\phi - 1} \exp(-b_\phi \phi_{jr}), \qquad \phi_{jr} > 0.$$

Its mean is $a_\phi/b_\phi = 1$ and its variance is $a_\phi/b_\phi^2 = 100$. The density is strongly concentrated near zero and has a long right tail; its large variance alone does not establish a weakly informative prior. Very small precision places the conditional beta variance close to $\mu_{jr}(1-\mu_{jr})$ and can yield pronounced density near the boundaries, whereas large precision gives tightly concentrated responses. The prior therefore affects the plausible shapes and overlap of the response regimes, especially when few units occupy a regime. Means and precisions are independent before observing data, but generally become dependent in the posterior through the beta likelihood.

Combining the prior blocks gives

$$p(\Omega, \pi) = p(\pi) \prod_{j=1}^{J} \left\{ 2\,\mathbf{1}(0 < \mu_{j0} < \mu_{j1} < 1) \prod_{r=0}^{1} p(\phi_{jr}) \right\}.$$

Let $\mathbf{Z} = (Z_1, \ldots, Z_N)$. The augmented posterior follows directly from Bayes' rule:

$$p(\Omega, \pi, \mathbf{Z} \mid Y, Q) \propto p(\Omega, \pi) \prod_{i=1}^{N} \left\{ \pi_{Z_i} \prod_{j=1}^{J} f_{j\eta_{Z_i,j}}(Y_{ij} \mid \Omega_j) \right\}.$$

The response factors contribute information about item regimes, the allocation factors link units to population profile proportions, and the priors specify support and regularization. Summing over allocations gives the marginal posterior of the continuous parameters:

$$p(\Omega, \pi \mid Y, Q) \propto p(\Omega, \pi) \prod_{i=1}^{N} \left\{ \sum_{c=1}^{C} \pi_c \prod_{j=1}^{J} f_{j\eta_{cj}}(Y_{ij} \mid \Omega_j) \right\}.$$

JAGS works with the augmented representation, whereas the marginalized representation is appropriate for a Stan implementation. Integrating the conditional membership probabilities from Section 3.2 against this marginal posterior yields $P(Z_i = c \mid Y, Q)$, incorporating uncertainty in all item parameters and profile proportions. Proper priors can regularize estimation, but do not establish likelihood identifiability or validate Q.

Before fitting, prior predictive simulation should draw the proportions, item parameters, allocations, and responses in that order and inspect the implied indicator distributions, regime contrasts, and variability. For municipal data, this assesses whether the assumed model generates plausible patterns on the coded scales. Sensitivity analyses should vary the precision prior, the prior on ordered means, and the Dirichlet concentrations, then compare posterior intervals and profile probabilities. After fitting, posterior predictive checks assess compatibility with the observed responses. The posterior normalizing constant is not available in a convenient closed form; the MCMC methods below approximate posterior summaries, subject to adequate convergence and Monte Carlo accuracy.

5.4.2 MCMC updates in JAGS

The paper reports slice sampling through JAGS and `R2jags` (Oliveira et al. 2026a; Neal 2003). JAGS constructs a directed graphical model and automatically assigns available samplers to stochastic nodes or blocks. Its updates may combine conjugate draws and nonconjugate sampling methods; the exact sampler assignment depends on the graph, loaded modules, and software version, and should not be inferred solely from the name JAGS (Plummer 2017). The following full conditionals

explain the statistical updates required by this model, without claiming that every node is updated by the same algorithm.

First, given the item parameters and profile proportions, each allocation has categorical full conditional probabilities

$$P(Z_i = c \mid Y, \Omega, \pi, Q) \propto \pi_c \prod_{j=1}^{J} f_{j\eta_{cj}}(Y_{ij} \mid \Omega_j), \qquad c = 1, \ldots, C.$$

These probabilities are normalized across profiles. The gate, conditional mean, and precision are then determined by the selected allocation; they are deterministic nodes and do not require separate sampling steps.

Second, with $n_c = \sum_i \mathbf{1}\{Z_i = c\}$, conjugacy gives

$$\pi \mid \mathbf{Z}, Y, \Omega, Q \sim \text{Dirichlet}(1 + n_1, \ldots, 1 + n_C).$$

Third, the continuous item parameters are updated conditional on current allocations and the other parameters. For item j, their joint full conditional is proportional to

$$p(\Omega_j \mid \mathbf{Z}, Y, Q) \propto p(\Omega_j) \prod_{i=1}^{N} f_{j\eta_{Z_i,j}}(Y_{ij} \mid \Omega_j).$$

The beta normalizing function depends on both the mean and precision, so these updates are nonconjugate. A Beta prior on a response mean does not produce a Beta full conditional, and a Gamma precision prior does not produce a Gamma full conditional. In the code, JAGS updates the auxiliary means and precisions, with the ordered means recalculated deterministically.

For a continuous scalar parameter x with unnormalized full conditional $q(x)$, slice sampling introduces an auxiliary height $u \mid x \sim \text{Uniform}(0, q(x))$. It then updates x on the slice $\{x : q(x) > u\}$, typically using interval expansion and shrinkage within the parameter's support. This avoids needing the normalizing constant or a closed-form conditional draw. Repeating the auxiliary-height and parameter updates preserves the target conditional distribution (Neal 2003). Adaptation tunes sampling behavior before retained posterior sampling; burn-in subsequently allows chains to move away from their initial states. These are different purposes, and neither alone demonstrates convergence.

5.4.3 A JAGS implementation

The model below implements the ordered mean–precision specification (Model A) used for Application 1 in Oliveira et al. (2026a). It corresponds to `BDINAmodelA.txt` in the repository's `Application1` directory (Bazán et al. 2026). Model B in the supplementary comparison uses a different log-shape parameterization motivated by Jiang et al. (2025); the comparison concerns these specific implementations and their priors.

The JAGS code is stored as an R character string, `bdina_model`. Here `u0` and `u1` correspond to μ_{j0} and μ_{j1}, `pai` corresponds to π, and `c[i]` corresponds to Z_i. The monitored `loglik` values are conditional on the sampled allocations; they are not the unit-level marginal log likelihood required for predicting new units.

```
bdina_model <- "model{
#likelihood
 for (i in 1:N) {
  for (j in 1:J) {
    for (k in 1:K) {
      w[i, j, k] <- pow(alpha[i, k], q[j, k])
                    }
    eta[i, j] <- prod(w[i, j, 1:K])
    u[i, j] <- ifelse(eta[i,j]==1, u1[j], u0[j])
    phif[i, j] <- ifelse(eta[i,j]==1, phi1[j], phi0[j])
#usual parameterization of the beta distribution
    a[i,j] <- u[i,j]*phif[i,j]
    b[i,j] <- (1-u[i,j])*phif[i,j]
    Y[i,j] ~ dbeta(a[i, j], b[i, j])
#log likelihood
    loglik[i, j] <- logdensity.beta(Y[i,j],a[i, j], b[i, j])
    }
#Priors for Respondent Parameters
  for (k in 1:K) {
alpha[i, k] <- all.patterns[c[i], k]
                  }
    c[i] ~ dcat(pai[1:C])}
  pai[1:C] ~ ddirch(delta[1:C])
#Priors for Item Parameters
  for (i in 1:2){
  for (j in 1:J) {
    muaux[i,j] ~ dbeta(1, 1)
     }
    }
  for (j in 1:J){
     mu[1:2,j] <- sort(muaux[1:2,j])
#ordering of the parameters uj0 and uj1
     u1[j] <- mu[2,j]
     u0[j] <- mu[1,j]
     phi0[j] ~ dgamma(0.01,0.01)
     phi1[j] ~ dgamma(0.01,0.01)
               }

 for (i in 1:N) {
  for (j in 1:J) {
    Y.rep[i,j] ~ dbeta(a[i, j], b[i, j])
    }
    }
sumyrep <- sum(Y.rep[1:N,1:J])
sumy <- sum(Y[1:N,1:J])
  ppp<- step(sumyrep-sumy)
}"
```

The following code loads the municipal data and Q-matrix from the local repository snapshot, checks the beta support, and enumerates the profiles. Run the displayed chunks in order after the helper definitions in Section 3.1. No change to the working directory or removal of existing R objects is required. R2jags requires a separately installed, working JAGS engine.

```
library("rstudioapi")
library("coda")
library("readxl")
library("R2jags")
library("loo")
```

```
library("psych")

#### Setting the directory and loading the functions

rm(list = ls())
setwd(dirname(rstudioapi::getActiveDocumentContext()$path))

#### Reading Q-Matrix and data set

Q <- read.csv("QApp1.csv")
str(Q)
Q<-Q[,-1]
Q
dim(Q)
#19x3

datac <- read.csv("dataApp1.csv")
str(datac)
datac<-datac[,-1]
str(datac)
dim(datac)
#1668x19
describe(datac)

#### Preparing data set to JAGS

N=nrow(datac)
J=ncol(datac)
K=ncol(Q)
C=2^K
delta0 <- rep(1,2^K) #vector with size of possible profiles

#Matrix containing all possible profiles
As <- rep(0,K)
for(j in 1:K){
  temp <- combn(1:K,m=j)
  tempmat <- matrix(0,ncol(temp),K)
  for(j in 1:ncol(temp))
    tempmat[j,temp[,j]] <- 1
  As <- rbind(As,tempmat)
}
As <- as.matrix(As)


para <- c("u0", "phi0", "u1", "phi1", "pai","loglik","ppp")
data <- list(Y=datac, N=N, J=J, K=K,
             C=C, q=Q, all.patterns=As, delta=delta0)
```

Now is possible to fit the B-DINA model for this application using the following code. Note that a full municipal fit can take substantial time.

```
#### Fitting Model with JAGS

start_time <- proc.time()
model_file <- tempfile(fileext = ".bug")
writeLines(bdina_model, model_file)
jagsfit <- R2jags::jags.parallel(data = data, inits = NULL,
                          parameters.to.save = para, model.file = model_file,
                          n.chains = 2,
```

```
                        n.iter = 5000,
                        n.burnin = 1000, n.thin=1, jags.seed = 123)
unlink(model_file)
end_time <- proc.time()
jags_time <- end_time-start_time

#### Saving results
results <- jagsfit$BUGSoutput$summary
results
dim(results)

write.csv(results[1:nrow(results),],"ResultsBDINAmodelAApp1.csv")
```

After the process of fitting, extract posterior summaries by parameter name rather than by fixed rowpositions, which depend on the dataset and monitored quantities.

```
#### Post Analysis
jags_time
jagsfit
summary(jagsfit)
jagsfit$DIC
jagsfit$parameters.to.save


#Computing Item parameters
#deviance
results[1,]
#loglik 2 31692 (1668*19)
results[2:31693]

#pai
results[31694:31701]

#phio
results[31702:31720]

#phi1
results[31721:31739]

#ppp
results[31740]

#muo
results[31741:31759]

#mu1
results[31760:31778]

#model comparison criteria
library(loo)
#extract log-likelihood
loglik_m1<-jagsfit$BUGSoutput$sims.list$loglik
#waic
waic_m1<-waic(loglik_m1)
waic_m1
#looic
looic_m1<-loo(loglik_m1)
looic_m1
```

```
#### Saving Rdata

save.image("FitBDINAModelAApp1.RData")
```

The main estimation description in Oliveira et al. (2026a) specifies two parallel chains, 25,000 iterations per chain, with the first 5,000 discarded. Taking 25,000 as the total and retaining every subsequent iteration gives 20,000 draws per chain and 40,000 pooled draws. The supplementary comparison (Oliveira et al. 2026b) and its repository scripts use two chains, 5,000 total iterations, 1,000 burn-in iterations, and thinning one, giving 8,000 pooled retained draws (Bazán et al. 2026).

The displayed call uses `n.chains = 2`, `n.iter = 5000`, `n.burnin = 1000`, and `n.thin = 1`. Here `n.iter` includes burn-in; adaptation is handled separately by the interface (Su and Yajima 2024). For this run, the pooled retained sample size is

$$M_{\text{retained}} = n_{\text{chains}} \frac{n_{\text{iter}} - n_{\text{burnin}}}{n_{\text{thin}}} = 8000.$$

This expression assumes that the post-burn-in iteration count is divisible by the thinning interval, as in the displayed settings. To use the main paper's total-iteration convention, set `n.iter = 25000` and `n.burnin = 5000`, giving 40,000 pooled retained draws with two chains and no thinning. Neither run length guarantees convergence. For a substantive fit, inspect adaptation warnings, use dispersed valid initial values, and record the software versions and all run settings.

After checking convergence, retained draws of μ_{j0}, μ_{j1}, ϕ_{j0}, ϕ_{j1}, and π_c support posterior means, standard deviations, and credible intervals. Item difficulty and discrimination should also be computed for each draw, retaining their posterior uncertainty. Profile probabilities can be averaged over parameter draws as in Section 3.2. These samples are autocorrelated, so the retained count is not the number of independent observations: effective sample size and Monte Carlo standard error quantify the accuracy of posterior summaries. Thinning reduces storage but does not establish convergence or create additional information.

Following the paper, inspect trace plots and between-chain convergence statistics near one; also examine mixing, effective sample sizes, and Monte Carlo error for the quantities actually reported. Increase run lengths or revise the parameterization when diagnostics are inadequate. Auxiliary unsorted means may exchange labels even though the ordered item parameters are anchored. A completed run alone does not demonstrate an adequate posterior approximation.

To obtain posterior class probabilities, normalize the weighted class likelihoods at each retained draw s:

$$r_{ic}^{(s)} = \frac{\pi_c^{(s)} \prod_j f_{j\eta_{cj}}(y_{ij} \mid \Omega_j^{(s)})}{\sum_d \pi_d^{(s)} \prod_j f_{j\eta_{dj}}(y_{ij} \mid \Omega_j^{(s)})}.$$

Averaging these responsibilities over posterior draws integrates parameter uncertainty. Population proportions π_c differ from the unit-specific probabilities $P(Z_i = c \mid Y, Q)$.

Oliveira et al. (2026a) use $T(Y) = \sum_{ij} Y_{ij}$ and $PPP = P\{T(Y^{rep}) \geq T(Y) \mid Y\}$ for posterior predictive checking. A value near .5 indicates agreement for this statistic, not proof of complete model fit. Also examine item variances, quantiles, and residual dependence.

For predicting new units, use the **person-marginal** log likelihood $\ell_i^{(s)} = \log \sum_c \pi_c^{(s)} \prod_j f_{j\eta_{cj}}(y_{ij})$. For predicting a missing item of an existing unit, a conditional response likelihood targets a different task. Criteria from the two targets cannot be compared interchangeably.

Following Vehtari et al. (2017), WAIC on the deviance scale is $-2(\text{lppd} - p_{WAIC})$, where $\text{lppd} = \sum_i \log\{S^{-1} \sum_s \exp(\ell_i^{(s)})\}$ and the usual variance penalty is $p_{WAIC} = \sum_i \text{Var}_s(\ell_i^{(s)})$. The supporting information (Oliveira et al. 2026b) instead prints the alternative penalty $2\sum_i[\log E_s\{\exp(\ell_i^{(s)})\} - E_s(\ell_i^{(s)})]$. Label the convention used; `loo::waic` uses the variance penalty. The DIC convention reported in the supporting information is $\bar{D} + \text{Var}(D)/2$ (Oliveira et al. 2026b).

For PSIS-LOO, use `loo` with a draws-by-unit matrix of marginal log likelihoods, inspect Pareto-k diagnostics, and refit or use K-fold validation when importance sampling is unreliable. Preserve chain information when estimating relative efficiency (Vehtari et al. 2017). Continuous densities can exceed one, so negative deviance or WAIC is possible.

5.5 Practical Estimation: Municipal Indicators: Published Application

5.5.1 Data and substantive coding

The application covers Southeastern Brazilian municipalities and 19 indicators from the Atlas of Human Development. Attribute order is Economy, Health, Education (Oliveira et al. 2026a).

Larger coded responses are intended to represent favorable conditions; adverse indicators require the documented reversal. For example, the variable GINI is used as equality, $1 - \text{GINI}$.

There is a source discrepancy: the introduction refers to 1,688 municipalities, whereas the supplement's comparison gives $N = 1,668$. Use the actual number of rows after documenting exclusions when reproducing the analysis. The repository data used below confirm 1,668 observations; no observations are inferred from published summaries.

Table 4: Municipal Q-matrix transcribed from Table 7.

	Economy	Health	Education
DEP	1	0	0
S60	0	1	0
ILIT	1	0	1
HIGH	1	0	1
GINI	1	0	0
PPOV	1	0	0
INC1	1	0	0
UNEM	1	0	0
BWAT	1	1	0
GARB	1	1	0
LIGHT	1	0	0
SANI	0	1	0
ELPOV	1	0	1
CHILD	0	1	1
HEAD	0	0	1
WORKS	1	0	1
HDEI	0	0	1
HDLI	0	1	0
HDII	1	0	0

	Economy	Health	Education

When preparing the real data, match columns to these names, retain municipality identifiers separately, and record theoretical rescaling and reversals. Inspect endpoints before fitting. Marginal normality tests or Cronbach's alpha do not validate a conditional beta likelihood, the DINA gate, or the proposed Q.

5.5.2 Application code and data from the repository

The application now uses the files supplied by 1. The original files are retained under `A-Beta-DINA-Model/Application1/`, together with the repository README and license files. The repository provides its own recommended citation, used in the references below.

File	Role
`dataApp1.csv`	Already coded municipal responses
`QApp1.csv`	Application Q-matrix
`BDINAmodelA.txt`	Original ordered mean–precision JAGS model
`BDINAModelB.txt`	Original alternative log-shape JAGS model
`FitBDINAModelAApp1.R`	Original Model A fitting and comparison script
`FitBDINAModelBApp1.R`	Original Model B fitting and comparison script
`ResultsBDINAmodelAApp1.csv`	Saved posterior summaries supplied by the authors

The original scripts drop the first column in both CSV files, generate profiles in increasing number of attributes, and use `jags.parallel` with two chains, 5,000 iterations, 1,000 burn-in iterations, thinning 1, and seed 123.

Table 5 summarizes the 19 coded indicators for the 1,668 municipalities in the repository dataset. It links each indicator abbreviation to its source column and reports the observed mean, standard deviation, minimum, and maximum. These are descriptive summaries across municipalities, before conditioning on latent attribute profiles; they are distinct from the regime-specific beta parameters estimated by B-DINA.

Table 5: Descriptive statistics computed from the repository's municipal data.

	indicator	source_column	mean	sd	min	max
DEP	DEP	RAZDEP	0.953	0.006	0.924	0.970
S60	S60	SOBRE60	0.823	0.025	0.720	0.881
ILIT	ILIT	T_ANALF18M	0.884	0.064	0.617	0.985
HIGH	HIGH	T_MED18M	0.287	0.092	0.080	0.662
GINI	GINI	GINI	0.534	0.055	0.220	0.680
PPOV	PPOV	PPOB	0.680	0.160	0.216	0.962
INC1	INC1	REN1	0.739	0.161	0.223	0.953
UNEM	UNEM	T_DES18M	0.940	0.027	0.792	0.995
BWAT	BWAT	T_BANAGUA	0.945	0.086	0.383	0.995
GARB	GARB	T_LIXO	0.972	0.053	0.440	0.995
LIGHT	LIGHT	T_LUZ	0.990	0.016	0.812	0.995
SANI	SANI	AGUA_ESGOTO	0.976	0.040	0.696	0.995
ELPOV	ELPOV	T_FUNDIN_TODOS_MMEIO	0.861	0.088	0.529	0.990
CHILD	CHILD	T_M10A17CF	0.939	0.038	0.594	0.995
HEAD	HEAD	T_MULCHEFEFIF014	0.842	0.074	0.444	0.995

	indicator	source_column	mean	sd	min	max
WORKS	WORKS	T_NESTUDA_NTRAB_MMEIO	0.891	0.061	0.596	0.995
HDEI	HDEI	IDHM_E	0.608	0.083	0.339	0.825
HDLI	HDLI	IDHM_L	0.828	0.027	0.723	0.890
HDII	HDII	IDHM_R	0.681	0.058	0.502	0.891

Table 5 shows substantial differences in both the level and variability of the indicators. LIGHT and SANI have high means (0.990 and 0.976) and small standard deviations (0.017 and 0.040), indicating that their coded values are concentrated near the upper bound. In contrast, PPOV and INC1 have standard deviations of approximately 0.160 and 0.161, revealing greater between-municipality variation. HIGH has the lowest mean (0.287), whereas HDEI averages 0.608. These comparisons describe the indicators on their respective coded scales; equal numerical values need not represent equal substantive conditions across indicators. All recorded values lie strictly inside $(0, 1)$, satisfying the support requirement of the beta likelihood for this supplied dataset.

Figure 3 complements Table 5 with histograms of six indicators selected to illustrate different distributional shapes: PPOV, INC1, GINI, LIGHT, SANI, and HDEI. Each panel shows municipality counts over the same unit-interval response scale, allowing the location, asymmetry, and concentration of the observed distributions to be compared.

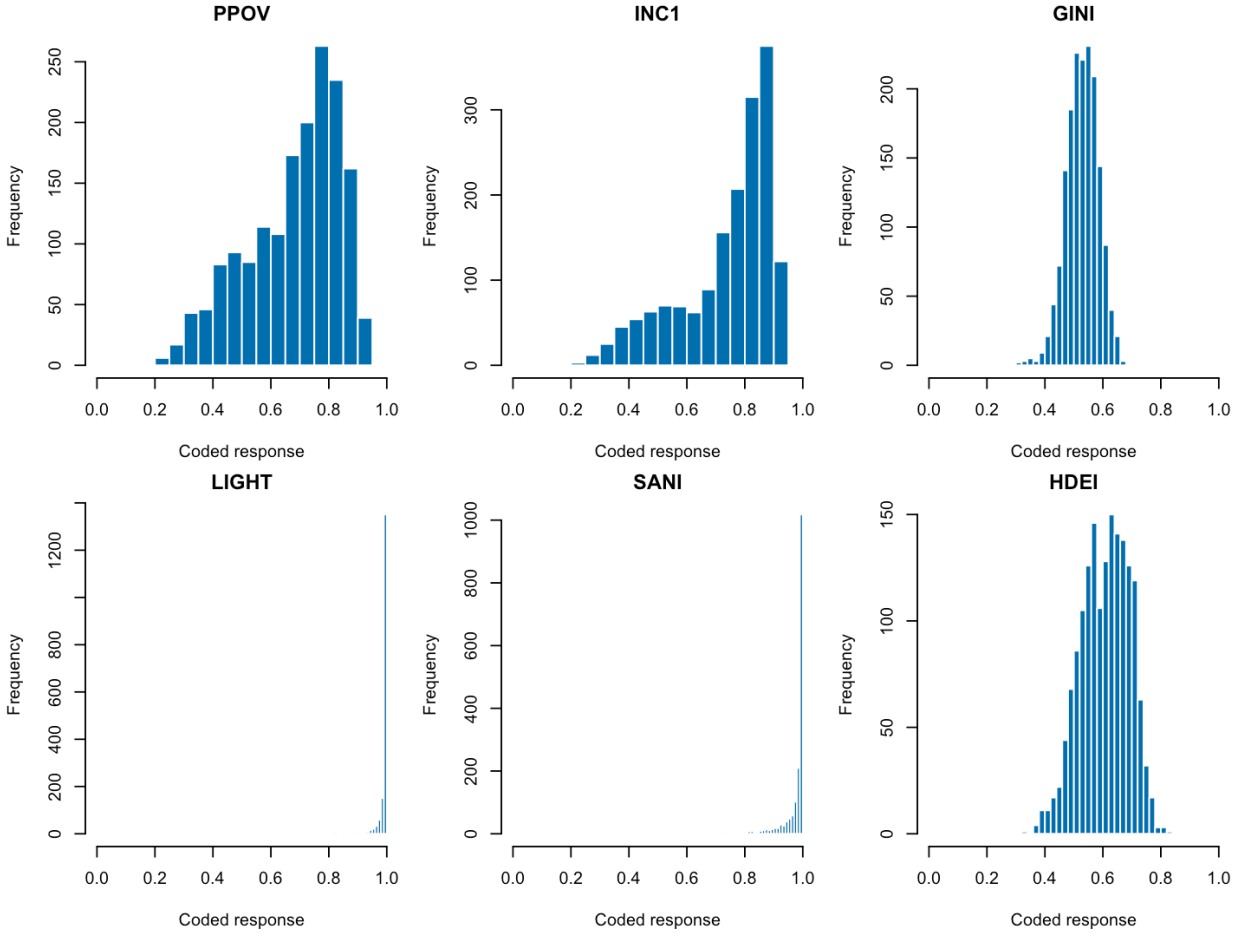


Figure 3: Observed distributions of six municipal indicators from the repository data.

PPOV and INC1 display long lower tails, with most municipalities toward the upper part of their coded scales. GINI and HDEI have more nearly symmetric, centrally concentrated distributions, while LIGHT and SANI show pronounced concentration near the recorded upper limit of 0.995 and relatively few lower observations. The repeated upper values warrant attention to the source's coding and recording rules; their concentration alone does not establish a distinct latent class. Together, Table 5 and Figure 3 motivate a bounded response model that permits different means, variances, and asymmetries across indicators. However, these marginal distributions mix over latent profiles: they do not establish that the conditional beta regimes, the conjunctive gate, or local independence fit the data. Those assumptions require assessment after fitting the diagnostic model.

In the following we show results based in the MCMC output considering the specifications give in the github explained in section 4 which are different of the given in the paper.

```
repository_summary <- read.csv(file.path(application_dir,
  "ResultsBDINAmodelAApp1.csv"), row.names = 1, check.names = FALSE)
item_nodes <- unlist(lapply(c("u0", "u1", "phi0", "phi1"),
  function(p) paste0(p, "[", seq_len(ncol(Y_municipal)), "]")))
stopifnot(all(item_nodes %in% rownames(repository_summary)))
repository_items <- repository_summary[item_nodes,
  c("mean", "sd", "2.5%", "97.5%", "Rhat", "n.eff")]
knitr::kable(repository_items, digits = 3,
 caption = "Saved repository item estimates, equal-tail intervals, and
    convergence diagnostics.")
```

Table 6: Saved repository item estimates, equal-tail intervals, and convergence diagnostics.

	mean	sd	2.5%	97.5%	Rhat	n.eff
u0[1]	0.948	0.000	0.947	0.948	1.021	77
u0[2]	0.810	0.001	0.809	0.812	1.001	3500
u0[3]	0.832	0.002	0.828	0.837	1.058	32
u0[4]	0.213	0.002	0.209	0.217	1.085	23
u0[5]	0.517	0.002	0.513	0.521	1.001	5600
u0[6]	0.505	0.005	0.494	0.515	1.121	17
u0[7]	0.573	0.006	0.561	0.586	1.075	26
u0[8]	0.936	0.001	0.933	0.938	1.003	610
u0[9]	0.906	0.003	0.899	0.912	1.012	140
u0[10]	0.952	0.002	0.948	0.955	1.002	940
u0[11]	0.980	0.001	0.978	0.981	1.007	250
u0[12]	0.956	0.002	0.953	0.959	1.001	8000
u0[13]	0.779	0.003	0.773	0.785	1.090	22
u0[14]	0.938	0.001	0.935	0.940	1.001	2600
u0[15]	0.804	0.003	0.798	0.810	1.007	250
u0[16]	0.841	0.002	0.837	0.846	1.040	44
u0[17]	0.533	0.003	0.528	0.538	1.031	54
u0[18]	0.816	0.001	0.814	0.818	1.001	8000
u0[19]	0.624	0.002	0.621	0.628	1.070	27
u1[1]	0.956	0.000	0.955	0.956	1.004	430
u1[2]	0.835	0.001	0.833	0.836	1.001	3800
u1[3]	0.921	0.001	0.919	0.923	1.006	290
u1[4]	0.341	0.003	0.336	0.346	1.023	73
u1[5]	0.543	0.002	0.540	0.546	1.005	330
u1[6]	0.774	0.003	0.768	0.778	1.001	6200
u1[7]	0.828	0.002	0.824	0.832	1.003	700
u1[8]	0.942	0.001	0.941	0.944	1.001	3700

	mean	sd	2.5%	97.5%	Rhat	n.eff
u1[9]	0.987	0.000	0.986	0.988	1.007	250
u1[10]	0.993	0.000	0.993	0.994	1.006	270
u1[11]	0.995	0.000	0.995	0.995	1.000	1
u1[12]	0.995	0.000	0.995	0.995	1.000	1
u1[13]	0.920	0.001	0.918	0.923	1.036	48
u1[14]	0.940	0.001	0.938	0.943	1.001	8000
u1[15]	0.869	0.002	0.865	0.872	1.006	310
u1[16]	0.927	0.001	0.925	0.929	1.006	290
u1[17]	0.660	0.002	0.657	0.664	1.022	76
u1[18]	0.840	0.001	0.838	0.841	1.002	1700
u1[19]	0.712	0.001	0.709	0.714	1.001	8000
phi0[1]	1375.666	83.217	1219.383	1544.089	1.006	310
phi0[2]	242.341	11.816	219.884	265.979	1.001	8000
phi0[3]	35.306	1.856	31.722	39.009	1.017	95
phi0[4]	61.379	3.335	55.185	68.141	1.069	27
phi0[5]	103.488	6.094	91.995	115.685	1.013	120
phi0[6]	16.483	0.985	14.629	18.502	1.204	12
phi0[7]	10.085	0.585	8.981	11.251	1.071	27
phi0[8]	60.107	3.694	53.137	67.583	1.013	120
phi0[9]	9.291	0.480	8.376	10.238	1.001	3200
phi0[10]	15.776	0.839	14.178	17.485	1.001	6700
phi0[11]	57.858	3.773	50.748	65.678	1.009	180
phi0[12]	22.707	1.227	20.345	25.117	1.001	5200
phi0[13]	32.805	1.790	29.390	36.426	1.071	27
phi0[14]	33.796	1.608	30.694	37.080	1.001	7100
phi0[15]	24.009	1.337	21.476	26.719	1.001	5200
phi0[16]	37.312	2.044	33.394	41.509	1.010	170
phi0[17]	77.312	4.856	68.179	87.422	1.010	160
phi0[18]	210.928	10.646	190.627	231.888	1.004	460
phi0[19]	120.528	7.531	106.050	135.481	1.145	15
phi1[1]	3712.377	174.966	3375.239	4066.428	1.017	100
phi1[2]	379.374	19.071	342.810	416.562	1.001	4300
phi1[3]	95.515	4.362	87.191	104.254	1.001	8000
phi1[4]	39.874	1.852	36.340	43.580	1.003	770
phi1[5]	78.128	3.370	71.572	84.766	1.006	280
phi1[6]	25.977	1.162	23.791	28.269	1.012	140
phi1[7]	31.859	1.526	28.933	34.951	1.014	120
phi1[8]	80.731	3.481	74.032	87.749	1.012	140
phi1[9]	146.878	8.308	131.468	163.859	1.003	600
phi1[10]	890.062	64.141	773.293	1021.534	1.014	110
phi1[11]	26700.074	7052.390	15350.625	35265.464	6.501	2
phi1[12]	14849.374	832.517	13541.331	16605.265	1.028	68
phi1[13]	60.792	2.975	55.095	66.750	1.011	150
phi1[14]	34.642	1.892	31.103	38.512	1.001	4100
phi1[15]	38.870	1.802	35.368	42.470	1.001	8000
phi1[16]	70.624	3.349	64.196	77.245	1.001	3700
phi1[17]	80.169	3.774	72.926	87.978	1.010	160
phi1[18]	288.851	14.161	261.532	316.959	1.002	1000
phi1[19]	130.907	5.742	120.111	142.568	1.001	8000

```
important_nodes <- c( paste0("pai[", 1:8, "]"), "ppp")
knitr::kable(repository_summary[important_nodes,
  c("mean", "sd", "2.5%", "97.5%", "Rhat", "n.eff")], digits = 3,
 caption = "Saved repository population, fit, and sampling summaries.")
```

Table 7: Saved repository population, fit, and sampling summaries.

	mean	sd	2.5%	97.5%	Rhat	n.eff
pai[1]	0.300	0.012	0.276	0.324	1.001	3600
pai[2]	0.026	0.005	0.017	0.036	1.053	34
pai[3]	0.038	0.006	0.026	0.051	1.053	35
pai[4]	0.006	0.004	0.001	0.016	1.016	100
pai[5]	0.046	0.006	0.034	0.059	1.003	700
pai[6]	0.159	0.010	0.141	0.178	1.022	73
pai[7]	0.006	0.003	0.001	0.013	1.010	210
pai[8]	0.419	0.013	0.394	0.444	1.025	66
ppp	0.372	0.483	0.000	1.000	1.003	810

Note that some parameters has conventional R-hat different of 1, and smaller effective sample size. These diagnostics indicate poor between-chain agreement in that run mainly explained because we are using in the example number of iterations and burnin less than the reported in the paper. Inspect item and proportion diagnostics before treating the saved estimates or comparison criteria as stable.

5.5.3 Comparison with results in the paper

Table 9 prints square roots of the estimated precisions reported in the paper. They must be squared before using the beta variance formula. The following values are published, rounded estimates; derived variances and distances here use those rounded values.

Table 8: Published Table 9 estimates; mean gaps recomputed from rounded means. H retains the source's label.

item	mu0	sqrt_phi0	mu1	sqrt_phi1	KS_published	H_published	mean_gap
DEP	0.947	36.523	0.955	60.512	0.62	0.33	0.008
S60	0.811	15.612	0.835	19.532	0.42	0.14	0.024
ILIT	0.834	5.935	0.922	9.847	0.69	0.40	0.088
HIGH	0.214	7.769	0.343	6.373	0.69	0.41	0.129
GINI	0.521	9.889	0.543	8.739	0.17	0.03	0.022
PPOV	0.500	3.939	0.765	4.727	0.78	0.53	0.265
INC1	0.606	3.048	0.841	6.701	0.75	0.49	0.235
UNEM	0.933	7.867	0.943	8.924	0.12	0.01	0.010
BWAT	0.876	2.627	0.977	7.054	0.56	0.27	0.101
GARB	0.952	4.006	0.994	34.377	0.71	0.43	0.042
LIGHT	0.979	7.505	0.995	190.939	0.80	0.57	0.016
SANI	0.958	4.756	0.995	123.062	0.85	0.57	0.037
ELPOV	0.778	5.869	0.921	7.951	0.84	0.63	0.143
CHILD	0.938	5.533	0.940	6.108	0.02	0.00	0.002
HEAD	0.818	5.035	0.876	6.320	0.34	0.10	0.058
WORKS	0.841	6.113	0.927	8.453	0.67	0.38	0.086
HDEI	0.534	8.705	0.662	9.035	0.75	0.49	0.128
HDLI	0.817	14.530	0.840	16.996	0.37	0.11	0.023
HDII	0.622	10.766	0.709	11.168	0.69	0.41	0.087

PPOV and INC1 have large mean gaps. CHILD, GINI, and UNEM have small published discrimination indices. SANI and LIGHT illustrate how precision can strongly distinguish distributions even when the means are close.Note that these results are comparable to those obtained previously, despite using a smaller MCMC sample size.

In the following we show the published profile proportions and fit

Table 9: Published Table 10 population proportions and 95% HPD intervals, transcribed without correction.

profile	Economy	Health	Education	proportion	lower	upper
1	0	0	0	0.279	0.256	0.302
2	1	0	0	0.059	0.045	0.075
3	0	1	0	0.029	0.018	0.040
4	0	0	1	0.010	0.005	0.017
5	1	1	0	0.055	0.043	0.068
6	1	0	1	0.157	0.139	0.176
7	0	1	1	0.004	0.001	0.009
8	1	1	1	0.407	0.407	0.432

Note that these results are comparable to those obtained previously, despite using a smaller MCMC sample size. For example, Table 10 reports .407 for the all-positive profile, whereas .419 was the result before. The published total-score PPP is .404 and the obtained before was .372, which is close.

5.6 Exercises

For each fitted model, document the data coding, estimation settings, and convergence diagnostics, and report uncertainty alongside parameter estimates.

1. Discuss as can be improved the estimation of the B-DINA model modifying the number of samples MCMC and burning.
2. Fit bounded IRT models to the municipal dataset and compare their item-level results with those obtained using B-DINA. Relate the IRT item parameters to the B-DINA item parameters, explaining why their numerical values are not directly interchangeable. What complementary information do continuous latent scores and discrete attribute profiles provide about municipal conditions?
3. Fit the B-DINA specification developed here to the visual analogue scale dataset analyzed by Jiang et al. (2025), provided as Application 2 in the repository of
    1. Describe the response coding, attribute definitions, and Q-matrix. Interpret the estimated item regimes and population profile proportions, and summarize uncertainty in individual profile assignments.
4. models to the dataset used in Exercise 2. Compare the item-level results and substantive conclusions with those from B-DINA. Explain what each approach reveals about response patterns and how the choice between continuous traits and discrete attributes changes the interpretation of individual differences.
5. Examine the repository code for Model A, developed here, and Model B, based on Jiang et al. (2025; Bazán et al. 2026). Express both models using beta means and precisions, and compare their parameterizations, ordering constraints, and priors. Does Model B guarantee a higher conditional mean in the mastery regime? Explain how the differences may affect estimation, interpretation, and model comparison.

6. Reproduce the comparisons between Models A and B for Applications 1 and 2 reported in the Supporting Information (Oliveira et al. 2026b) of Oliveira et al. (2026a). State the sampling settings and the definitions of the reported fit criteria. Distinguish criteria based on conditional likelihoods from those based on likelihoods marginalized over latent profiles, and identify the prediction task each targets. Compare your results with the published summaries and discuss any discrepancies in light of Monte Carlo uncertainty, convergence, and prior specification.
7. Discuss as improve the computational time for the B-DINA model by considering other MCMC algorithms.

5.7 Conclusions

Beta B-DINA (or Bounded DINA) is a Cognitive Diagnostic Model (CDM) and a latent variable model that extends the traditional DINA (Deterministic Inputs, Noisy "And" Gate) model to handle bounded continuous responses (such as proportions, rates, or index scores on a 0–1 or 0–100 scale) rather than strict binary (0 or 1) or categorical outcomes.

The model studied here follows Oliveira et al. (2026a) and parameterizes the beta distribution in terms of its mean and precision, linking each item's conditional mean to the latent attribute profile. Its relationship to the model proposed by Jiang et al. (2025) is examined in Oliveira et al. (2026b). The model, like the standard DINA model, it is conjunctive: a respondent typically needs to master all required attributes (specified via the Q-matrix) to achieve the higher expected response state, while lacking required attributes shifts the expected Beta distribution mean lower. In the present proposal, it is typically estimated using a Bayesian framework (such as MCMC), incorporating techniques to handle issues like label-switching nonidentifiability and evaluating model fit via posterior predictive (p)-values.

The model links bounded continuous indicators to discrete attribute profiles through a conjunctive Q-matrix gate. Means and precisions characterize each item's two regimes, while posterior profile probabilities express uncertainty in classification. Strong distributional separation and informative Q designs are central to reliable diagnosis.

By considering a application with municipal indicator, the proposed B-DINA model effectively classifies municipalities and reveals relationships between observed indicators and latent attributes. As bounded continuous variables are common across the social sciences and policy analysis, the proposed B-DINA could offer a broadly applicable classification tool in the practice.

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